

Idle Currents

Turning Pakistan's Underused Power into a Global Data Centre Hub

What Pakistan Must Change in Its Electricity Laws, Regulations, Contracts and Institutions

By Reza Khan and Barrister Waqas Aziz Qureshi

17 September 2026

Suggested citation: Khan, R. and Qureshi, W. A. (2026). Idle Currents: Turning Pakistan's Underused Power into a Global Data Centre Hub, ARS Special Report ARS-SR-2026-001. Ardent Research Solutions, Islamabad. www.arspakistan.pk

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About This Report

Idle Currents asks whether Pakistan can convert underused electricity into globally traded computing capacity, and what would have to change in law, regulation, contract and institutional practice before it could proceed with it. It treats the question as a legal and commercial one rather than a promotional one. Electricity is abundant on paper in Pakistan. The right to buy it, at a known price, for a known term, with a known remedy if it is not delivered, is not.

The report is organised in six parts. Part I measures what Pakistan actually has rather than what it has installed, and sets out how the independent power producer model allocates payment and risk. Part II asks who under existing Pakistani law may lawfully sell electricity to a very large private consumer, examining captive generation and hydropower, the resource most often invoked in this debate and the one most often misdescribed. Part III turns to demand: what a hyperscale data centre requires, what Pakistan would actually be selling, and the five power procurement structures available. Part IV analyses the NEPRA Act, 1997 and the surrounding regulatory framework provision by provision, identifying precisely where the transaction breaks. Part V sets out the ARS reform package, the proposed legal model, a bankability scorecard applying the tests a hyperscaler's lenders and lawyers would apply, and the contractual architecture required. Part VI addresses siting, scenario economics, an implementation sequence and the risks that remain, including the risk that Pakistan solves one stranded asset problem only to create another.

This is the first report in a continuing ARS series on Pakistan's digital infrastructure economy. Part Two will follow.

A NOTE ON METHOD

This report distinguishes throughout between observed data, calculations derived by ARS from observed data, and illustrative scenarios. Charts and tables state which category applies. Where a legal proposition is advanced, the act, section, regulation or notification is cited. Where the law is discretionary rather than mandatory, the report says so rather than describing a gap that does not exist. Nothing in this report is legal advice or a legal opinion, and no part of it should be relied upon as such.

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Executive Summary

Pakistan closed the 2024-25 financial year with 41,121 MW of installed generation capacity, a reported system peak of just above 33,000 MW, and net generation of 135,078 GWh.^{1,2} Divide that energy across the hours of the year and the fleet ran at an average load of roughly 15,400 MW. The country therefore maintains, insures, indexes and pays for a generation system more than two and a half times larger than the load it actually serves on average. This is not a shortage. It is a utilisation failure with a contractual cause.

The cost of that failure is visible in the tariff. For FY 2025-26 the National Electric Power Regulatory Authority determined a power purchase price of Rs 3,342.369 billion for the ex-WAPDA distribution companies, of which capacity charges were projected at between Rs 2.03 trillion and Rs 2.16 trillion, or approximately 63% of the total.³ Almost two thirds of what Pakistani consumers pay for electricity is not payment for electricity. It is payment for the availability of plant, much of which is not called upon.

The conventional response has been to retire capacity. In FY 2024-25 the government terminated or renegotiated contracts covering 2,829 MW, which NEPRA estimated could save more than Rs 900 billion over time, and installed capacity fell from 45,888 MW to 41,121 MW.^{2,4} That is one way to close the gap between capacity and load. The other way is to add load. A hyperscale data centre is the only category of new industrial demand available to Pakistan that is large, flat, growing globally, priced in foreign currency and largely indifferent to where on earth it sits, provided the power is reliable, the price is predictable and the contract is enforceable. The International Energy Agency puts global data centre electricity consumption at 415 TWh in 2024, rising to approximately 945 TWh by 2030, with AI-optimised facilities the fastest-growing component.⁵ Its updated assessment puts 2025 consumption at 485 TWh and 2030 at 950 TWh, and records that data centre electricity demand grew 17% in 2025 against 3% growth in global electricity demand overall.⁶

The arithmetic of adding load is more favourable than the arithmetic of removing plant. A plant carrying a fixed capacity charge delivers electricity at a cost per unit that falls steeply with utilisation. On an illustrative basis, moving a plant from roughly 12% utilisation to roughly 85% utilisation cuts the delivered cost per unit by around 60%, because the same fixed charge is spread across seven times as many units. Retiring capacity removes the charge. Utilising capacity removes the charge and generates revenue.

The obstacle is not physics and it is not, in the main, an absence of law. Pakistan has a competitive market architecture. Sections 23A to 23H of the Regulation of Generation, Transmission and Distribution of Electric Power Act, 1997, inserted by the 2018 Amendment Act, create licence categories for market operators, traders, suppliers and system operators.⁷ Section 22 permits sales to bulk power consumers. Section 14C confers a right of open access on captive generators. The NEPRA (Wheeling of Electric Power) Regulations, 2016 allow a licensed generator to sell to a consumer of 1 MW or more anywhere in the country.⁸ The Competitive Trading Bilateral Contract Market was

formally launched in January 2026, with a first competitive auction of approximately 800 MW scheduled for September 2026.^{9,10}

What Pakistan has is a permission architecture, not a rights architecture. Section 22 provides that the Authority "may permit" the sale of electric power to bulk power consumers. Section 14C(2) grants a right of open access to captive generators, but subjects it to availability of adequate transmission facility "determined by the national grid company or the provincial grid company", with disputes going to the regulator. Wheeling charges have moved from above Rs 22 per unit to below Rs 9 per unit, but they moved by administrative decision, and what moves by administrative decision can move back.^{9,11} A lender underwriting a fifteen-year facility against a Pakistani data centre does not ask whether the transaction is permitted today. It asks who holds the discretion, on what criteria it is exercised, and what happens when it is exercised differently.

Part One reaches four findings.

First, Pakistan's surplus is a surplus of capacity, not a surplus of energy, and the two are routinely conflated. Underused capacity is real and large. Cheap surplus electricity available on demand at a fixed low price is not a thing that presently exists, because the plants that are idle are largely the expensive ones that sit low in the merit order, and the plants that are cheap are largely dispatched already. Any proposition that Pakistan can sell a hyperscaler cheap power simply because plant is idle mistakes an accounting surplus for an economic one.

Second, the independent power producer model contains a genuine and underexploited commercial opportunity, but it is an opportunity in contract restructuring rather than in generation. The value lies in converting a capacity payment that is currently a pure liability into a payment partly recovered from a new, creditworthy, dollar-earning load, and in doing so without creating any new capacity obligation. That requires a bilateral contract between a generator and a consumer, wheeled over a network neither controls, at a use of system charge neither sets.

Third, captive generation cannot carry a hyperscale campus on its own and Parliament has recently made the gas-fired route substantially worse. The Off the Grid (Captive Power Plants) Levy Act, 2025 imposes a levy on natural gas and RLNG consumed by captive plants, calibrated to the gap between NEPRA's industrial B3 tariff and the plant's own generation cost, stepping up to 20% from August 2026.^{12,13} The notified rate reached Rs 1,406 per MMBtu by April 2026.¹⁴ This is deliberate policy and it is working as intended. It also means that any data centre power strategy built on gas self-generation is building on a base that the legislature has committed itself to eroding.

Fourth, hydropower is Pakistan's strongest long-run card and its weakest short-run one. WAPDA generated 33.12 billion units in calendar year 2025 at an average cost of Rs 3.83 per unit.¹⁵ That is genuinely cheap energy. But ARS calculation from WAPDA's own generation statement shows that the major stations ran at annual capacity factors between 36% and 51%. Ghazi Barotha, the best performer, reached 51.2%. Those are utilisation figures rather than availability figures, and the annual average is itself misleading: NEPRA's monthly data for FY 2024-25 shows the six largest WAPDA stations running at

80% of nameplate in August and 12% in January, and below their own full year average for six consecutive months. The 969 MW Neelum-Jhelum plant has been out of service since a tunnel collapse on 2 April 2024 and WAPDA told the Senate Standing Committee on Water Resources in July 2026 that it hopes to restore generation by March 2028.^{16,17} Hydro in Pakistan delivers low-cost energy on a seasonal profile. It does not deliver firm power, and selling it as though it does would be a straightforward misrepresentation to an investor.

THE ARS POSITION, IN ONE SENTENCE

Pakistan does not need to invent a data centre electricity market from scratch; it needs to convert a discretionary permission architecture into a bankable rights architecture, and the first step is to stop describing idle capacity as though it were cheap available energy.

PAKISTAN'S POWER SURPLUS

1. Pakistan's Power Paradox

1.1 Too Much Generation, Too Little Demand

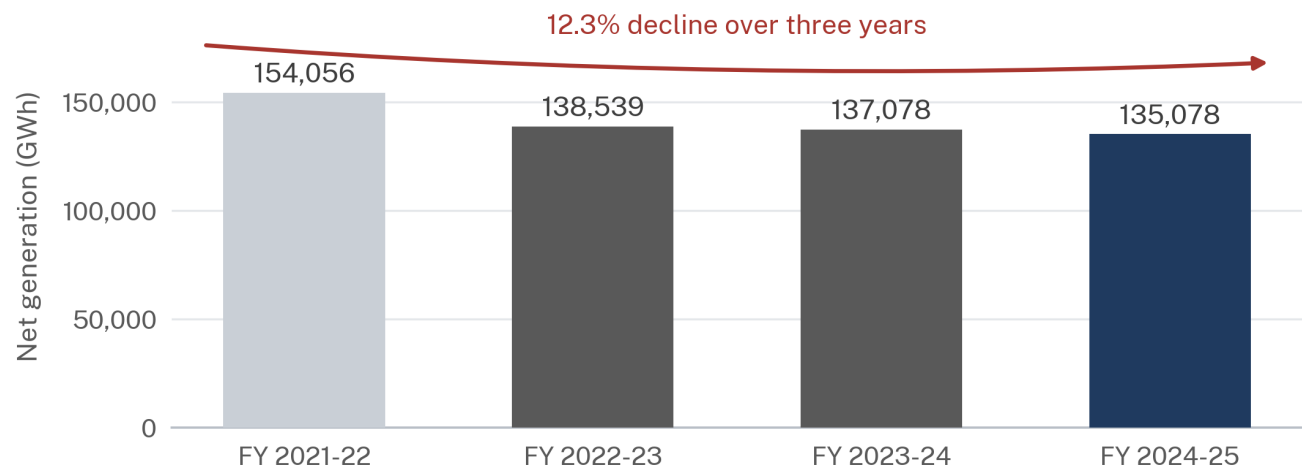
NEPRA's State of Industry Report 2025, released in January 2026 and covering the year ended 30 June 2025, records installed generation capacity of 41,121 MW, down from 45,888 MW twelve months earlier following the retirement or decommissioning of 2,829 MW of inefficient plant.² Reported system peak demand was just above 33,000 MW.⁴ Total net generation was 135,078 GWh, the fourth consecutive annual decline from 154,056 GWh in FY 2021-22.¹

The demand picture is worse than the headline suggests because consumer numbers have continued to grow. Grid sales in FY 2024-25 reached 111 TWh, an increase of just 1.7% against consumer growth of 5.3%.¹⁸ Over a five-year comparison, real GDP grew at roughly 3% per annum and population at roughly 2%, while grid electricity sales contracted at approximately 1% per annum.¹⁸ The link between economic activity and grid demand has broken.

It has broken for an identifiable reason. Pakistan added approximately 5.3 GW of cumulative net-metered rooftop solar capacity by April 2025, having installed 2.8 GW in the first ten months of FY 2024-25 alone, a tenfold increase in two years.¹⁹ Cumulative net metering capacity reached roughly 6.1 GW by June 2025. Consumers facing a national average tariff of Rs 34 per unit did what rational consumers do, and the grid lost the customers who used the most electricity at the highest tariff. Those customers were precisely the ones cross-subsidising the rest of the system.

Figure 1. Grid Generation Has Fallen for Three Consecutive Years While Capacity Was Still Being Added

Total net electricity generation, Pakistan, GWh



Source: NEPRA, State of Industry Report 2025. Figures are net generation on the national grid and exclude off-grid and net-metered self-generation.

This is the paradox in its simplest form. Pakistan built generation capacity for a demand curve that did not arrive, financed it with contracts that pay whether or not the plant runs, priced the resulting cost into a tariff, and thereby drove away the demand that would have justified the capacity. Each turn of the cycle makes the next turn worse.

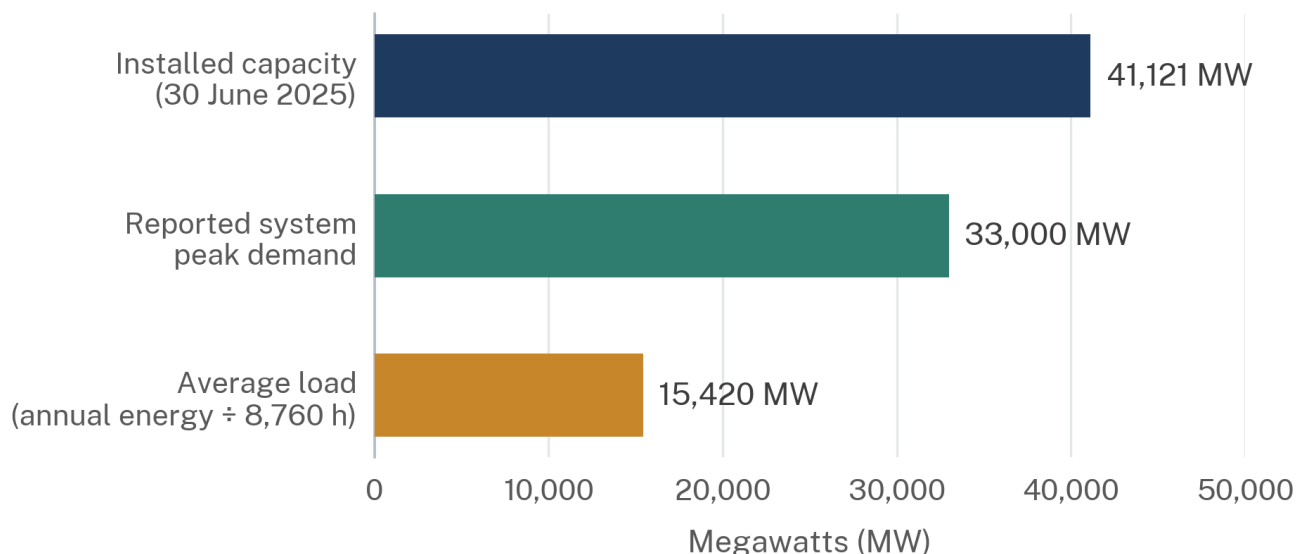
1.2 Installed Capacity versus Actual Utilisation

Installed capacity is a nameplate figure. It is what the plates on the machines say the machines could produce if everything worked, the fuel arrived, the water flowed, the network could carry the output and someone wanted to buy it. It is the least useful number in the Pakistani power debate and the most frequently quoted.

A more informative measure is average load. Net generation of 135,078 GWh spread across the 8,760 hours of a year corresponds to an average load of approximately 15,420 MW. Against installed capacity of 41,121 MW, that is a system-wide load factor of about 37.5%. Even measured against the reported peak of 33,000 MW, average load is under half. Pakistan is not running a power system at capacity. It is running a power system at slightly more than a third of capacity while paying for all of it.

Figure 2. Pakistan Runs a 41 GW Fleet at an Average Load of Roughly 15 GW

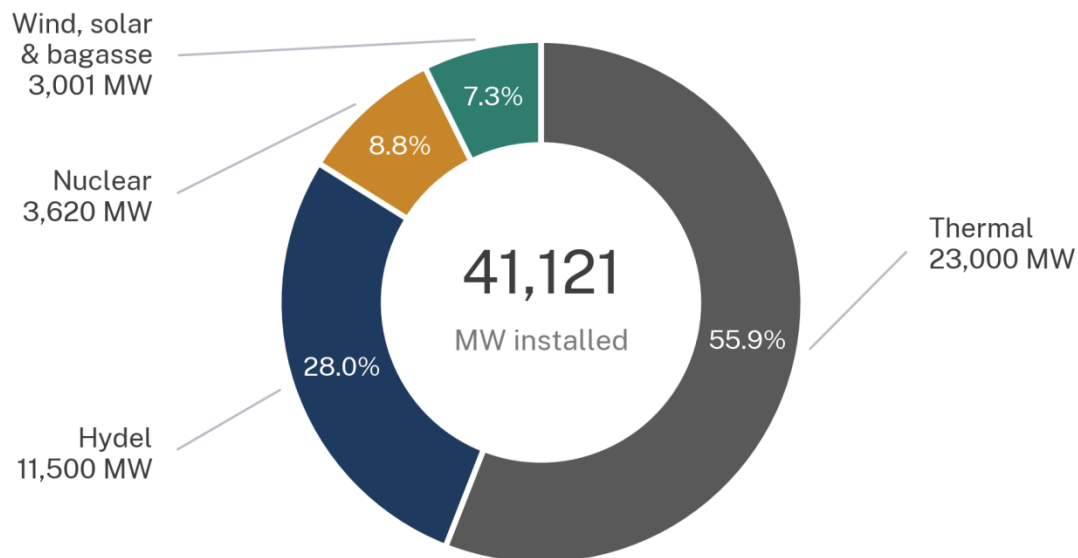
Installed generation capacity, reported peak demand and implied average load, FY 2024-25, MW



Source: NEPRA, State of Industry Report 2025, for installed capacity and peak demand. Average load is an ARS calculation from NEPRA's reported net generation of 135,078 GWh divided by 8,760 hours. Excludes captive and net-metered capacity.

Two qualifications matter. Average load is not a target: no thermal system runs at a 100% load factor, because reserve margin, maintenance and the daily and seasonal shape of demand all require headroom. A well-run system might target a load factor in the region of 55% to 65%. Pakistan is well below any such benchmark. Second, published capacity figures vary by basis. Analysis presented by the Sustainable Development Policy Institute puts FY 2025 installed capacity at 46,605 MW against dependable capacity of 41,000 MW and peak demand of 28,306 MW, a nameplate-to-dependable gap of about 26%.²⁰ The discrepancies arise from whether K-Electric, captive plant and net-metered capacity are included, and from whether nameplate or dependable capacity is used. This report uses NEPRA's State of Industry Report 2025 figures as its primary basis and flags where an alternative basis would change the conclusion. On the central point, no basis changes the conclusion.

Figure 3. Fifty-Six Per Cent of the Fleet Is Thermal and Contracted on Availability
 Installed generation capacity by source, 30 June 2025, MW and per cent of total



Source: NEPRA, State of Industry Report 2025; Renewables First, Pakistan Electricity Review 2026. Wind, solar and bagasse is an ARS residual after thermal, hydel and nuclear, reconciled to the reported 41,121 MW total.

1.3 Peak Demand versus Base Load

A power system is dimensioned for its peak, not its average, which is why the ratio between the two matters so much. Pakistan's peak is summer-afternoon air conditioning load, concentrated in a few weeks, and its trough is a winter night. The system must hold plant available for a peak that occurs for a small number of hours and is then idle for most of the year, and the contracts under which that plant is held available pay for the holding rather than for the output.

Rooftop solar has now added a second problem to the first. Distributed solar generation hollows out midday net demand and leaves a steep evening ramp when output falls away, the pattern familiar internationally as the duck curve. Analysis of Pakistani dispatch data indicates that midday net load is depressed by several gigawatts with marginal cost collapsing into a low range, while the grid must then ramp several gigawatts within a few hours as solar drops in the late afternoon, with marginal cost spiking.²⁰ A system that is simultaneously over-capacitated on an annual basis and ramp-constrained on a daily basis is not a system with spare energy lying around. It is a system with a shape problem.

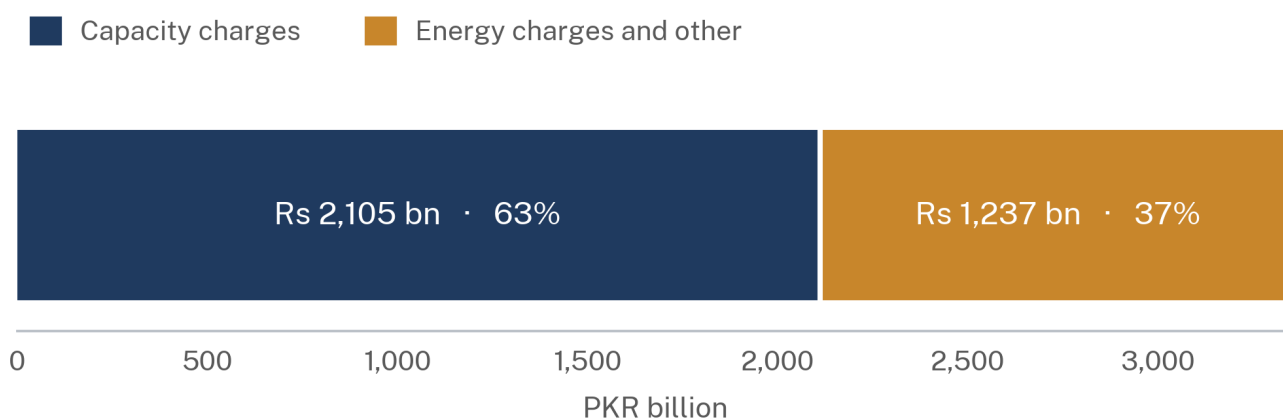
This is precisely why a data centre is a different kind of customer. A hyperscale campus is close to a pure base load: flat, predictable, running at a high load factor around the clock and around the year, with no seasonal peak and no ramp. It is the load shape that Pakistan's fleet was designed for and did not get. That is the strongest single argument for the proposition this report examines, and it is an argument about load shape rather than about surplus.

1.4 The Cost of Idle Capacity

The cost is quantifiable. For FY 2025-26, NEPRA determined a power purchase price of Rs 3,342.369 billion for the ex-WAPDA distribution companies. CPPA-G's petition projected capacity charges in a range of Rs 2.03 trillion to Rs 2.16 trillion, equivalent to between Rs 16.04 and Rs 16.45 per unit, and representing approximately 63% of the projected power purchase price.³ Capacity payments had risen 46% year on year to Rs 1.9 trillion in the preceding cycle, with coal and nuclear plants accounting for the largest share of the increase.²¹

Figure 4. Nearly Two Thirds of What Consumers Pay for Power Is Not for Electricity at All

NEPRA-determined power purchase price for the ex-WAPDA distribution companies, FY 2025-26, PKR billion



Source: NEPRA determination of the CPPA-G power purchase price forecast for FY 2025-26 of Rs 3,342.369 bn; CPPA-G projected capacity charge range of Rs 2.03-2.16 trn, approximately 63% of the projected power purchase price. The capacity figure shown is the midpoint of the CPPA-G scenario band.

Two subsidiary costs deserve mention because they are specific to underutilisation. Part Load Adjustment Charges compensate plants required to run below their efficient output, and NEPRA has repeatedly flagged them: Rs 46.59 billion was incurred on PLAC in FY 2022-23 against Rs 38.20 billion the previous year.²² Payments on account of non-project missed volume arise where a plant is available but not dispatched for reasons outside its control. Neither charge buys a single additional unit of electricity for a consumer. Both are the price of holding a fleet in a state of readiness that the demand curve does not require.

A further Rs 265 billion was attributable in FY 2024-25 to distribution company transmission and distribution losses, with a recovery shortfall of approximately Rs 132.46 billion against a bill recovery rate of 96.62%.² Distribution company performance contributed roughly Rs 397 billion to circular debt during the year.⁴ NEPRA reported bill recovery of 96.62% against an allowed 100%.

1.5 Capacity Payments and Circular Debt

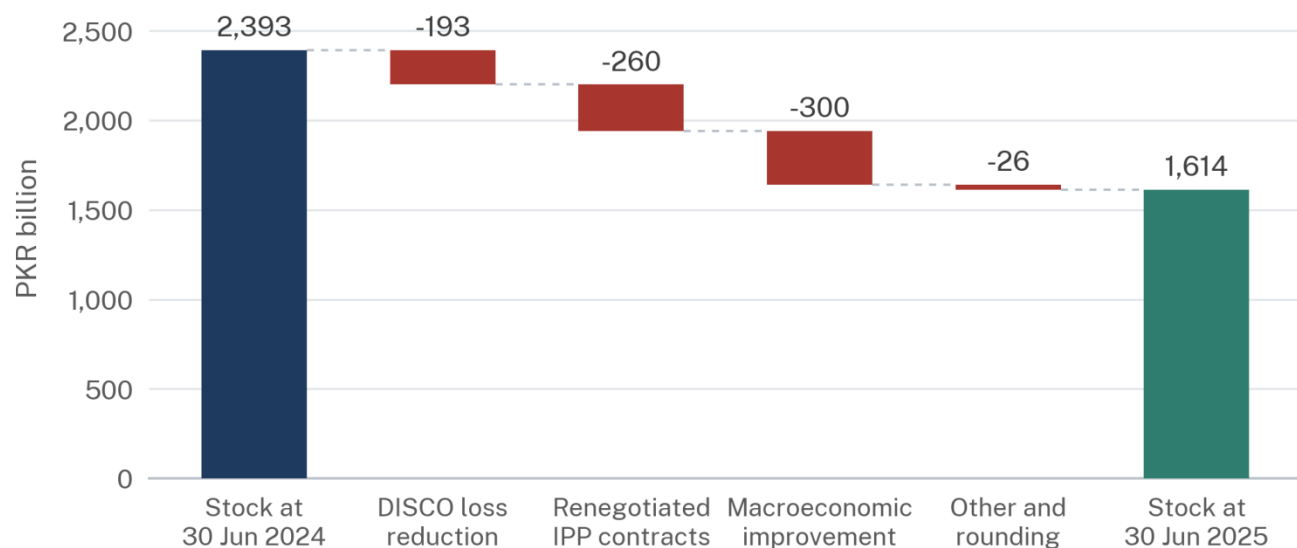
Circular debt is the residue of a system in which the contracted cost of power exceeds the revenue actually collected from consumers plus the subsidy actually paid by government. As at 30 June 2025 the stock stood at approximately Rs 1.614 trillion, down from Rs 2.393

trillion a year earlier.² That reduction of roughly Rs 780 billion has been the government's principal claim of reform success.

NEPRA was careful not to endorse the interpretation. The regulator cautioned that the decline did not reflect a structural improvement and noted that comparable falls in the past had been driven by temporary financial measures rather than lasting reform.² The Federal Minister for Energy disputed NEPRA's characterisation and offered a decomposition of the reduction: approximately Rs 193 billion from reduced distribution company losses, Rs 260 billion from renegotiated independent power producer contracts and around Rs 300 billion linked to macroeconomic improvement.²³ The Prime Minister's Office subsequently sought the Power Division's comments on the report following the ministry's objections.

Figure 5. The Rs 780 Billion Fall in Circular Debt Came Largely from Contract Renegotiation, Not Structural Repair

Movement in the circular debt stock, FY 2024-25, PKR billion



Source: NEPRA, State of Industry Report 2025, for the opening and closing stock. The decomposition is as attributed to the Federal Minister for Energy (Power Division) in response to the report; NEPRA itself cautioned that the reduction did not reflect a structural improvement. The residual is shown as rounding.

For an investor the decomposition matters more than the headline. Two thirds of the reduction came from contract renegotiation and favourable macroeconomic movement, neither of which is repeatable at the same scale, and both of which are the kind of measure that a counterparty notices. A generator whose tariff was renegotiated once has learned something about the durability of its contract. So has the lender who financed it. This is directly relevant to Section 12: a state that has recently renegotiated its way out of Rs 260 billion of contractual obligation is a state that must work harder to persuade the next investor that the next contract will hold.

Servicing the remaining stock now runs through a Debt Servicing Surcharge, with NEPRA citing Rs 233 billion in this connection, and the government has indicated a six-year

settlement plan.²³ The surcharge is levied on consumers. It therefore raises the tariff that drove the demand loss that created the problem.

1.6 Underused Capacity versus Actual Surplus Electricity

This distinction is the most important analytical point in Part One, and it is the one most often lost. Four different quantities are routinely described using the same word.

Table 1. Four Quantities That Are Not the Same Thing

Definitions applied consistently throughout this report

Quantity	What it measures	FY 2024-25	Relevance to a data centre
Installed capacity	Nameplate rating of all connected plant, irrespective of condition, fuel availability or network access	41,121 MW	Almost none. It is an accounting figure.
Available capacity	Plant that can actually be dispatched at a given moment after outages, fuel constraints and hydrology	Varies hourly; materially below nameplate	The starting point for any reliability discussion.
Underutilised capacity	Available plant not dispatched because demand, merit order or network constraints do not call for it	Large; the Power Division has publicly referred to about 8,700 MW of surplus capacity ²³	This is the real opportunity, and it is a contractual opportunity.
Surplus electricity	Energy that can be delivered to a new load at a price at or near short-run marginal cost	Does not exist as a standing quantity	This is what promoters imply exists. It does not.

Source: ARS definitions, applied to NEPRA State of Industry Report 2025 figures for FY 2024-25. Figures rounded.

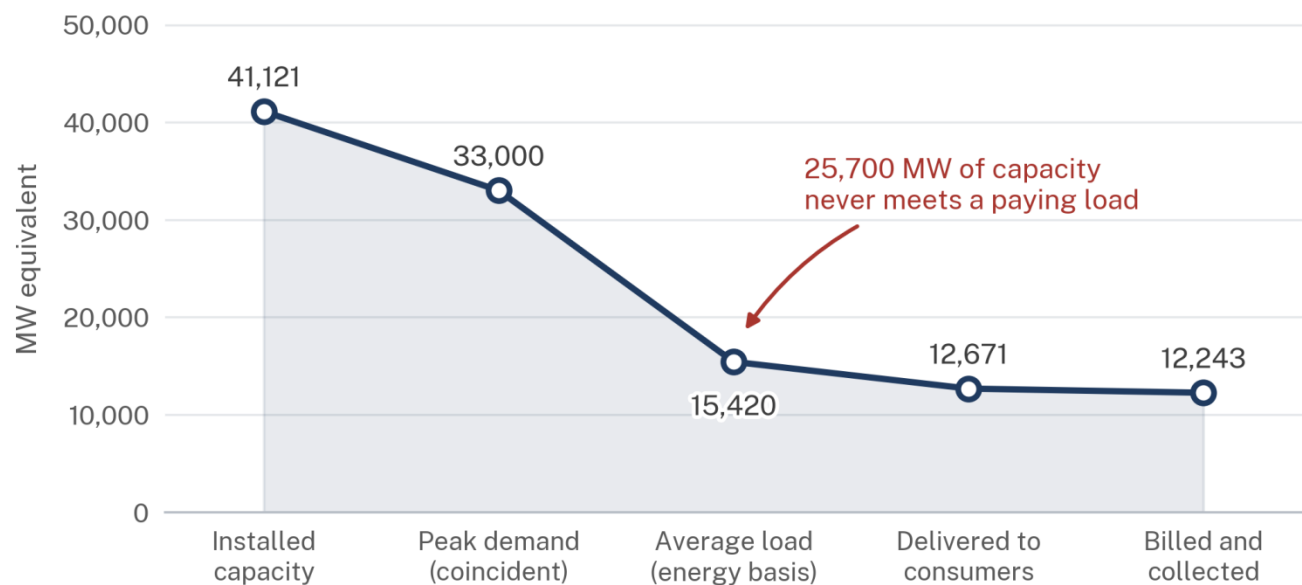
The plants that sit idle in Pakistan are, with limited exceptions, idle because they are expensive. Merit order dispatch calls the cheapest available plant first. A plant that is not being dispatched is, by construction, one whose variable cost exceeds that of the plant being dispatched ahead of it. Adding a new 300 MW load to the system therefore does not draw on free electricity. It draws on the marginal plant, and the marginal plant in Pakistan is frequently imported-fuel thermal.

That does not defeat the argument. It reframes it. The gain from a new base load is not that the energy is free. The gain is that the capacity charge already being paid on the idle plant gets spread across a larger volume of units, so the average cost of the whole system falls even though the marginal cost of the new units is not low. The beneficiary of that spreading is the existing consumer base. The question of how much of that benefit is passed to the new consumer as a discount, and how much is retained to reduce the general tariff, is a regulatory design question that Section 10.2 addresses directly.

The Federal Government's own framing of its 2,000 MW allocation for bitcoin mining and AI data centres, announced in 2025, used the language of monetising surplus electricity and repurposing underutilised capacity into a revenue-generating asset, identifying underutilised coal plants operating at low capacity factors as the intended source.²⁴ The instinct is sound. The vocabulary is imprecise in ways that matter to a counterparty conducting due diligence, because a hyperscaler's energy team will immediately ask which specific plant, under which contract, at what tariff, delivered over which circuit, and with what remedy if it is not delivered. Part IV of this report takes those questions in order.

Figure 6. Where the Fleet Actually Ends Up: From 41 GW of Plant to About 12 GW of Collected Revenue

FY 2024-25, expressed on a common MW-equivalent basis



Source: NEPRA, State of Industry Report 2025; Renewables First, Pakistan Electricity Review 2026, for grid sales of 111 TWh. ARS calculation: energy quantities converted to average MW by dividing GWh by 8,760 hours; the collection stage applies NEPRA's reported recovery ratio of 96.62%. Illustrative reconciliation, not a NEPRA energy balance.

2. The Independent Power Producer Model

2.1 How Pakistan's IPP System Works

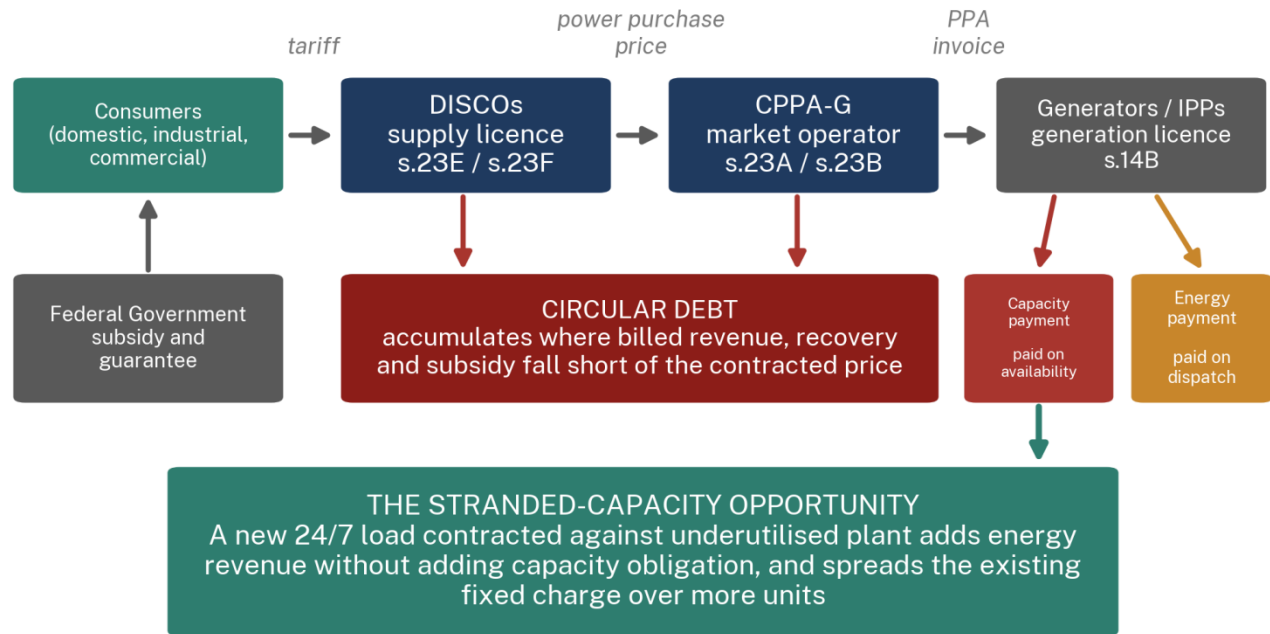
Pakistan opened generation to private capital in 1994 and did so on terms designed to overcome a specific problem: a country with no regulatory track record, a history of currency depreciation and a state-owned offtaker of uncertain creditworthiness could not attract long-term foreign investment on ordinary commercial terms. The solution was to remove almost all commercial risk from the generator. Successive policies in 1994, 2002 and 2015 refined the structure without changing its logic.

The structure has three legs. A generator obtains a licence, now under section 14B of the NEPRA Act, to construct, own and operate a generation facility.⁷ It signs a long-term power purchase agreement with a single buyer, historically WAPDA and subsequently the Central Power Purchasing Agency (Guarantee) Limited, which holds a market operator licence issued under section 23A and is governed under sections 23A and 23B.²⁵ The Federal Government provides an implementation agreement and, in most cases, a sovereign guarantee of the offtaker's payment obligations. NEPRA determines the tariff under section 31.

The distribution companies then buy from CPPA-G and sell to consumers. Since the 2018 Amendment Act, a distribution licence under section 20 and a supply licence under sections 23E and 23F are conceptually distinct, with existing distribution licensees deemed to hold supply licences for a transitional period under the proviso to section 23E(1).⁷ The market operator, system operator, trader and supplier categories inserted by that Act are the statutory scaffolding of a competitive market. The scaffolding was erected in 2018. The building has been under construction ever since.

Figure 7. How Money Moves Through Pakistan's Single Buyer Power Market, and Where It Leaks

Contractual and payment architecture under the NEPRA Act, 1997 as amended; schematic, not to scale



Source: ARS analysis of the Regulation of Generation, Transmission and Distribution of Electric Power Act, 1997 (as amended by Act XII of 2018), and NEPRA licensing and tariff determinations.

2.2 Power Purchase Agreements and Capacity Payments

A Pakistani power purchase agreement splits the generator's remuneration into two components with entirely different economic logics, and understanding the split is essential to understanding everything that follows.

Table 2. The Two Halves of an IPP Tariff

Structural comparison of the capacity and energy components under Pakistan's standard PPA architecture

Feature	Capacity payment (capacity purchase price)	Energy payment (energy purchase price)
Trigger	Plant declared available for dispatch	Units actually dispatched and delivered
Recovers	Debt service, return on equity, fixed operation and maintenance, insurance	Fuel cost, variable operation and maintenance
Sensitivity to output	None. Payable at zero dispatch if availability is maintained	Directly proportional to units generated
Currency exposure	Substantially indexed to the US dollar for foreign-financed plant	Indexed to fuel price, itself often dollar-denominated for imported fuel
Effect of adding a new load	Unchanged in total; falls sharply per unit delivered	Rises in proportion to units taken
Who bears the risk of low demand	The consumer, through tariff, and the state, through subsidy and circular debt	Nobody. The cost simply is not incurred

Source: ARS analysis of the standard PPA structure applied under Pakistan's 1994, 2002 and 2015 power policies and NEPRA tariff determinations. Indicative and structural; specific PPAs vary materially by vintage, technology and sponsor.

The consequence is that Pakistan's electricity cost structure is dominated by a charge that is entirely insensitive to how much electricity anyone uses. This is why the tariff cannot be reduced by using less electricity, and why every reduction in demand raises the per-unit cost for those who remain. It is also why a large new base load is worth more to the system than its own consumption would suggest.

2.3 Energy Payments versus Capacity Payments in Practice

At the FY 2025-26 projection, capacity charges of Rs 2.03 trillion to Rs 2.16 trillion translate to between Rs 16.04 and Rs 16.45 per unit against a determined national average tariff of Rs 34 per unit.^{3,26} Close to half the headline tariff is the capacity charge alone, before energy, before transmission and distribution, before the distribution margin, before the debt servicing surcharge and before general sales tax.

A hyperscaler evaluating Pakistan will read that as follows. The variable cost of electricity in Pakistan is not internationally uncompetitive. The all-in tariff is. The gap between the two is a fixed charge levied to recover the cost of plant that exists because of a demand forecast that failed. Whether any portion of that gap can be lawfully and durably excluded from the price offered to a new large load is, in the end, the single commercial question on which this entire proposition turns. It is a question of regulatory design and statutory authority, not of engineering, and Section 10 of this report answers it.

2.4 Take-or-Pay: What It Actually Means

The phrase take-or-pay is used loosely in Pakistani public debate and the looseness obscures the legal position. In the standard Pakistani structure, the offtaker's obligation is not to take a specified volume of energy. It is to pay the capacity charge in respect of capacity that the generator has declared and made available, whether or not the system operator calls upon that capacity. The generator's obligation is availability, not output. Where the plant is unavailable, liquidated damages apply and the capacity payment is reduced or forfeited in accordance with the agreed availability mechanics.

This distinction has practical consequences. It means the offtaker cannot escape the charge by declining to dispatch, but it also means the generator earns nothing extra by being dispatched more, beyond recovery of its variable costs. Neither party has a contractual interest in higher utilisation. The generator is indifferent. The offtaker prefers lower utilisation, because dispatch adds fuel cost. Only the consumer and the state, who bear the fixed charge, have an interest in the plant running more. Neither of them is a party to the agreement.

That misalignment is the contractual root of the stranded capacity problem, and it explains why the problem has proved so durable. It also explains why the government's route out has been renegotiation and termination rather than utilisation. In FY 2024-25 contracts covering 2,829 MW were terminated, with NEPRA estimating potential savings above Rs 900 billion, and the Power Division has stated that negotiations with independent power producers produced terminations or revisions across several contracts, with roughly Rs 260 billion of the circular debt reduction attributed to renegotiated IPP contracts and around 8,000 MW of expensive future projects removed from the plan.^{2,23}

2.5 Dispatch and Merit Order

Dispatch in Pakistan is not a matter of contract but of statute and system operation. Section 14B(4) of the NEPRA Act requires a generation licensee connected to the national grid to make the facility available to the national grid company for safe, reliable, non-discriminatory and economic dispatch, subject to compensation fixed by the Authority for voltage support and uneconomic dispatch. Section 14D(2) imposes a materially identical duty by reference to the system operator.⁷ Section 23G, in force following the 2018 Amendment Act, establishes the system operator licence and lists generation scheduling, commitment and dispatch, transmission congestion management and procurement of ancillary services among its functions.

Economic dispatch means merit order: the system operator calls plant in ascending order of variable cost, subject to must-run obligations, network constraints and technical minimums. The critical implication for this report is that a data centre cannot, merely by contracting with a particular generator, cause that generator to be dispatched. Dispatch is a system function. A bilateral contract between a generator and a consumer is, in a wheeling or bilateral contract market, a financial and commercial arrangement settled against physical flows that the system operator determines. Confusing the contract with

the physics is one of the most common errors in this debate, and it has direct consequences for how a curtailment clause must be drafted.

2.6 Why Plants Are Paid When They Are Not Fully Dispatched

Several mechanisms operate simultaneously and it is worth separating them, because they attract different remedies.

Table 3. Why Payment and Output Diverge

Mechanisms by which Pakistani generators receive payment without corresponding dispatch

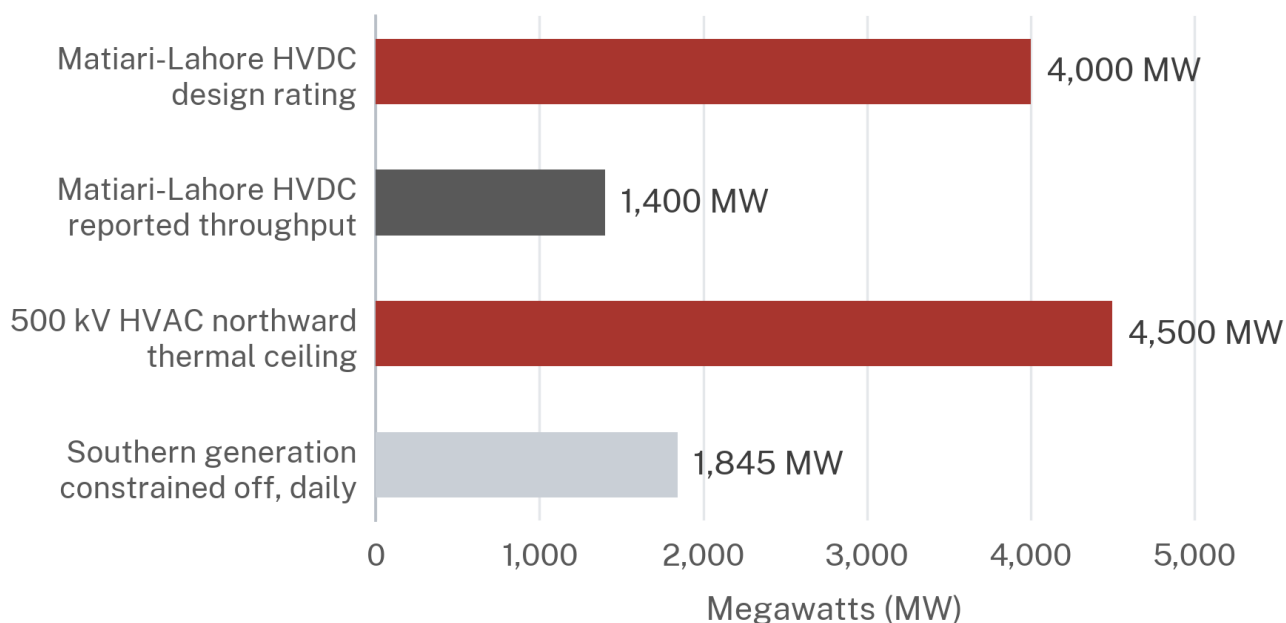
Mechanism	What happens	Character of the remedy
Capacity payment on available but undischpatched plant	The plant declares availability, is not called, and is paid the full capacity charge	Contractual. Requires renegotiation or new load
Part Load Adjustment Charges	A plant is dispatched below efficient output and compensated for the efficiency penalty. Rs 46.59 bn in FY 2022-23	Operational. Better load management and time-of-use tariffs
Non-project missed volume	Plant available but not dispatched for reasons outside the generator's control, including network constraint	Network. Requires transmission investment
Must-run status	Nuclear, certain hydro and some contracted plant dispatched ahead of merit order	Structural. Reflects technology and contract, not inefficiency
Transmission constraint	Cheap southern generation cannot reach northern load; costlier northern plant runs instead	Network. Requires corridor reinforcement

Source: ARS analysis of NEPRA State of Industry Reports 2023, 2024 and 2025, and the NEPRA Act, 1997 as amended. PLAC figure is for FY 2022-23, the most recent year for which a discrete published figure was located.

The last of these is the one most often overlooked and it is the one most relevant to siting. Pakistan's cheapest thermal generation is concentrated in the south and its largest hydro in the north, while load is distributed across both. The north-south corridor is the constraint. The Matiari to Lahore HVDC link was designed for 4,000 MW but has been reported operating at around 1,400 MW, roughly 35% of design rating, on account of converter station faults, while three 500 kV HVAC circuits reach a thermal ceiling of approximately 4,500 MW northward, with the result that a reported 1,845 MW of southern generation is constrained off daily.²⁰ NEPRA's own Performance Evaluation Report for transmission in FY 2024-25 records congestion and overloading of 500 kV and 220 kV grid stations and lines, and attributes delays to right-of-way disputes, land acquisition, foreign exchange constraints, funding shortfalls and security concerns.²⁷

Figure 8. The North-South Corridor Is Where Cheap Southern Power Stops Being Deliverable

Reported transmission capability on Pakistan's principal evacuation corridor, MW



Source: Sustainable Development Policy Institute analysis of the NTDC Transmission System Expansion Plan 2025-35 and NEPRA State of Industry Report 2025, presented at UNESCAP; NEPRA Performance Evaluation Report, Transmission, FY 2024-25. HVDC throughput reflects reported converter station faults and is an operational, not a design, limitation.

The Asian Development Bank approved a USD 330 million Second Power Transmission Strengthening Project in November 2025, comprising a new 500 kV line of approximately 290 km together with grid upgrades serving Islamabad and Faisalabad, intended to enable evacuation and transfer of up to 3,200 MW of hydropower from northern generation clusters.²⁸ The World Bank has separately emphasised that unbundling of the National Transmission and Despatch Company, announced by the government in November 2024, is central to establishing a well-governed and creditworthy transmission system operator.²⁹ NTDC now operates as the National Grid Company. These are the right measures. They are also multi-year measures, and a hyperscaler siting a campus in 2027 cannot underwrite against a corridor that will be reinforced in 2030.

2.7 The Stranded Capacity Opportunity

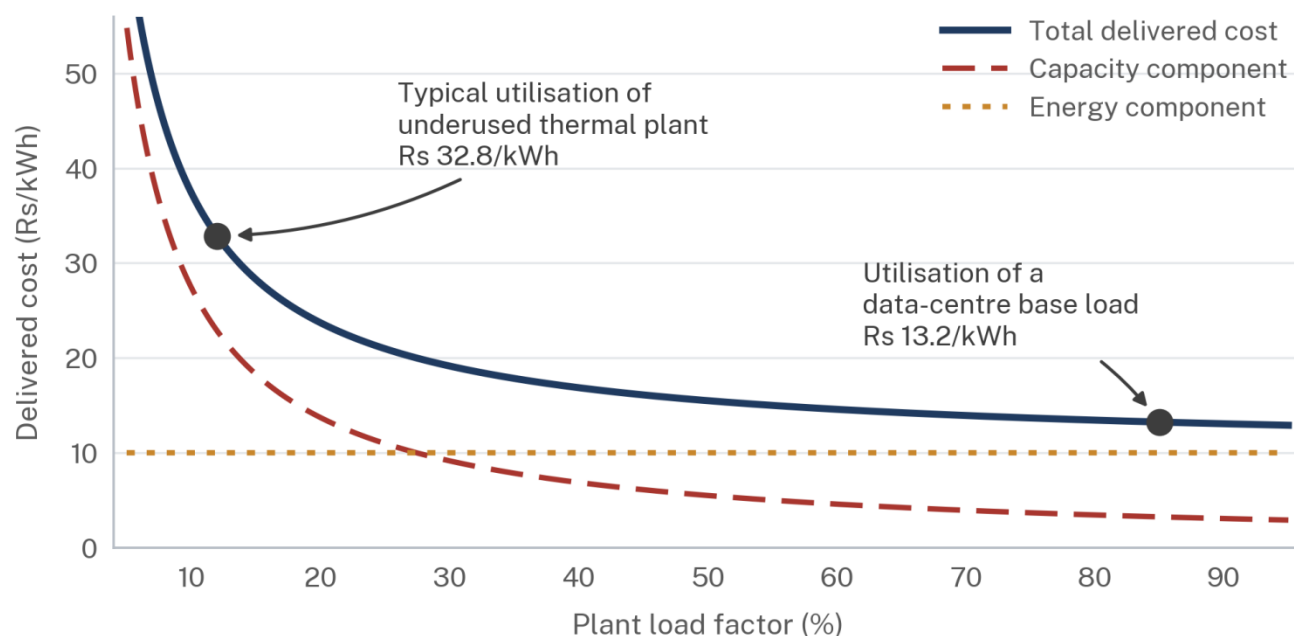
With the mechanics established, the opportunity can be stated precisely rather than rhetorically.

A plant carrying a fixed capacity charge delivers energy at a total cost per unit equal to the variable energy cost plus the fixed charge divided by units delivered. The second term is a hyperbola. At low utilisation it dominates completely; at high utilisation it approaches insignificance. On an illustrative basis, using a capacity charge of Rs 2.0 million per MW per month and a variable energy charge of Rs 10 per kWh, a plant running at 12% load factor delivers at roughly Rs 32.8 per unit, while the same plant at 85% load factor delivers

at roughly Rs 13.2 per unit. The plant is identical. The contract is identical. Only the utilisation has changed.

Figure 9. Adding Utilisation, Not Adding Plant, Is What Lowers the Cost of Pakistani Electricity

Delivered cost per unit for a plant carrying a fixed capacity charge, by load factor, Rs/kWh



Source: ARS illustrative model, not a plant-specific tariff. Assumes a capacity charge of Rs 2.0 million per MW per month and a variable energy charge of Rs 10 per kWh, both held constant. Real Pakistani tariffs are plant-specific, partly indexed to the US dollar and determined by NEPRA.

Four conditions must hold for this arithmetic to translate into a real gain, and each of them is a legal or regulatory condition rather than a commercial one.

The first is that the new load must not trigger any new capacity obligation. If serving a 300 MW data centre requires new plant to be contracted, the system has simply repeated its original mistake at a new address. The load must be served from existing available capacity, which means the contracting structure must be capable of identifying and committing that capacity.

The second is that the new load must pay a price that exceeds short-run marginal cost by enough to make a real contribution to the fixed charge. A tariff set at or near marginal cost transfers the benefit entirely to the data centre and leaves the general consumer no better off, which is politically unsustainable and, given NEPRA's obligations under section 31(2) to protect consumers against monopolistic pricing and to eliminate economic distortions, regulatorily contestable.

The third is that the energy must be physically deliverable to the site. Contracting against a southern plant to serve a northern campus across a constrained corridor produces a financial arrangement that the system cannot honour physically, and the resulting curtailment risk sits with somebody.

The fourth is that the arrangement must be durable for the term of the investment. A data centre is a fifteen to twenty-five year asset financed on long-term debt. An arrangement that depends on a tariff concession renewable annually, or on a wheeling charge set by administrative decision, is not an arrangement a lender will underwrite.

WHERE THIS LEAVES THE ARGUMENT

The stranded capacity opportunity is real, and it is larger than most public discussion recognises. But it is not an opportunity in electricity. It is an opportunity in contract design, and it can only be captured if Pakistani law allows a specific generator and a specific consumer to commit to each other for a specific term at a specific price, across a network at a charge that neither of them controls. Sections 8 and 9 examine whether it does.

WHO CAN SUPPLY POWER?

3. Captive Power: Can Data Centres Do It Alone?

3.1 What Captive Power Is

The NEPRA Act defines a captive generating plant, at section 2(ia) inserted by the 2018 Amendment Act, as a power plant set up by any person to generate electricity primarily for his own use, including a plant set up by a cooperative society or association of persons for the use of its members.⁷ The definitional pivot is the words "primarily for his own use". A captive plant is not a business selling electricity. It is a consumer generating for itself, and the whole regulatory treatment follows from that.

The attraction for a large industrial user has always been the same. Self-generation escapes the tariff, and therefore escapes the cross-subsidies, the capacity charge on plant the user did not commission, the debt servicing surcharge and the general sales tax embedded in the electricity bill. In a country where the national average tariff for FY 2025-26 was determined at Rs 34 per unit, that escape has been worth a great deal, which is precisely why the state has moved to close it.

3.2 Pakistan's Captive Generation Framework

Section 14C is short and unusually important, and it is worth reading closely because it does two different things.

Subsection (1) provides that notwithstanding anything contained in the Act, a person may construct, maintain or operate a captive generating plant and dedicated transmission lines, with a proviso that supply from the captive plant through the grid is regulated in the same manner as the generating facility of a generating company. The effect is that behind-the-meter self-generation sits largely outside the licensing regime, but the moment the plant exports through the public network it is regulated as an ordinary generator.

Subsection (2) is the provision that matters commercially. Every person who has constructed and operates a captive generating plant "shall have the right to open access for the purposes of carrying electricity from his captive generating plant to the destination of his use". This is drafted as a right, in mandatory language, which is materially stronger than the permissive drafting of section 22. But it is immediately qualified. The right is subject to availability of adequate transmission facility, and that availability "shall be determined by the national grid company or the provincial grid company, as the case may be". Any dispute regarding availability is to be adjudicated by the Authority.⁷

For a project lawyer this is the crux. Section 14C(2) creates a statutory right whose exercise is conditioned on a determination made by the very entity whose network is being accessed, with recourse only to a regulator whose determination is then appealable under section 12G to the Appellate Tribunal and thereafter to the High Court. The right exists. The timeline for vindicating it does not. A data centre developer holding a construction programme cannot wait out a determination, an appeal to the Appellate Tribunal within its three-month statutory window under section 12G(1), and a further appeal to the High Court under section 12G(6). Section 10.1 proposes a specific remedy directed at this precise sequencing problem.

3.3 Industrial Self-Generation and the New Fiscal Position

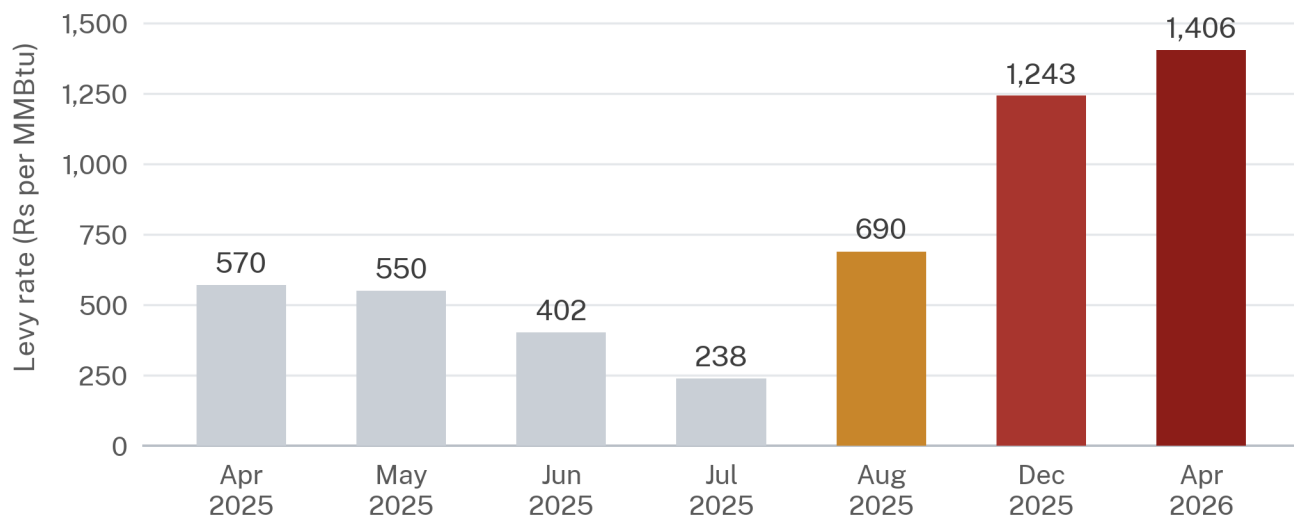
Captive generation in Pakistan has historically been gas-fired, concentrated in textiles and heavy manufacturing, and supplied through SNGPL and SSGCL. That position has been changed by statute rather than by regulation, and the change is recent, deliberate and escalating.

The Off the Grid (Captive Power Plants) Levy Ordinance, 2025 was promulgated on 30 January 2025 and re-promulgated in March 2025, and was replaced by the Off the Grid (Captive Power Plants) Levy Act, 2025 (Act XIV of 2025), notified in the Gazette on 10 June 2025.^{12,13} Section 2(e) defines the levy as an off-the-grid charge on natural gas or RLNG-based captive power plants. Section 3 requires every captive power plant to pay the levy to the Federal Government on consumption of natural gas or RLNG, over and above the sale price notified under sections 8 and 43B of the Oil and Gas Regulatory Authority Ordinance, 2002.

Section 4 is the operative pricing provision and it is unusually explicit about intent. Before notifying the levy, the relevant Divisions are to calculate the rate by taking into account the difference between the industrial B3 power tariff notified by NEPRA and the self-generation cost of the captive plant at OGRA-notified gas tariffs. In other words, the levy is calibrated by statute to close the arbitrage between self-generation and grid supply. The rate steps up by 5% immediately, 10% from July 2025, 15% from February 2026 and 20% from August 2026.¹³ Section 5 directs the proceeds to reducing generation tariffs for consumer categories, with an annual report to both Houses three months after the close of each financial year, and section 6 makes unpaid levy recoverable under the Public Finance Management Act, 2019.

Figure 10. Parliament Has Legislated Gas-Fired Captive Generation Into Retreat
Notified off-the-grid levy on natural gas and RLNG consumed by captive power plants, Rs per MMBtu

Statutory step-up: 5% to 10% (Jul 2025), to 15% (Feb 2026), to 20% (Aug 2026)



Source: Off the Grid (Captive Power Plants) Levy Act, 2025 (Act XIV of 2025), sections 3 and 4; Petroleum Division notifications. The Act sets the levy by reference to the gap between NEPRA's B3 industrial tariff and the captive plant's own generation cost at OGRA-notified gas prices, stepping up to 20% from August 2026.

The notified rates trace the escalation. Rs 238 per MMBtu was notified in July 2025 in fulfilment of an International Monetary Fund structural benchmark, rising to Rs 690 in August 2025, Rs 1,243 for December 2025, and Rs 1,406 by April 2026 following a Rs 163 increase.^{12,14,30} In January 2026 the levy was extended to third-party gas suppliers after consultation with the Ministry of Law and Justice, closing the route by which a captive plant could source gas outside the SNGPL and SSGCL networks.¹³

THE LEGAL POINT, STATED PRECISELY

Gas-fired captive generation for a data centre in Pakistan is lawful. Sections 2(ia) and 14C of the NEPRA Act permit it and no licence is required for behind-the-meter operation. But Parliament has enacted a levy whose stated statutory formula is the gap between grid and self-generation cost, and which steps up to 20% from August 2026. The commercial case is being legislated away by design, not by accident. Any financial model that projects gas-fired captive economics across a fifteen-year data centre life is modelling against a declared legislative trajectory.

3.4 Solar, Gas, Storage and Hybrid Generation

The non-gas captive routes are worth assessing on their own terms rather than by analogy to rooftop solar.

Solar photovoltaic generation is now genuinely cheap in Pakistan and the deployment figures prove that the market works. Cumulative net-metered capacity reached approximately 5.3 GW by April 2025 and a projected 6.3 GW by the close of FY 2024-25.¹⁹ But net metering is a settlement arrangement for a consumer whose consumption and generation both sit behind a distribution meter, and the regime has been shifting toward net billing, under which exports are compensated at a materially lower rate than imports. That shift changes the economics of oversizing an array, and it is not a framework designed for a 300 MW industrial load.

The physical requirement is the point most often understated. A solar array sized to serve a 100 MW continuous IT load must be sized not for 100 MW but for the load divided by its capacity factor, which in most of Pakistan means an array in the region of 400 MW to 500 MW of direct-current capacity, occupying something on the order of 1,200 to 2,000 acres depending on technology and layout, together with battery storage sufficient to carry the load through the night and through consecutive low-irradiance days. The storage requirement, not the array, is what breaks the budget. A 100 MW load requires on the order of 1,400 MWh of usable storage to cover a fourteen-hour non-generating period at full load, before any allowance for degradation, round trip losses or reserve.

This is not an argument against solar. It is an argument that solar plus storage is a firming technology and a cost reduction technology within a hybrid architecture, and not a standalone supply solution for a hyperscale campus at Pakistani capital costs. The same conclusion holds in most jurisdictions. It holds more strongly in Pakistan because of the cost of capital.

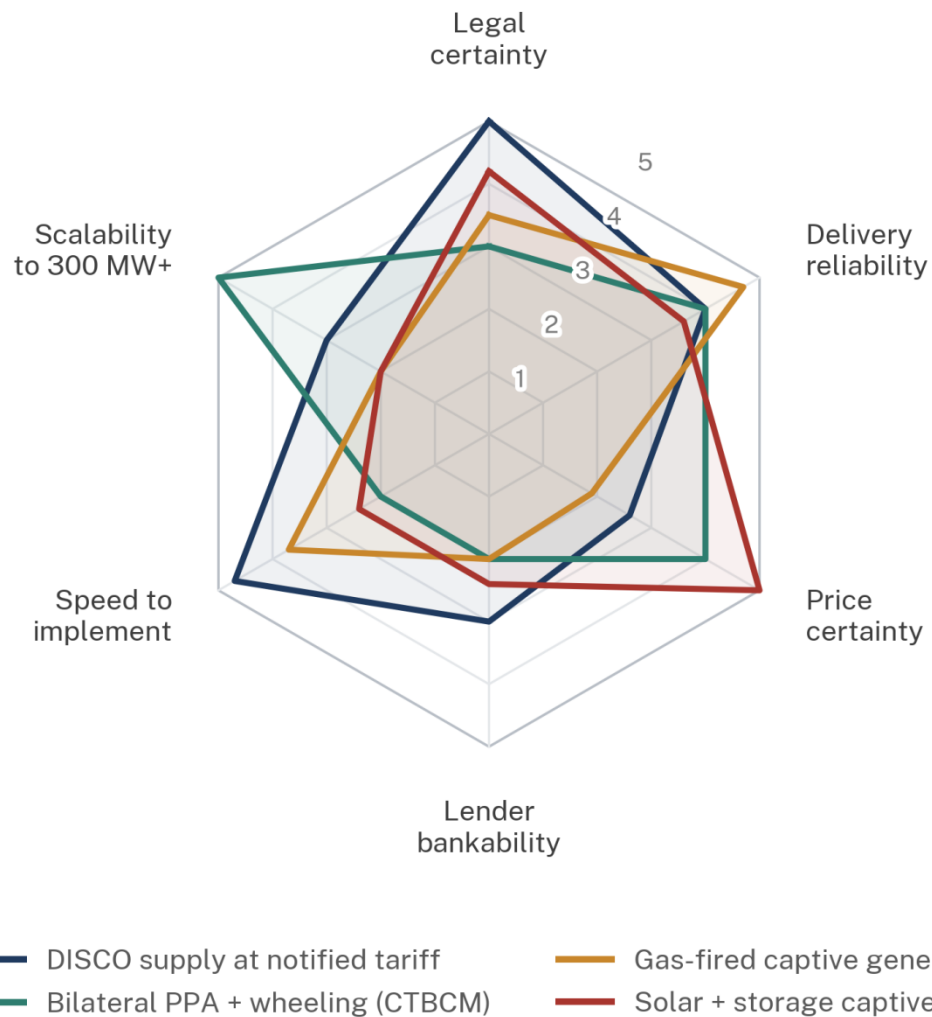
3.5 Captive versus Grid Procurement: A Structured Comparison

The following assessment scores four routes on six dimensions a hyperscaler's investment committee would apply. The scoring is an ARS analytical judgement applied consistently across routes on a common one-to-five scale. It is not a measurement, and the report states the criteria so the reader can disagree with a specific score rather than with the exercise.

Price certainty asks whether the developer can know the cost per unit for the term of its investment. Delivery reliability asks whether the route can meet a contracted availability above 99.9% at the meter. Legal certainty asks whether the right to procure rests on statute, on subordinate regulation, or on the exercise of administrative discretion. Scalability asks whether the route extends from 100 MW to 300 MW and beyond without a change of legal basis. Speed to implement asks how long from decision to energisation. Lender bankability asks whether an international project finance lender would accept the arrangement as security without a sovereign wrap.

Figure 11. No Single Procurement Route Scores Well on Every Dimension a Hyperscaler Tests

ARS assessment of four routes to power a large load in Pakistan, scored 1 (weak) to 5 (strong)



Source: ARS assessment. Scores are analytical judgements applied consistently across routes on a common 1-5 scale, not measured values. Legal certainty reflects whether the right rests on statute, on regulation, or on regulatory discretion. The methodology is set out in Section 3.5.

Table 4. Power Procurement Routes for a Large Load in Pakistan

Legal basis, commercial character and principal constraint of each available route

Route	Principal legal basis	What it delivers today	Binding constraint
DISCO or supplier at notified tariff	ss. 20, 23E, 23F; s. 31 tariff determination	Immediate, legally uncomplicated supply at the regulated industrial tariff	Price. Carries the full capacity charge, cross-subsidy and surcharges
Direct sale to a bulk power consumer	s. 22, read with the proviso to s. 21(2)(a)	A statutory pathway for generator-to-consumer sale above 1 MW	Section 22 says the Authority "may permit". It is a discretion, not a right
Bilateral contract with wheeling	Wheeling Regulations, 2016 (SRO 549(I)/2016); CTBCM market rules	Competitive energy pricing from a chosen generator, wheeled to site	Use of system charges set administratively; capacity allocation and curtailment risk unresolved
Supply via a licensed electric power trader	ss. 23C and 23D	A licensed intermediary able to contract bilaterally and hold wheeling agreements	Thin market. s. 23D(2)(d) requires agreements with transmission and distribution licensees
Captive generation on site	ss. 2(ia) and 14C	Full control of the asset and escape from the tariff stack	Gas route penalised by Act XIV of 2025; solar route constrained by land and storage cost
Hybrid of contracted, grid and on-site supply	All of the above simultaneously	The only configuration meeting both price and reliability tests	Requires every one of the above legal bases to work at once

Source: ARS analysis of the NEPRA Act, 1997 as amended, the NEPRA (Wheeling of Electric Power) Regulations, 2016, and the CTBCM design as approved by NEPRA. Assessment of bankability is an ARS judgement, not a measurement.

3.6 Advantages and Limitations for a Data Centre Specifically

Three characteristics of data centre load distinguish it from ordinary industrial load and change the captive calculus.

The first is that the reliability requirement is extreme and asymmetric. An industrial plant that loses power loses a shift. A data centre that loses power loses customer trust, contractual service credits and, in the case of AI training workloads, potentially weeks of computation. Design availability is typically expressed in the region of 99.98% or higher at the point of delivery, which is a permitted outage measured in tens of minutes per year. No single supply route in Pakistan meets that standard on its own. Redundancy is not optional and it is expensive.

The second is that the load is constant, which is unusual and valuable. It means the campus is a perfect base load customer, but it also means there is no operational flexibility to offer the system in exchange for a better price, at least for training and

inference workloads under service level commitments. Demand response, which is often the currency in which large loads buy favourable terms elsewhere, is largely unavailable here.

The third is that a hyperscale operator does not want to be a power company. Building and operating 400 MW of captive generation is a separate business with separate skills, separate regulatory exposure and separate capital. The commercial preference, everywhere in the world, is to contract for power rather than to build it. Pakistan will not attract hyperscale investment by offering an investor the opportunity to become a Pakistani independent power producer. It will attract it by offering a contract.

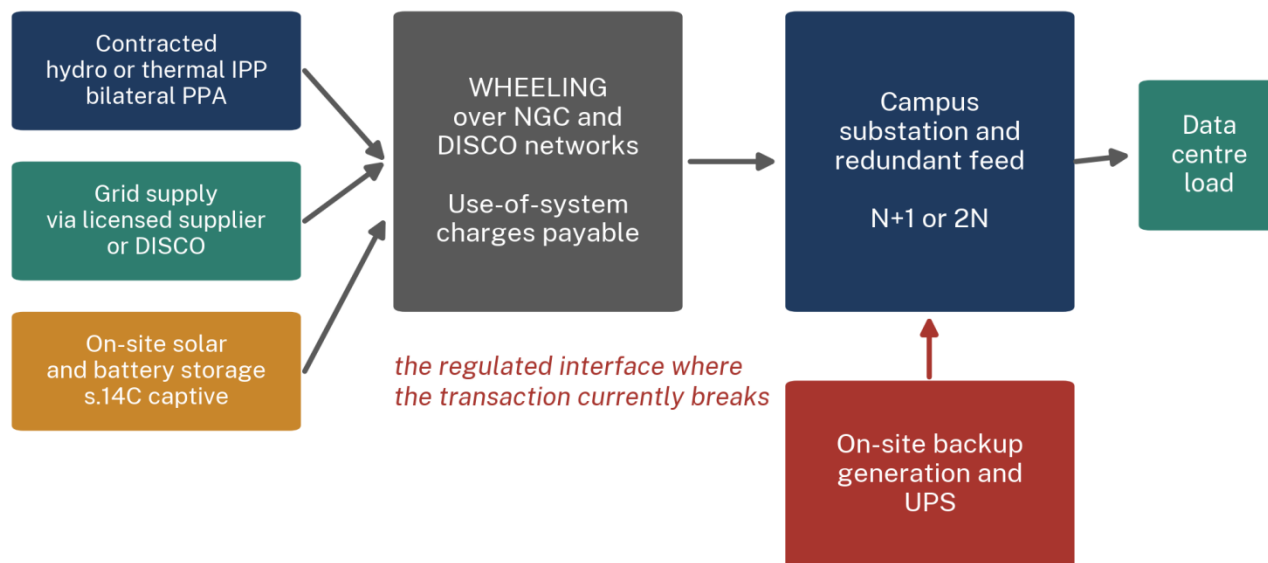
3.7 The Case for a Hybrid Architecture

The conclusion of this section is that no single route works and the correct configuration is a hybrid in which each leg carries the risk it is best placed to carry.

Contracted generation, whether hydro or efficient thermal, provides the bulk of the energy at a negotiated price under a long-term bilateral agreement. Grid supply from a licensed supplier or distribution company provides firming, backup and the balancing service that makes the contracted leg deliverable at constant load. On-site solar with storage reduces the average cost, provides ride-through for short interruptions and improves the emissions profile that a hyperscaler's own procurement policy will require. On-site backup generation and uninterruptible power supply provide the final layer of protection at the campus boundary.

Figure 12. A Hybrid Architecture Is the Only Configuration That Meets Both the Price and the Reliability Test

Indicative supply architecture for a 100 MW-plus campus in Pakistan; schematic, not an engineering design



Source: ARS analysis. Each leg rests on a different legal instrument: the bilateral PPA on sections 14D and 23C to 23F of the NEPRA Act, wheeling on the NEPRA (Wheeling of Electric Power) Regulations, 2016, and on-site generation on section 14C.

Each leg of that architecture rests on a different legal instrument, and this is the substantive difficulty. The bilateral leg depends on sections 14D and 23C to 23F and on the wheeling regulations. The firming leg depends on a supply licensee's willingness to contract for standby at a defined charge. The on-site leg depends on section 14C. The interface between them, and specifically the point at which wheeled power crosses onto the National Grid Company and distribution networks, is where the transaction currently breaks. That interface is the subject of Part IV.

4. Hydropower: Pakistan's Natural Advantage, Correctly Stated

4.1 The Endowment

Pakistan's hydropower endowment is genuine, large and geographically concentrated. The Indus and its western tributaries fall from the Karakoram, Himalaya and Hindu Kush across some of the steepest exploitable gradients in Asia. Installed hydel capacity stood at approximately 11,500 MW at 30 June 2025, roughly 28% of the national fleet, and rose during FY 2024-25 with new plant additions even as thermal capacity fell by 19.5%.¹⁸ NEPRA's own assessment of hydel potential records the resource as heavily concentrated

in Khyber Pakhtunkhwa, Azad Jammu and Kashmir, Gilgit-Baltistan and northern Punjab.³¹

The strategic attraction is straightforward. Hydro generation has no fuel cost, is not exposed to imported fuel prices, and is not exposed to the exchange rate risk that runs through the fuel component of Pakistan's thermal tariff. In a country whose electricity cost problem is substantially a foreign currency problem, that matters.

4.2 Existing Projects

Table 5. Principal Hydropower Assets and the Construction Pipeline

Installed and under-construction capacity, by owner category and location

Project	Capacity (MW)	Location	Type	Status
Tarbela (Units 1-14)	3,478	KP / Punjab	Storage	Operating since 1976 onward
Ghazi Barotha	1,450	Punjab	Run-of-river	Operating
Tarbela 4th Extension	1,410	KP	Storage	Operating
Mangla	1,000	AJK	Storage	Operating; refurbishment adding 310 MW
Neelum-Jhelum	969	AJK	Run-of-river	Shut since 2 April 2024; restoration targeted March 2028
Suki Kinari	884	KP	Run-of-river	Operating since September 2024; CPEC IPP
Karot	720	Punjab	Run-of-river	Operating since June 2022; CPEC IPP
Warsak	243	KP	Run-of-river	Operating
Chashma	184	Punjab	Canal / low head	Operating
Diamer Basha	4,500	KP / GB	Storage	Under construction; target February 2029
Dasu Stage-I	2,160	KP	Run-of-river	Under construction; target December 2027
Tarbela 5th Extension	1,530	KP	Storage	Under construction; target December 2026
Mohmand	800	KP	Storage	Under construction; target 2027

Source: Sources: WAPDA project and generation reporting for calendar year 2025; NEPRA hydel potential assessment; Senate Standing Committee on Water Resources briefing of 10 July 2026 for Neelum-Jhelum. Target dates are as stated by the executing agency. Neelum-Jhelum is a restoration of existing capacity, not a net addition.

4.3 Run-of-River versus Storage

The distinction determines whether a hydro plant can sell firm power, and it is routinely elided in Pakistani policy discussion.

A storage plant impounds water behind a dam and can therefore shift generation across days, weeks or seasons. Tarbela and Mangla are storage plants, although both are operated primarily as irrigation reservoirs, which means their generation follows the irrigation release schedule set by the Indus River System Authority rather than the merit order. Their operator does not fully control their output.

A run-of-river plant generates from the flow available at the moment. It has limited or no meaningful storage and its output tracks the hydrograph. Karot, Suki Kinari and Neelum-Jhelum are run-of-river schemes, as is Dasu Stage-I. Their annual energy is substantial and their winter output is not. A run-of-river plant cannot contract to deliver a flat 100 MW in January because in January the river does not carry the water.

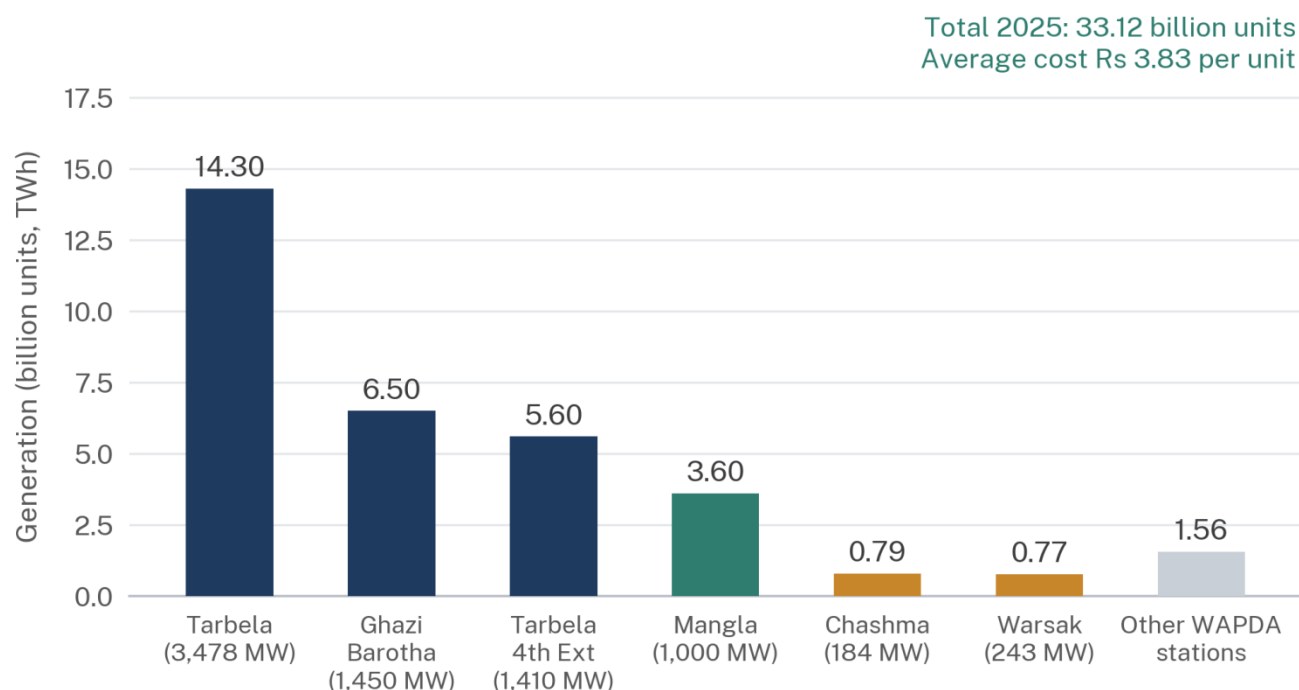
For a data centre power contract this is decisive. Annual energy is not what a hyperscaler buys. It buys delivered power in every hour. A generator can sell that only if it can either store water or buy firming from elsewhere, and the cost of that firming, not the cost of the hydro energy, sets the price of the delivered product.

4.4 Hydropower Economics

WAPDA generated 33.12 billion units of hydel electricity in calendar year 2025, at an average cost stated by the authority at Rs 3.83 per unit.¹⁵ Tarbela contributed 14.3 billion units, Ghazi Barotha 6.5 billion, Tarbela 4th Extension 5.6 billion, Mangla 3.6 billion, Chashma 0.79 billion, Warsak 0.77 billion, with the balance of 1.56 billion units from high-head, Jinnah and small stations.

Figure 13. Two Stations on the Indus Produce Well Over Half of WAPDA's Hydel Output

WAPDA hydel electricity generation by station, calendar year 2025, billion units (TWh)



Source: Water and Power Development Authority (WAPDA), generation statement for calendar year 2025. Excludes hydropower owned by independent producers and provincial utilities.

Two cautions attach to the Rs 3.83 per unit figure and both matter to any investor reading it. First, it is WAPDA's cost of generation on legacy public assets whose capital cost was largely written off decades ago. It is not a market price and it is not available to a private buyer as a purchase price. Second, it is not the price of new hydro. Karot and Suki Kinari are privately financed independent power producers with dollar-linked tariffs determined by NEPRA on the same capacity-and-energy architecture as thermal IPPs, and Neelum-Jhelum's first unit was commissioned in 2018 at a levelised tariff of Rs 13.50 per unit. New Pakistani hydro is cheap in fuel and expensive in capital, and the capital is priced in dollars.

The capital cost record reinforces the point. The Auditor-General's report on Neelum-Jhelum records that the fourth revised PC-I was approved at Rs 506.808 billion against an original 1989 approval of Rs 15.012 billion, a cost overrun of Rs 334.952 billion against the first revised PC-I and a delay of almost eight years.³² That is not a marginal variance. It is a category of execution risk that any lender will price.

4.5 Seasonal Water Flows and the Firm Power Problem

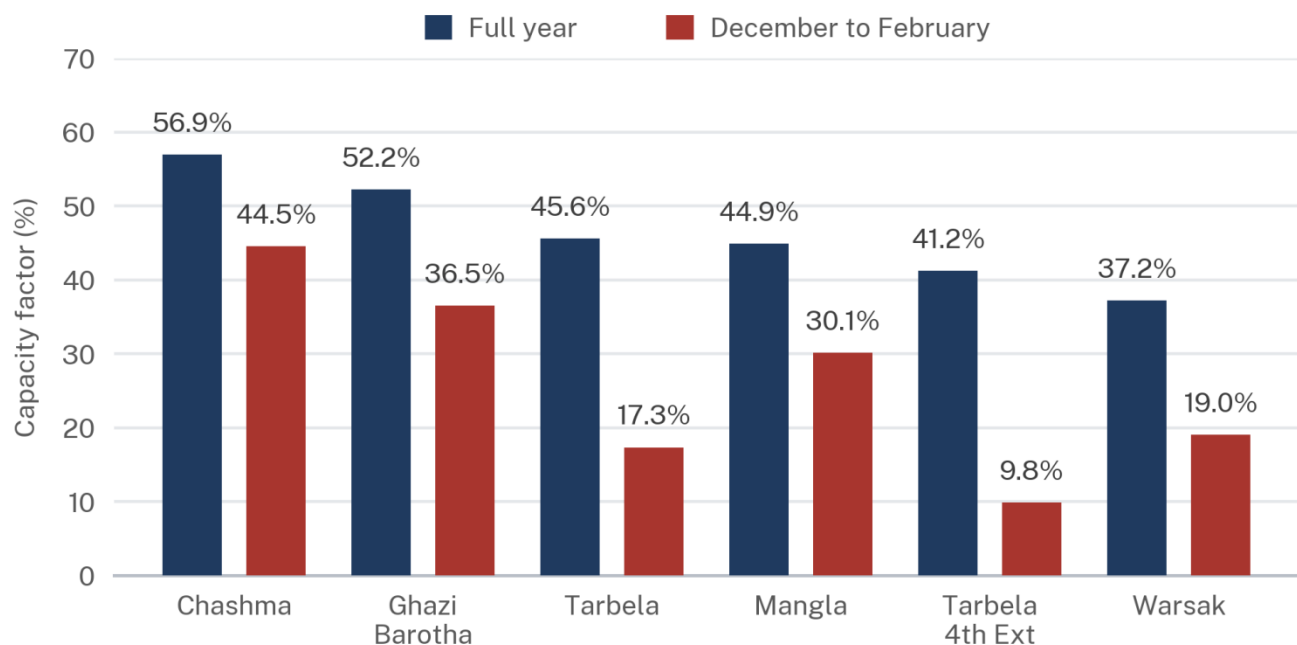
This is the section where the hydro case must be stated honestly, because the temptation to overstate it is considerable and an investor will discover the truth in the first week of diligence.

ARS has calculated annual capacity factors for the major WAPDA stations by dividing calendar year 2025 generation by installed capacity multiplied by 8,760 hours. Ghazi Barotha achieved 51.2%, Chashma 49.0%, Tarbela 46.9%, Tarbela 4th Extension 45.3%, Mangla 41.1% and Warsak 36.2%. These figures reflect hydrology, planned outages, irrigation operation and system dispatch together, and they are not a judgement on plant condition. But they are the numbers.

They are, however, the wrong resolution at which to ask whether hydro can serve a flat load. NEPRA's State of Industry Report 2025 publishes company-wise generation month by month, and running the same calculation across the fiscal year to 30 June 2025 gives a materially different picture.³³ Measured over December, January and February, Tarbela converts 17.3% of its nameplate and Tarbela 4th Extension 9.8%, against full year figures of 45.6% and 41.2%. The annual capacity factor is not a poor number because it is inaccurate. It is a poor number because it averages away the three months that decide whether a supply contract can be honoured.

Figure 14. The Annual Capacity Factor Hides What Happens in Winter

Capacity factor of the six largest WAPDA hydel stations, full year against December to February, FY 2024-25, per cent



Source: ARS calculation from NEPRA, State of Industry Report 2025, company-wise monthly generation data for FY 2024-25, and the installed capacity of each station as recorded in the same source. Capacity factor is generation divided by installed capacity multiplied by the hours in the period. Figures reflect hydrology, irrigation operation, planned outages and system dispatch together, and are not a measure of plant condition. Capacity factor measures energy converted, not availability.

These are utilisation figures rather than availability figures, and the distinction is the one on which the hydro case turns. A station may be fully available and still convert 45% of its nameplate, because the river does not carry the water. A thermal generator contracting under a Pakistani PPA commits to availability of roughly 85% to 92% precisely because availability is something it controls; a run-of-river station cannot make that commitment

in the dry season at any price. What a hyperscale campus buys is a flat quantity of megawatts in every hour, and what decides whether hydro can sell it is not the annual capacity factor but the station's dependable output in the dry season.

The shape behind those figures is seasonal and it is extreme. Indus basin flows are driven by snowmelt from spring and by the summer monsoon, peaking between June and August and falling to their minimum between December and February. Taken together, the six largest WAPDA stations represent 7,835 MW of installed capacity. That fleet ran at 80.0% of nameplate in August 2024 and at 12.1% in January 2025, a swing of more than six to one, and it produced below its own full year average for six consecutive months from November to April.

Figure 15. A Contract Struck at Hydro's Annual Average Would Fail for Six Months of the Year

Combined monthly output of the six largest WAPDA hydel stations, 7,835 MW installed, FY 2024-25, per cent of installed capacity



Source: ARS calculation from NEPRA, State of Industry Report 2025, company-wise monthly generation data for FY 2024-25. The stations are Tarbela, Tarbela 4th Extension, Ghazi Barotha, Mangla, Warsak and Chashma. Monthly output reflects hydrology, irrigation operation, planned outages and system dispatch together. The reference line is the same fleet's full year average of 45.9%.

The Neelum-Jhelum experience makes the availability point more sharply than any hydrological analysis. The 969 MW plant has been out of service since a tunnel collapse on 2 April 2024. WAPDA's chairman told the Senate Standing Committee on Water Resources on 10 July 2026 that restoration work was under way and that the authority hoped to resume generation by March 2028, nearly four years after closure, and the committee pressed for a transparent and independent inquiry into the causes of the collapse and the cost escalations.^{16,17} This followed an earlier tailrace tunnel failure whose repair cost approximately Rs 6 billion together with roughly Rs 37 billion in lost energy over twenty

months. A single-tunnel run-of-river scheme in a seismic zone is a concentrated engineering risk, and the record shows what happens when that risk materialises.

4.6 Hydro Plus Thermal Firming

The workable configuration is hydro for energy and thermal or storage for firm capacity, which is precisely the configuration Pakistan's existing fleet is already capable of providing. The country's underutilised thermal plant is, in this framing, not a liability to be retired but a firming resource to be contracted.

That reframing has a specific contractual implication. A hydro-anchored supply arrangement for a data centre requires at least two agreements running in parallel: an energy contract with the hydro generator, priced per unit and seasonal in volume, and a capacity or standby contract with a firming provider, priced per MW of committed availability and drawn upon when the hydro leg falls short. The second contract is the expensive one and it is the one Pakistani market design has not yet standardised. Replacement power obligations, the mechanism by which a supplier undertakes to source alternative energy when its contracted generator underdelivers, do not currently exist as a standard product in the Pakistani market. Section 11 sets out what such a product would need to contain.

4.7 Northern Pakistan as a Compute Geography

The northern hydro corridor has real advantages for compute siting. Ambient temperatures across Khyber Pakhtunkhwa's northern districts, Azad Jammu and Kashmir and Gilgit-Baltistan permit free cooling for a substantially larger share of the year than Karachi or Lahore, which reduces power usage effectiveness and therefore the total electricity bill for a given IT load. Proximity to generation reduces transmission losses and reduces exposure to the north-south corridor constraint discussed in Section 2.6. Land is available and comparatively inexpensive.

Four constraints qualify the advantage and each will appear in an investor's risk register.

Fibre is the first. Compute capacity is worthless without low-latency, diverse, high-capacity connectivity to the customers who buy it. Pakistan's international capacity improved materially with the landing of the Africa-2 cable, a 45,000 km system connecting 33 countries through 46 landing stations,²⁴ but international capacity lands on the coast. The domestic backhaul from Karachi to a northern campus, and the requirement for genuinely diverse routing rather than nominally diverse routing sharing a common duct, is an unresolved question that Section 5.6 examines in detail.

Seismicity is the second. The northern corridor sits across the collision zone between the Indian and Eurasian plates. Reports highlighting Neelum-Jhelum's location in a seismic zone existed before the project was launched.¹⁷ Data centres can be engineered for seismic risk, and are, in Japan and California. Doing so costs money and it changes the insurance position.

Constitutional and jurisdictional status is the third, and it is the one most often overlooked by promoters. Azad Jammu and Kashmir and Gilgit-Baltistan are not provinces of Pakistan

within the meaning of Article 1(2) of the Constitution and are governed under their own distinct constitutional instruments. That affects which legislature may enact what, which courts have jurisdiction over a dispute, how land is acquired, how a security interest is perfected and registered, and whether a foreign investor's protections operate identically to those available in Punjab or Sindh. This is not an insuperable obstacle. It is a diligence question with a real answer, and the answer must be worked out before a site is selected rather than after.

Transmission is the fourth. The ADB's Second Power Transmission Strengthening Project is intended to enable evacuation of up to 3,200 MW from northern generation clusters, which is an acknowledgement that the corridor cannot presently carry what the north can generate.²⁸ A campus sited in the north to be close to hydro is well placed if it consumes locally and poorly placed if it must draw firming power from the south.

4.8 Water versus Electricity: Two Different Resource Questions

Data centres consume water for evaporative cooling, and the volumes are not trivial. Locating a large campus in the Indus basin therefore raises a water question that is entirely separate from the electricity question, and which in Pakistan carries a political charge that electricity does not.

Water allocation between provinces operates under the 1991 Water Apportionment Accord and is administered by the Indus River System Authority. Any material industrial abstraction in a basin already under stress, in a country where inter-provincial water allocation is among the most sensitive subjects in federal politics, will attract scrutiny that a purely electrical project would not. The technical answer is closed-loop or air-cooled design, which raises capital cost and worsens power usage effectiveness but largely removes the water question. That trade-off should be made deliberately and disclosed early, not discovered during permitting.

There is also a treaty dimension that any foreign investor's country risk assessment will identify. India placed the Indus Waters Treaty in abeyance on 23 April 2025 following the Pahalgam attack and has reaffirmed that position since, including at the United Nations in March 2026 and again in July 2026.^{34,35} The Permanent Court of Arbitration issued a binding award on general interpretation of the treaty on 8 August 2025 and a supplemental award concerning maximum pondage on 15 May 2026, both of which India rejected as null and void on the basis that the court is not properly constituted.^{34,36} Pakistan's position is that the treaty remains valid and operational.

ARS takes no position on the merits of that dispute, which is a matter of international law between two states. The relevant observation for this report is narrower and unavoidable: an investor being asked to place a multi-billion-dollar asset whose power supply is anchored to Indus basin hydrology will read that the treaty governing the upstream regulation of those rivers is currently contested, that one party denies the jurisdiction of the tribunal seized of it, and that hydrological data sharing has been affected. That is a country risk factor and it should be addressed in the offer document rather than left for the investor to find.

4.9 What Hydropower Can and Cannot Do

Table 6. Hydropower Against the Requirements of a Hyperscale Load

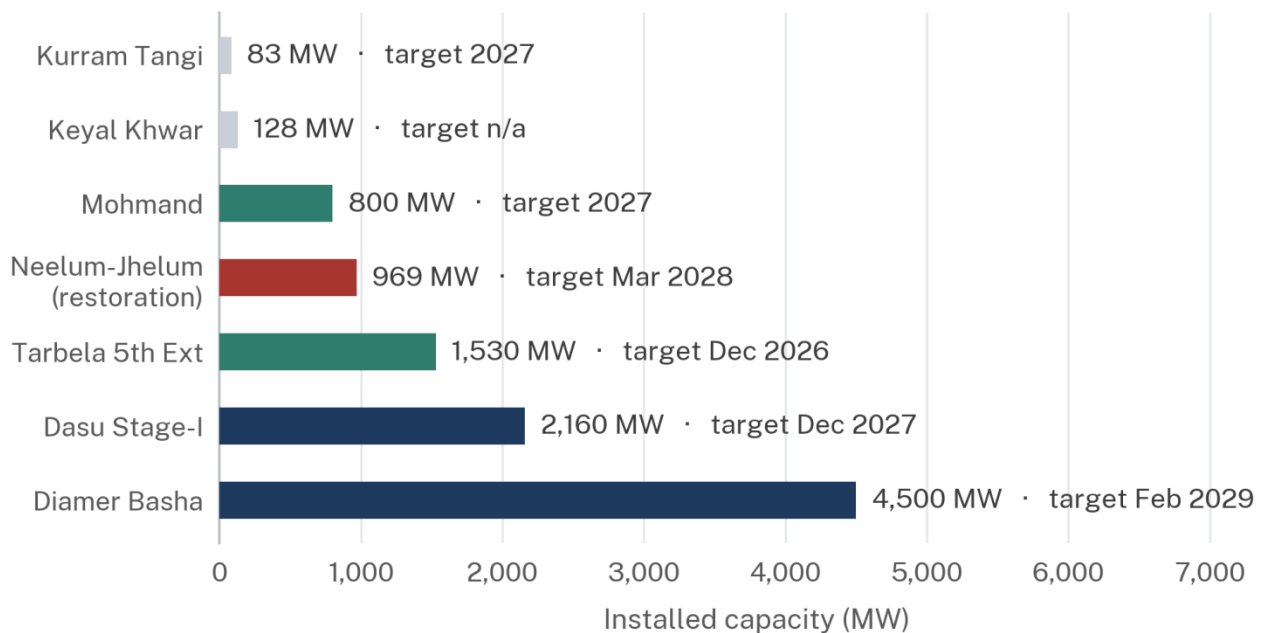
An honest assessment of fit, requirement by requirement

Requirement	What hydro delivers	Verdict	What closes the gap
Low energy cost	Genuinely low variable cost; no fuel exposure	Meets the requirement	Nothing required
Foreign currency resilience	No imported fuel; but new hydro carries dollar-linked capital charges	Partially meets	Local currency financing; longer tenors
Firm output in every hour	Combined output of the six largest stations falls to 12% of nameplate in January against 80% in August, and sits below its full year average for six months	Does not meet	Thermal or storage firming under a separate contract
Winter firmness	Run-of-river output falls to its minimum December to February	Does not meet	Storage hydro, thermal firming, or a seasonal contract structure
Scalability to 300 MW+	Ample installed and pipeline capacity in aggregate	Meets in principle	Transmission corridor reinforcement
Delivery certainty	Neelum-Jhelum out of service since April 2024	Materially qualified	N-1 contracting across multiple generators, not a single asset
Low-carbon profile	Strong, and increasingly a procurement requirement for hyperscalers	Meets the requirement	Certification and tracking mechanism

Source: ARS assessment, drawing on WAPDA generation reporting for calendar year 2025 and the capacity factor calculation set out in Section 4.5.

Figure 16. The Hydro Pipeline Is Real but Arrives After the Window in Which Compute Capacity Is Being Sited

Major hydropower capacity under construction or restoration, with stated target dates, MW



Source: WAPDA project reporting for calendar year 2025; Senate Standing Committee on Water Resources briefing, 10 July 2026, for Neelum-Jhelum restoration to March 2028. Target dates are as stated by the executing agency and have historically slipped. Neelum-Jhelum is a restoration of existing capacity shut since the tunnel collapse of 2 April 2024, not a net addition.

The pipeline chart states the timing problem plainly. Diamer Basha at 4,500 MW is targeted for February 2029, Dasu Stage-I at 2,160 MW for December 2027, Tarbela 5th Extension at 1,530 MW for December 2026 and Mohmand at 800 MW for 2027. These are meaningful additions. They also arrive after the period in which global compute capacity is being sited, and Pakistani hydro target dates have historically slipped by years rather than months. A strategy that requires Diamer Basha to be commissioned on schedule is not a strategy. It is a hope.

4.10 The Correct Statement of the Hydro Case

Hydropower gives Pakistan a durable structural advantage in the cost of energy and a durable structural disadvantage in the firmness of supply. Both statements are true simultaneously, and a serious offer to an international investor must make both.

The correct proposition is not that Pakistan can power data centres with cheap hydroelectricity. It is that Pakistan can offer a blended product in which hydro supplies the energy at low variable cost, existing underutilised thermal capacity supplies the firmness that hydro cannot, and the combination is delivered under a single contract at a single blended price with a single counterparty answerable for delivery. That product does not currently exist in Pakistani law or in Pakistani market practice. Constructing it is the subject of Parts IV and V.

THE DATA CENTRE OPPORTUNITY

5. Why Data Centres Change the Equation

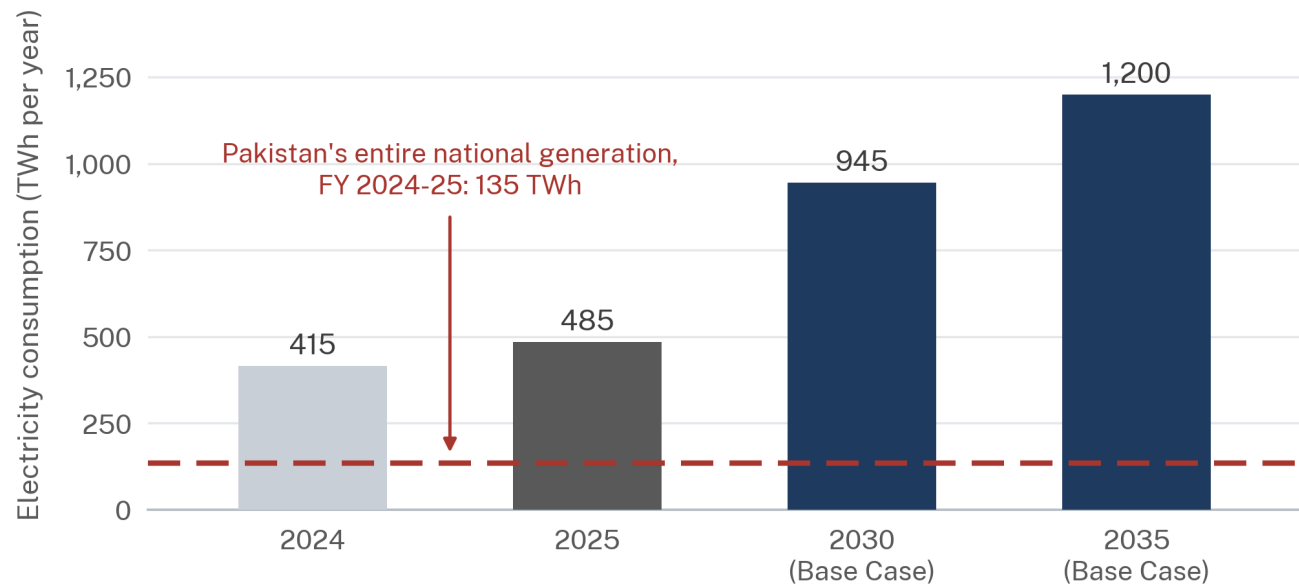
5.1 Artificial Intelligence Has Made Electricity a Strategic Input

For most of the internet era, the location of computing capacity was determined by proximity to users and by the cost of bandwidth. Electricity was a line item. Since 2023 that has changed. Training and serving large models requires accelerator hardware with power densities several times those of conventional server racks, and the binding constraint on new capacity in most developed markets is now the availability of grid connections rather than the availability of capital, land or chips.

The International Energy Agency puts global data centre electricity consumption at 415 TWh in 2024, approximately 1.5% of world electricity demand, rising to around 945 TWh by 2030 under its Base Case.⁵ Its updated assessment records consumption of 485 TWh in 2025, growth of 17% against overall global electricity demand growth of 3%, and projects roughly 950 TWh by 2030 and 1,200 TWh by 2035, with global data centre capacity approaching 100 GW.⁶ The increment between 2024 and 2030 alone is more than four times Pakistan's entire annual generation.

Figure 17. Global Data Centres Will Add Roughly Four Times Pakistan's Total Generation by 2030

World data centre electricity consumption, IEA Base Case, TWh per year



Source: International Energy Agency, Energy and AI, for 2024 and 2030; IEA, Key Questions on Energy and AI, for the 2025 estimate. The Pakistan comparison is from NEPRA State of Industry Report 2025. The 2035 figure is the IEA Base Case, not a forecast.

That growth has to be sited somewhere, and the queue for grid connections in Northern Virginia, Ireland, the Netherlands, Singapore and parts of the United States is now measured in years. This is the window Pakistan is being invited to enter. It is also a window that closes: capacity sited elsewhere between 2026 and 2030 does not come back.

5.2 Hyperscale versus Conventional Data Centres

The distinction matters because the two categories buy electricity in entirely different ways. A conventional enterprise or colocation facility in Pakistan today draws a few megawatts on an ordinary commercial connection and treats power as an operating cost. Transworld Associates, the landing partner for several of Pakistan's submarine cables, has been developing a Tier III certified facility in Karachi connected directly to the subsea systems.³⁷ Facilities of that class are valuable and they do not require a change in the law.

A hyperscale campus is a different proposition. It draws between 100 MW and 1,000 MW, is built in phases over several years, is financed on long-term project debt or corporate balance sheet against a fifteen to twenty-five year asset life, and negotiates its power supply directly rather than accepting a published tariff. It is the second category that requires legal reform, and it is the second category that would make a material difference to Pakistan's electricity economics and to its export earnings.

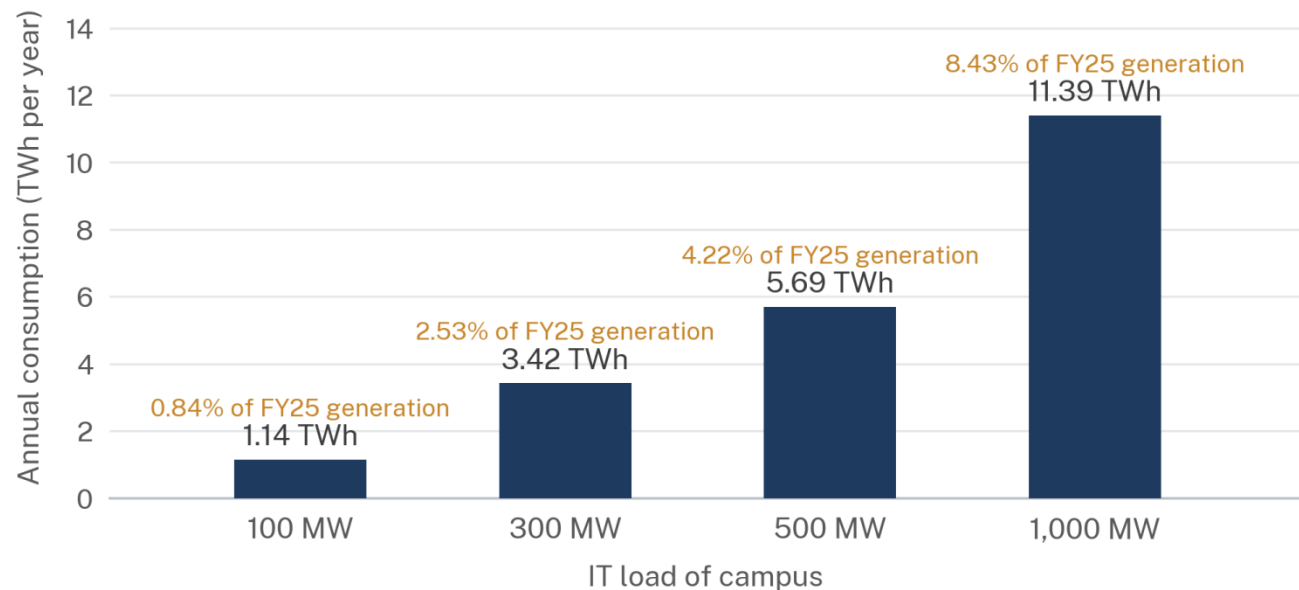
5.3 What 100 MW, 300 MW, 500 MW and 1 GW Actually Mean

These figures are quoted loosely and it is worth fixing what they represent. The number normally refers to critical IT load, the power delivered to the servers themselves. Total

facility load is higher because cooling, power conversion and lighting consume additional energy. The ratio is power usage effectiveness. A well-designed modern facility in a temperate climate achieves a PUE around 1.15 to 1.25; a facility in a hot climate using conventional mechanical cooling will be higher. This report uses 1.30 throughout as a conservative planning assumption for Pakistani conditions and states it wherever it is applied.

Figure 18. A Single Gigawatt Campus Would Consume More Than Eight Per Cent of Pakistan's Electricity

Annual electricity consumption by campus size, at an assumed power usage effectiveness of 1.30



Source: ARS calculation. Total facility load equals IT load multiplied by a PUE of 1.30; annual energy equals facility load multiplied by 8,760 hours at 100% utilisation. The share is measured against NEPRA's reported FY 2024-25 net generation of 135,078 GWh. Illustrative, not a forecast.

Table 7. What Each Campus Size Means in Pakistani Terms

Facility load, annual consumption and system significance, at PUE 1.30 and continuous operation

IT load	Facility load	Annual energy	Share of national generation	Share of average national load	Annual electricity purchase at 8 US cents/kWh
100 MW	130 MW	1.14 TWh	0.84%	0.84%	Rs 25 billion
300 MW	390 MW	3.42 TWh	2.53%	2.53%	Rs 76 billion
500 MW	650 MW	5.69 TWh	4.22%	4.22%	Rs 127 billion
1,000 MW	1,300 MW	11.39 TWh	8.43%	8.43%	Rs 255 billion
2,000 MW (the allocation announced in 2025)	2,600 MW	22.78 TWh	16.86%	16.86%	Rs 510 billion

Source: ARS calculation. Facility load equals IT load multiplied by 1.30. Annual energy equals facility load multiplied by 8,760 hours. National comparison uses NEPRA's reported FY 2024-25 net generation of 135,078 GWh and average load of approximately 15,420 MW. Rupee value assumes a delivered price of 8 US cents per kWh converted at Rs 280 to the US dollar, the same basis as Table 14, and is illustrative.

The last row deserves attention because it is the government's own stated figure. An allocation of 2,000 MW, if it were fully built and fully utilised, would consume close to 17% of Pakistan's current annual generation and would generate electricity purchases on the order of Rs 510 billion a year at an assumed delivered price of 8 US cents per unit, equivalent to approximately Rs 22.4 at Rs 280 to the US dollar. Set against a projected capacity charge of Rs 2.03 trillion to Rs 2.16 trillion, that is a contribution of real significance to the fixed-cost recovery problem described in Part I. It is also, on any realistic assessment, a decade of construction rather than a single announcement.

5.4 Reliability Requirements

Reliability is where the Pakistani proposition is weakest and where the conversation with an investor will begin. The Uptime Institute tier classification is the industry reference: Tier III facilities are concurrently maintainable and Tier IV are fault tolerant, with corresponding design availability targets that translate into permitted annual downtime measured in hours for Tier III and in minutes for Tier IV. A hyperscale operator building at 300 MW will typically design to Tier III equivalence with N+1 or 2N redundancy in the power train, and will contract for an availability standard at the point of delivery that Pakistan's distribution network does not currently meet at any voltage level.

The practical consequence is that the campus must be designed as though the grid will fail, with on-site uninterruptible power supply and generation capable of carrying the full load. That is normal practice everywhere. What is not normal is designing for the frequency of interruption that Pakistani distribution feeders experience, and the

additional redundancy required is a cost that must be counted against whatever electricity price advantage Pakistan offers.

5.5 Cooling, Water and Site Conditions

Cooling is the second largest electrical load after the servers themselves and its cost is driven by ambient conditions. Karachi's coastal humidity, Lahore's summer dry-bulb temperatures and the northern corridor's cooler climate produce materially different PUE outcomes for the same equipment. The northern hydro corridor's advantage on this dimension is real and is one of the few unambiguous Pakistani advantages identified in this report.

The cooling technology choice is also a water choice. Evaporative cooling reduces electricity consumption and consumes water. Closed-loop and air-cooled designs consume little or no water and raise electricity consumption. In the Indus basin, where inter-provincial allocation under the 1991 Water Apportionment Accord is administered by the Indus River System Authority and is among the most politically sensitive subjects in Pakistani federalism, this trade-off should be resolved in favour of low water consumption and disclosed at the outset. Section 14 returns to this in the site analysis.

5.6 Fibre and Network Requirements

Compute capacity is worthless without connectivity to the customers who buy it, and Pakistan's position here has improved sharply and remains structurally limited.

Pakistan is served by submarine systems including SMW-4, SMW-5, IMEWE, AAE-1, TW-1 and PEACE, which together provided approximately 13 Tbps of international bandwidth before recent additions.³⁸ Africa-1 landed at Karachi in early 2025. The 2Africa system, extending 45,000 km across 46 landing stations in 33 countries with a design capacity of up to 180 Tbps, lands at Karachi through Transworld Associates as part of the 2Africa Pearls branch.³⁹ SEA-ME-WE 6 was connected in November 2025, with the landing partner initially providing 4 Tbps of additional capacity, an increase of nearly 40% on the then-current national total.³⁸ Three international systems were added within a single year.

Two qualifications limit what this means for a hyperscale campus. First, all of it lands at Karachi. There is no diversity of landing geography, and a campus sited in the north depends entirely on domestic backhaul from the coast. Second, the requirement for a hyperscaler is not headline capacity but genuinely diverse routing: two paths that do not share a duct, a bridge crossing or a single point of failure. Whether Pakistan can offer that between Karachi and an inland site is an engineering question that has not been publicly answered, and it is one of the first questions a network architect will ask.

5.7 Site Selection Criteria

Bringing these together, a hyperscale site selection team applies a standard filter in roughly this order: is there deliverable power at the required scale, is the price predictable for the contract term, is there diverse fibre, is the climate suitable, is water available or avoidable, is the land acquirable with clear title, is the seismic and flood risk manageable, is the jurisdiction one in which the investment can be financed and insured,

and can permits be obtained on a known timetable. Pakistan currently produces a defensible answer to the first, a weak answer to the second and no settled answer to the last. Section 14 scores six candidate locations against these criteria.

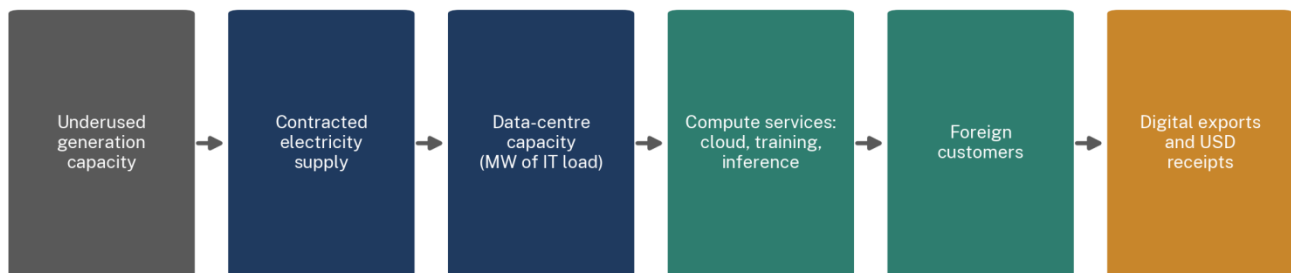
6. What Would Pakistan Actually Be Selling?

6.1 Electricity Is the Input, Not the Export

The most common analytical error in this debate is to treat the proposition as electricity export. It is not. Pakistan cannot export electrons to a customer in Frankfurt. What it can do is convert electricity into a service that is delivered over fibre and billed in foreign currency, which is a fundamentally different transaction with different economics and a different legal architecture.

Figure 19. What Pakistan Would Actually Be Selling Is Compute Time, Not Kilowatt-Hours

The conversion chain from idle generation to foreign exchange earnings



Electricity is the input. It is not the export. What Pakistan sells is compute time, priced in dollars.

Source: ARS analysis. Each link in the chain requires a distinct legal instrument, from the power purchase agreement at the left to the data transfer and tax framework at the right.

The distinction has three practical consequences. First, the value captured depends on where in the chain Pakistan participates: leasing space and power to a foreign operator captures far less than operating the compute service, which in turn captures less than owning the models or the platform. Second, the foreign exchange benefit accrues only if the revenue is billed abroad and repatriated, which is a State Bank and tax question rather than an energy question. Third, the electricity itself is sold domestically in rupees at a domestic tariff, which means the electricity transaction and the export transaction are legally and commercially separate even though they are physically the same activity.

6.2 The Layers of the Offer

Table 8. What Pakistan Could Sell, and What Each Layer Is Worth

Ascending order of value capture and of institutional difficulty

Layer	What is sold	Value capture	Difficulty	Foreign exchange effect
Wholesale capacity	Powered shell and MW of IT capacity leased to a foreign operator	Lowest	Lowest	Indirect; rent received in USD
Colocation	Rack space, power and cooling to enterprise customers	Low	Low	Partial; mixed domestic and foreign customers
Enterprise hosting	Managed hosting and disaster recovery for regional customers	Moderate	Moderate	Direct where customers are foreign
Cloud services	Compute, storage and networking sold as a regional cloud region	High	High	Direct and recurring
AI inference	Serving trained models to customers by the token or the request	High	High	Direct; latency-sensitive, favours regional demand
AI training	Large-scale training runs sold as reserved capacity	Highest	Highest	Direct; latency-insensitive, favours cheap power

Source: ARS analysis. Value capture and difficulty are relative rankings, not measured values. Foreign exchange treatment reflects whether revenue is ordinarily billed to a foreign customer and repatriated.

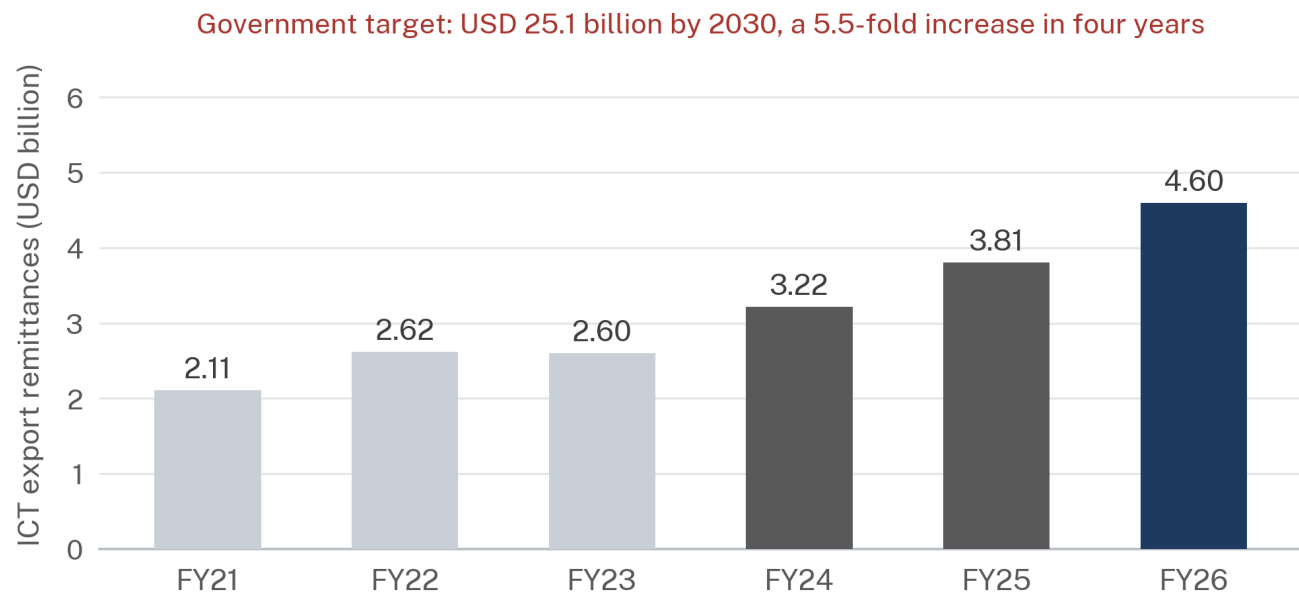
The bottom two rows deserve emphasis because they point in different directions. Inference is latency-sensitive and therefore wants to be near its users, which favours Pakistan serving Pakistani and regional demand. Training is largely latency-insensitive and cares almost entirely about the cost of power and the availability of capacity, which is exactly the arbitrage Pakistan is positioned to offer. A realistic Pakistani proposition is training and batch workloads first, inference for the domestic and regional market second, and full cloud region status only once the first two are established.

6.3 Digital Services Exports and Foreign Exchange

Pakistan already exports digital services at scale and the trajectory is strong. IT and telecom export remittances reached a record USD 4.6 billion in FY 2025-26, up 20.6% from USD 3.814 billion in FY 2024-25, having grown from USD 2.108 billion in FY 2020-21.⁴⁰ The Ministry of Information Technology and Telecommunication has set a target of USD 25.1 billion by 2030.⁴¹

Figure 20. Pakistan's Digital Exports Are Growing Fast and Are Nowhere Near the Official Target

IT and telecom export remittances, FY 2020-21 to FY 2025-26, USD billion



Source: State Bank of Pakistan export remittance data as reported; the official target is as stated by the Ministry of Information Technology and Telecommunication. The target lies above the plotted range and is shown as an annotation only.

That target is a 5.5-fold increase in four years and it will not be reached by extending the current model, which is dominated by services delivered by people rather than by machines. Freelancing and software services scale with the number of trained workers. Data centre capacity scales with megawatts and capital. The strategic case for compute infrastructure is precisely that it is the only element of Pakistan's digital export mix whose growth is not constrained by the rate at which the country can train software engineers.

STATING THE SCENARIO HONESTLY

ARS does not forecast what data centre exports would be worth to Pakistan, and this report contains no such forecast. The arithmetic that can be stated is narrower and is set out in Section 15: what a campus of a given size consumes, what it costs to build, and what it would pay for electricity. Revenue depends on commercial terms with customers who do not yet exist, and any figure attached to it would be a construction rather than an estimate.

7. What Would a Pakistani Data Centre Power Deal Look Like?

Five structures are available in principle under existing Pakistani law. Only one of them is in routine commercial use. This section sets each out, identifies its legal basis, and assesses it against the criteria a project finance lender applies.

Figure 21. Five Ways to Get Electricity to a Pakistani Data Centre, and Only One of Them Is Currently Routine

Transaction architecture of each procurement model

MODEL A

Regulated tariff supply



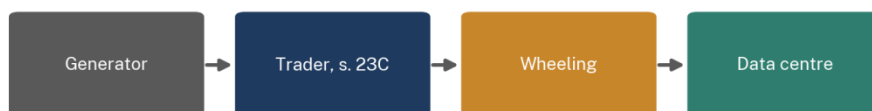
MODEL B

Bilateral PPA, wheeled



MODEL C

Supply via licensed trader



MODEL D

On-site captive plant



MODEL E

Hydro PPA plus firming



Source: ARS analysis of the NEPRA Act, 1997 as amended, and the NEPRA (Wheeling of Electric Power) Regulations, 2016. Model A is the only route in routine commercial use today. Models B, C and E depend on the wheeling and bilateral contract framework; Model D depends on section 14C.

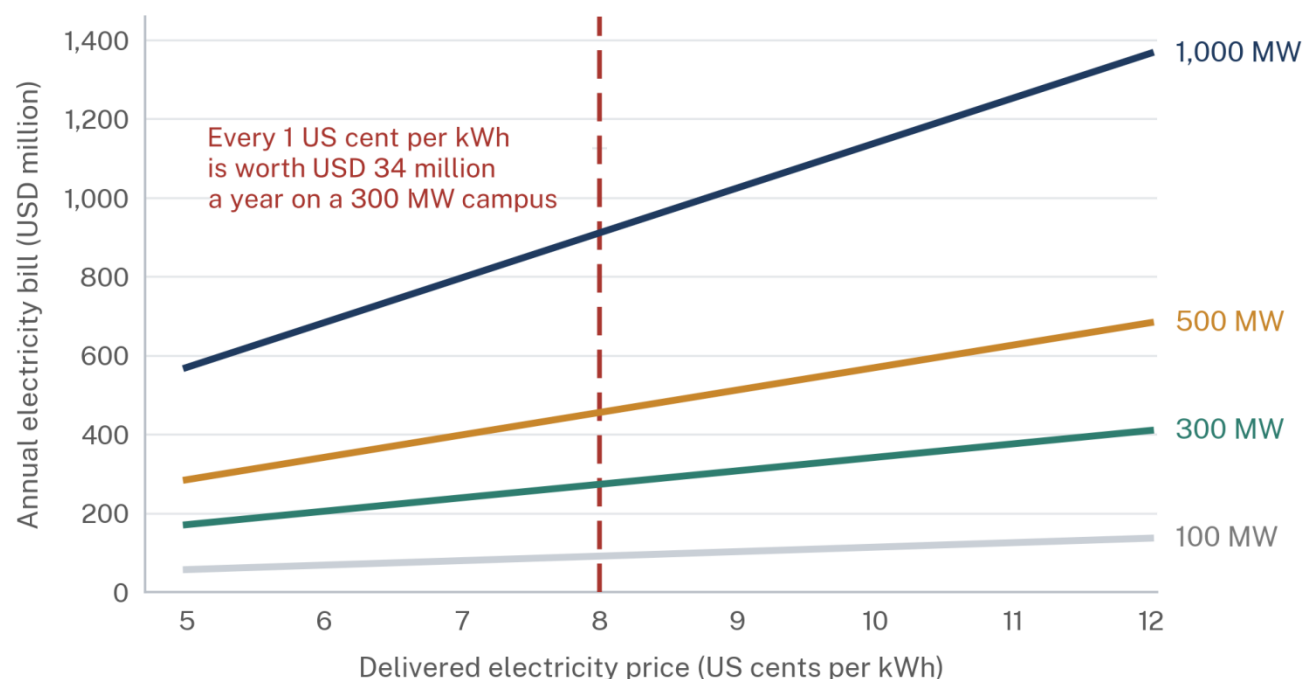
7.1 Model A: Supply from a DISCO or Licensed Supplier at the Notified Tariff

The investor takes a connection from the distribution company in whose service territory the site sits and pays the tariff determined by NEPRA under section 31, as notified by the Federal Government. The legal basis is settled: a distribution licence under section 20 and a supply licence under sections 23E and 23F, with existing distribution licensees deemed to hold supply licences under the proviso to section 23E(1).⁷

This route works today and requires no reform. Its defect is price. The consumer pays the full capacity charge, the cross-subsidy, the debt servicing surcharge and general sales tax, which is the Rs 34 per unit national average tariff described in Part I. At a delivered price around 12 US cents per kWh, a 300 MW campus would face an annual electricity bill of roughly USD 410 million, against roughly USD 205 million at 6 cents. On a fifteen-year view that difference is larger than the entire construction cost of the facility.

Figure 22. Price, Not Availability, Is the Variable a Hyperscaler Optimises

Annual electricity cost by campus size and delivered tariff, USD million



Source: ARS calculation on the same PUE 1.30 and 8,760-hour basis as Figure 18. Delivered price is the all-in price at the campus meter including energy, capacity, use of system charges and taxes. Illustrative sensitivity, not a price quotation.

7.2 Model B: Generator, Bilateral PPA, Wheeling, Data Centre

The investor contracts directly with a chosen generator and pays the network owners to transport the energy. The legal basis is section 14D(3), under which a generation company may supply electric power to any consumer subject to section 23E, read with the proviso to section 21(2)(a) preserving sales to bulk power consumers, section 22 permitting such sales, and the NEPRA (Wheeling of Electric Power) Regulations, 2016, notified vide SRO 549(I)/2016, which allow a licensed generator to sell to a consumer of 1 MW or more anywhere in Pakistan.^{7,8}

This is the structure the whole proposition depends on and it is the structure with the greatest legal uncertainty. Section 22 provides that the Authority "may permit" the sale. Wheeling charges are set administratively and have moved from above Rs 22 per unit to below Rs 9 per unit.⁹ The Framework Guidelines for Wheeling Auctions 2025 introduced a competitive allocation of a defined wheeling quantum, with NEPRA determining uniform wheeling charges and the market operator running the auction, and approximately 800 MW to be allocated.¹¹ Competitive allocation is an improvement on discretionary allocation. It is not the same as a right, and an auction that allocates a fixed quantum is by definition a rationing mechanism.

7.3 Model C: Generator, Licensed Trader, Data Centre

An electric power trader licensed under section 23C intermediates between generators and the consumer, aggregating supply and managing balancing. Section 23D(2) sets out

the trader's obligations, including at subsection (2)(d) the requirement to enter into agreements with transmission and distribution licensees for use of their systems.⁷

The advantage of this model is that the trader, not the data centre, carries the burden of managing multiple generators and the network relationships, which is exactly the specialisation a hyperscaler wants. The obstacle is that the Pakistani trading market is thin, and a trader's ability to offer a firm fifteen-year price depends on its own upstream contracts and its own balance sheet. A hyperscaler will look through the trader to the underlying credit.

7.4 Model D: Captive Generation

The investor builds its own generation on or adjacent to the site under sections 2(ia) and 14C. Section 3 of this report set out why this route cannot carry a hyperscale campus alone: the Off the Grid (Captive Power Plants) Levy Act, 2025 has removed the commercial basis for the gas-fired version, and the solar-plus-storage version requires land and storage at a scale that does not clear the capital cost test at Pakistani financing rates.^{12,13}

7.5 Model E: Hydro PPA plus Grid plus Thermal Firming plus Backup

This is the configuration ARS regards as the only one that satisfies both the price test and the reliability test, and it is the configuration this report recommends. Hydro or efficient thermal generation supplies the bulk of the energy under a long-term bilateral agreement. Grid supply from a licensed supplier provides firming and balancing. On-site solar with storage reduces average cost and provides ride-through. On-site backup generation provides the final layer.

Its difficulty is that it requires every one of the preceding legal bases to work simultaneously, and it requires one instrument that does not exist in Pakistan at all: a replacement power obligation under which the supplier undertakes to source alternative energy when the contracted generator underdelivers, at a defined price and with a defined remedy. Section 11 sets out what that instrument would need to contain.

Table 9. The Five Models Compared Against the Criteria a Lender Applies

Legal basis, commercial character, and readiness of each structure

Model	Legal basis	Indicative delivered price	Reliability	Scalability to 300 MW+	Regulatory complexity	Bankable today
A. DISCO or supplier	ss. 20, 23E, 23F, 31	Highest; full tariff stack	Grid-dependent	Constrained by feeder capacity	Low	Yes, but uneconomic
B. Bilateral PPA with wheeling	s. 14D(3), s. 22, Wheeling Regs 2016	Lower; energy plus use of system	Depends on curtailment terms	Good, subject to corridor	High	No
C. Licensed trader	ss. 23C, 23D	Lower; trader margin applies	Depends on trader's portfolio	Good in principle	High	No
D. Captive generation	ss. 2(ia), 14C	Gas route penalised; solar high capex	High, self-controlled	Poor at hyperscale	Moderate	Partially
E. Hydro PPA plus firming	All of the above combined	Lowest achievable blended price	Highest, if firming contracted	Best	Highest	No

Source: ARS analysis of the NEPRA Act, 1997 as amended, the NEPRA (Wheeling of Electric Power) Regulations, 2016, the Framework Guidelines for Wheeling Auctions 2025 and the Off the Grid (Captive Power Plants) Levy Act, 2025. Price bands are illustrative planning ranges, not quotations. Bankability is an ARS judgement.

The table states the central finding of Part III. The model that works commercially is the model that is least ready legally, and the model that is ready legally is the model that does not work commercially. Closing that gap is the subject of the remainder of this report.

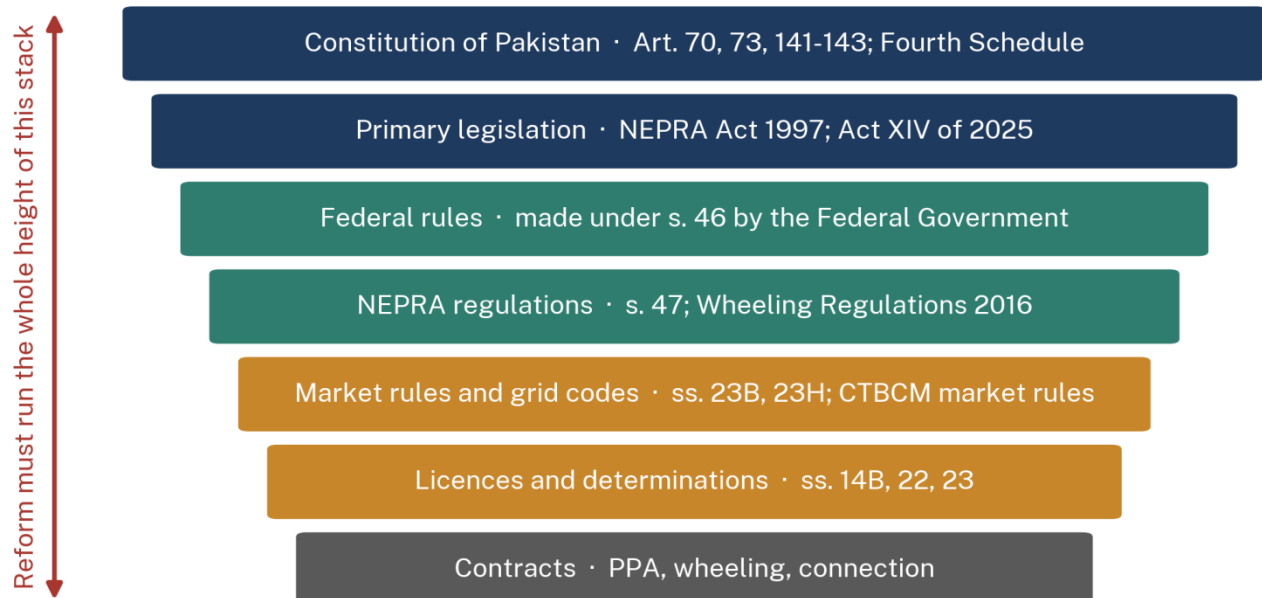
WHAT CAN THE LAW DO TODAY?

8. Pakistan's Current Electricity Law Framework

This section takes the relevant provisions in turn. For each it states what the law says, what it means in practice, what a data centre can do today on the strength of it, and what remains uncertain. It is important to be precise about a point made in Part I: Pakistan does not have a legal vacuum in electricity. It has a reasonably complete legal architecture that was designed for a transition it has not yet completed.

Figure 23. A Bankable Power Contract Requires Every Tier of This Stack to Be Aligned

Hierarchy of legal instruments governing a Pakistani electricity transaction



Source: ARS analysis. A change at one tier that is not carried through the tiers below it does not change what an investor can actually contract for.

8.1 The NEPA Act, 1997 and Its 2018 Amendment

The governing statute is the Regulation of Generation, Transmission and Distribution of Electric Power Act, 1997 (Act XL of 1997), substantially amended by Act XII of 2018.⁷ The 1997 Act established NEPA as an independent regulator with exclusive responsibility for licensing and tariff determination. The 2018 Amendment inserted the architecture of a

competitive market: captive generation and open access at section 14C, the market operator at sections 23A and 23B, the electric power trader at sections 23C and 23D, the electric power supplier at sections 23E and 23F, and the system operator at sections 23G and 23H.

Section 45 gives the Act overriding effect over inconsistent laws. Section 46 empowers the Federal Government to make rules and section 47 empowers the Authority to make regulations. Section 12G provides an appeal to the Appellate Tribunal within three months, with a further appeal to the High Court. Section 7(4) preserves a Province's power to construct power houses and grid stations and to determine tariff for distribution within the Province, a provision whose significance for provincially sponsored compute zones has not been tested.

8.2 Bulk Power Consumers and Section 22

Section 2(ii) defines a bulk power consumer as a consumer who purchases or receives electric power at one premises in an amount of one megawatt or more, or such other amount as the Authority may determine. Any data centre in the scope of this report clears that threshold many times over. The definition confers no procurement right by itself; it identifies a class.

Section 22 is the provision that matters and its drafting is decisive. It provides that the Authority "may permit" the sale of electric power to bulk power consumers located within the service territory of a distribution or supply licensee, with subsection (2) requiring one year's notice before a consumer exits an existing supply arrangement. The verb is permissive. There is no statutory criterion the Authority must apply, no timeline within which it must decide, and no deemed approval if it does not. An investor cannot tell from the face of the statute whether its application will succeed, how long it will take, or on what basis a refusal could be challenged beyond ordinary administrative law grounds.

8.3 Sections 23C to 23H: Traders, Suppliers, Market and System Operators

Sections 23C and 23D create the electric power trader. Section 23D(2) sets out the trader's obligations, including at subsection (2)(d) the requirement to enter into agreements with transmission and distribution licensees for the use of their systems. A trader is therefore only as capable as the network agreements it can obtain.

Sections 23E and 23F create the electric power supplier. The proviso to section 23E(1) deems existing distribution licensees to hold supply licences for a transitional period, which was intended to allow continuity while competition developed. A supplier may decline to serve a consumer who has contracted elsewhere, which is the counterpart of the consumer's freedom to choose and is a provision a data centre relying on grid firming must read carefully.

Sections 23A and 23B create the market operator and sections 23G and 23H the system operator. The system operator's functions include generation scheduling, commitment and dispatch, transmission congestion management and the procurement of ancillary services. NEPRA has supported the transfer of the market operator and system operator functions into an Independent System and Market Operator.²⁵ The significance for this

report is that congestion management and curtailment ranking, which determine who loses power first when the network is constrained, are functions of the system operator and are governed by market rules rather than by contract. A data centre cannot negotiate its own curtailment position bilaterally; it can only be protected by the rules.

8.4 Captive Generation and Open Access: Section 14C

Section 14C(1) permits any person to construct, maintain or operate a captive generating plant and dedicated transmission lines, with a proviso that supply through the grid is regulated as though the plant were that of a generating company. Section 14C(2) provides that every person who has constructed and operates a captive generating plant "shall have the right to open access" for carrying electricity to the destination of his use, subject to availability of adequate transmission facility as determined by the national grid company or the provincial grid company, with disputes adjudicated by the Authority.

This is the only place in the Act where a procurement-related entitlement is drafted as a right rather than a permission, and the drafting is therefore instructive for the reform proposed in Section 11. Its weakness is the condition. The determination of transmission availability is made by the entity whose network is being accessed, on criteria the Act does not specify, within no stated period, and the only remedy is adjudication by NEPRA followed by appeal. A right whose exercise depends on an unreviewable-in-practice determination by the counterparty is a right in form.

8.5 Wheeling and the 2016 Regulations

The NEPRA (Wheeling of Electric Power) Regulations, 2016, notified vide SRO 549(I)/2016, provide the operative framework under which a licensed generator may sell to a bulk power consumer anywhere in Pakistan using the transmission and distribution networks of others.⁸ The regulations establish the requirement for a wheeling agreement and for the payment of use of system charges.

Three features limit their usefulness for a hyperscale transaction. The use of system charge is determined administratively and has been revised materially, falling from above Rs 22 per unit to below Rs 9 per unit.⁹ There is no published methodology binding the charge for the term of a long-term contract. And there is no NEPRA-approved standard wheeling agreement, which means every transaction is negotiated from first principles with counterparties who have no commercial incentive to conclude it quickly.

8.6 CTBCM and the State of the Market in 2026

The Competitive Trading Bilateral Contract Market is the market design NEPRA approved to move Pakistan from a single buyer model to bilateral contracting. It was formally launched in January 2026, with an initial trading tranche and a first competitive auction of approximately 800 MW scheduled for September 2026.^{9,10} The Framework Guidelines for Wheeling Auctions 2025 were approved in December 2025, with NEPRA to determine uniform wheeling charges and the market operator to conduct the auction.¹¹

This is genuine progress and the report should say so plainly. Pakistan is further along than most commentary allows. But an auction that allocates a fixed quantum of wheeling

capacity is a rationing instrument, and a hyperscaler cannot build a fifteen-year investment case on winning successive auctions. The distinction between a market that allocates scarcity and a market that confers rights is the distinction on which this entire report turns.

8.7 The Legal Matrix

Table 10. Provision by Provision: What the Law Permits and What Must Change

The principal analytical table of this report

Provision	What the law says	What a data centre can do today	Limitation	Required change	Instrument
s. 2(ii)	Bulk power consumer means 1 MW or more at one premises	Qualify automatically as a bulk power consumer	Status confers no entitlement to procure	Create a defined class of strategic large load with objective criteria	Primary legislation
s. 2(ia)	Captive plant means one set up primarily for own use	Build behind-the-meter generation without a licence	"Primarily" undefined; multi-tenant campus status unclear	Define permitted self-use share and treatment of co-located tenants	NEPRA regulation
s. 7(4)	A Province may build power houses and set intra-provincial distribution tariff	Explore a provincially sponsored zone supply arrangement	Untested for large industrial loads; interaction with federal market unclear	Clarify the interface between provincial supply and the national market	Rules under s. 46
s. 14B(4)	Generator must be available for non-discriminatory economic dispatch	Understand that dispatch is a system function	A bilateral contract cannot compel dispatch of the chosen plant	Define financial settlement against system dispatch in the market rules	ISMO market rules
s. 14C(1)	Any person may build a captive plant and dedicated lines	Build on-site generation	Grid export is regulated as ordinary generation	No change required	None
s. 14C(2)	Right of open access, subject to availability determined by the grid company	Apply for open access from a captive plant	Determination by the counterparty; no timeline; no deemed approval	Statutory time limit with deemed approval and published capacity register	Primary legislation

Provision	What the law says	What a data centre can do today	Limitation	Required change	Instrument
s. 14D(3)	Generator may supply any licensee and, subject to s. 23E, any consumer	Contract bilaterally with a generator in principle	Cross-reference to s. 23E imports the supply licensing regime	Clarify that a qualifying large load may contract without a supply licence	Primary legislation
s. 21(2) (a) proviso	Preserves generation company sales to bulk power consumers	Rely on it as a supporting basis	Operates subject to s. 22	Align with the new statutory right	Primary legislation
s. 22	The Authority may permit sale to bulk power consumers; one year's notice to exit	Apply for permission	Discretionary. No criteria, no timeline, no deemed approval	Convert to a right for qualifying loads with objective eligibility	Primary legislation
ss. 23A-23B	Market operator licence and commercial code	Transact through CPPA-G	Transition to ISMO incomplete	Complete the licence transfer and publish the market code	Notification and licence
ss. 23C-23D	Electric power trader; must hold network agreements	Contract through a licensed trader	Thin market; trader depends on network agreements it may not obtain	Standard network access agreement available to all traders on equal terms	NEPRA regulation
ss. 23E-23F	Electric power supplier; may decline a consumer contracted elsewhere	Buy firming from a supplier	No standby or replacement power product exists	Create a licensed standby and replacement power product with a determined tariff	NEPRA regulation
ss. 23G-23H	System operator; congestion management and ancillary services	Nothing directly	Curtailment ranking unpublished; no compensation regime	Publish curtailment ranking and a compensation mechanism	ISMO market rules
s. 27	No assignment or transfer of a licence without the Authority's approval	Nothing directly	Blocks pre-agreed lender step-in; approval is case by case	Pre-approved assignment and step-in for registered strategic projects	Primary legislation

Provision	What the law says	What a data centre can do today	Limitation	Required change	Instrument
s. 31	Tariff determination; s. 31(3)(f) prefers competition to regulation	Rely on the stated policy preference	No long-term tariff methodology for a 15-year bilateral contract	Publish a binding long-term tariff and use of system methodology	NEPRA regulation
s. 45	The Act overrides inconsistent laws	Rely on federal primacy in electricity	Does not resolve conflicts with provincial land, water and environmental law	Inter-governmental coordination protocol for designated zones	Administrative
Wheeling Regs 2016	Licensed generator may sell to a 1 MW+ consumer anywhere in Pakistan	Wheel energy from a chosen generator	Charges set administratively ; no standard agreement	Published methodology, term certainty and a standard wheeling agreement	NEPRA regulation
Act XIV of 2025	Levy on gas and RLNG used by captive plants, rising to 20% from Aug 2026	Nothing helpful	Removes the commercial basis for gas-fired self-generation	Exempt or carve out generation dedicated to a designated compute zone	Primary legislation

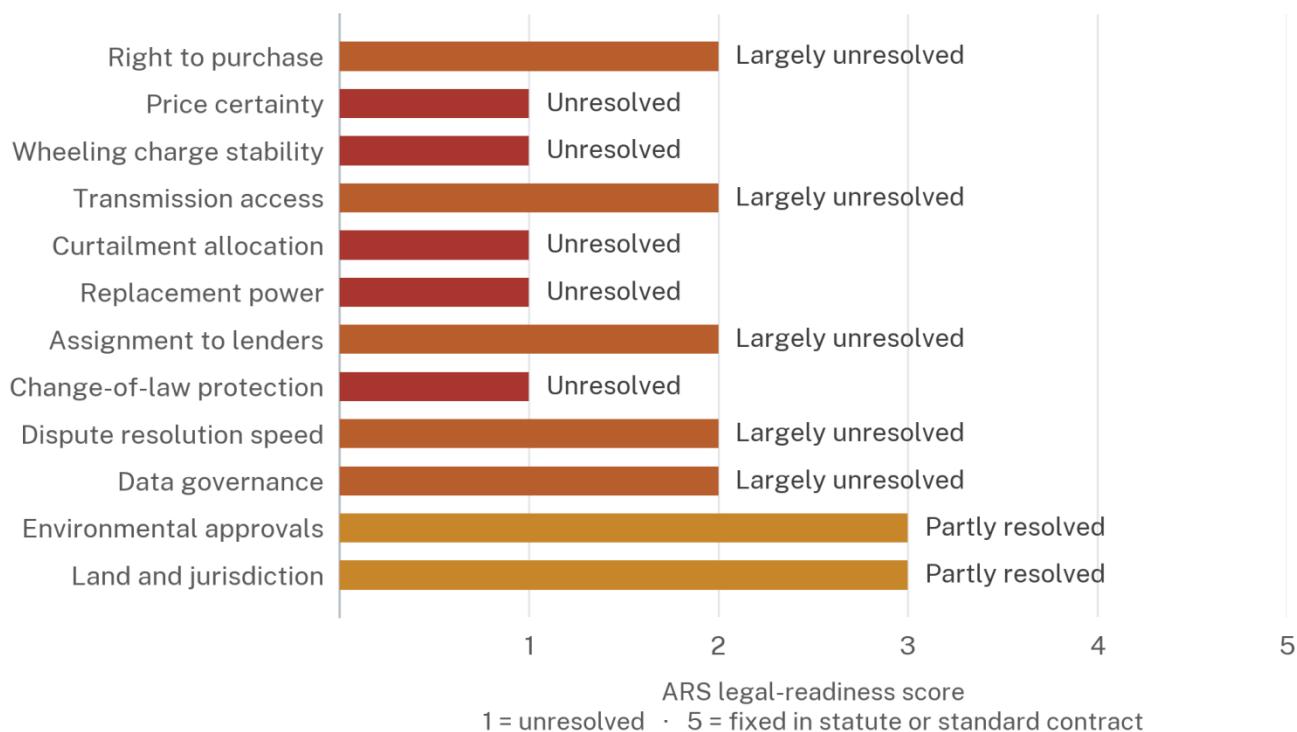
Source: Regulation of Generation, Transmission and Distribution of Electric Power Act, 1997 (Act XL of 1997) as amended by Act XII of 2018; NEPRA (Wheeling of Electric Power) Regulations, 2016 (SRO 549(I)/2016); Off the Grid (Captive Power Plants) Levy Act, 2025 (Act XIV of 2025); Framework Guidelines for Wheeling Auctions 2025. Columns four to six are ARS analysis and constitute a policy proposal, not a legal opinion.

9. Where the Existing Legal Framework Falls Short

Section 8 established that Pakistan has most of the legal components. This section identifies why they do not yet assemble into a financeable transaction. Throughout, the report distinguishes three categories of obstacle that are routinely conflated: legal restrictions, which arise from what the statute or regulation says; engineering restrictions, which arise from the physical network; and administrative restrictions, which arise from how institutions behave within a framework that would otherwise permit the transaction. The remedies are different in each case, and proposing a legislative amendment to solve an engineering problem wastes the amendment.

Figure 24. Every Question a Project Finance Lawyer Asks Is Currently Answered by Discretion or Not at All

ARS assessment of the legal readiness of each element of a hyperscale power transaction in Pakistan



Source: ARS assessment on a five-point scale from resolved to unresolved, applied consistently across dimensions. Resolved means the answer is fixed in statute or standard contract and is not subject to unilateral administrative change. This is an analytical judgement, not a measurement.

9.1 A Right to Buy or a Permission to Buy?

This is the foundational defect and everything else follows from it. Section 22 says the Authority "may permit". Section 14C(2) says a captive generator "shall have the right", then conditions that right on a determination by the network owner. Neither provision gives an investor an entitlement it can plead, price and finance.

The comparator is instructive. India's Electricity Act, 2003 provides for open access at section 42, historically with a 1 MW threshold. The Electricity (Promoting Renewable Energy Through Green Energy Open Access) Rules, 2022, notified on 6 June 2022 under section 176, reduced that threshold to 100 kW for green energy, established a central nodal agency operating a single-window national portal, and, critically, prescribed a fifteen-day timeline for approval of each application failing which the application is deemed approved subject to technical requirements.^{42,43} Indian implementation is uneven and the Karnataka High Court has held that regulating open access falls within the exclusive domain of the State Electricity Regulatory Commissions under sections 42(2) and 181.⁴⁴ But the central design point stands: India converted open access from a discretion into a time-bound, deemed approval procedure. Pakistan has not.

9.2 Statutory versus Regulatory Certainty

An entitlement created by regulation can be withdrawn by regulation. NEPRA may make regulations under section 47 and the Federal Government may make rules under section 46, and neither process requires the deliberation that primary legislation attracts. A wheeling charge that fell from Rs 22 to Rs 9 by administrative decision can rise by the same route without any change in the underlying statute.

This is not an argument that regulation is illegitimate. It is an argument about where a fifteen-year commitment must sit. The elements a lender requires to be immovable are the right to procure, the eligibility criteria, the methodology by which network charges are set, and the assignability of the contract. Those four belong in the Act. The detailed rates, procedures and technical standards belong in regulation, where they should be capable of adjustment.

9.3 Open Access Eligibility and Transmission Capacity

Eligibility is unclear because it is decided case by case rather than by published criterion, and capacity is unclear because there is no public register of available transmission capacity at each node. An investor evaluating a site cannot determine, before committing to land, whether the network can carry the load it intends to connect. In every mature market that information is published. Its absence in Pakistan is an administrative failure rather than a legal one, and it can therefore be remedied quickly and cheaply.

The underlying engineering constraint is real and was set out in Section 2.6: the Matiari to Lahore HVDC link operating well below its 4,000 MW design rating, a northward HVAC thermal ceiling of approximately 4,500 MW, and a reported 1,845 MW of southern generation constrained off daily.²⁰ NEPRA's own transmission performance evaluation records congestion and overloading of 500 kV and 220 kV assets.²⁷ No statutory amendment relieves a thermal limit. This is the clearest example in the report of a constraint that must be solved with conductor and steel rather than with drafting.

9.4 Curtailment, Grid Reliability and Replacement Power

Three connected gaps sit here and together they are the single largest obstacle to bankability after the right to procure itself.

Curtailment allocation is unpublished. When the network is constrained, the system operator reduces load or generation according to criteria that are not available to an investor in advance. A data centre cannot price a risk whose incidence it cannot observe.

Grid reliability at the point of delivery is not contractually guaranteed at a standard a hyperscale facility requires, and there is no mechanism by which a consumer can purchase a higher standard at a determined price.

Replacement power does not exist as a product. If a contracted generator fails to deliver, there is no supplier obligation to source alternative energy at a defined price. The consumer falls back to the grid at whatever the applicable tariff is, which converts a fixed-price contract into a variable-price exposure precisely at the moment the investor most

needs certainty. This is a regulatory gap and it can be closed by NEPRA regulation without amending the Act.

9.5 Supplier Default, Change in Law and Contract Enforcement

Pakistani commercial contract enforcement is slow and both foreign investors and their insurers price that fact. The relevant question for this report is narrower: whether a dispute about network access, wheeling charges or curtailment can be resolved on a commercial timescale. Under the current framework such a dispute goes to NEPRA under section 14C(2) or the general dispute provisions, then on appeal to the Appellate Tribunal under section 12G within three months, then to the High Court. That sequence is measured in years.

Change in law is a specific and acute concern in Pakistan given the recent record. The Off the Grid (Captive Power Plants) Levy Act, 2025 imposed a new fiscal burden on an existing category of investment with a legislated escalation schedule.^{12,13} IPP contracts covering 2,829 MW were terminated or renegotiated in FY 2024-25, with roughly Rs 260 billion of circular debt reduction attributed to renegotiation.^{2,23} Both measures may be defensible on their merits. Their combined effect on an investor's assessment of contractual durability is nonetheless significant and cannot be argued away. It must be answered with an express change-of-law provision.

9.6 Assignment and Financing

Section 27 prohibits the surrender, assignment or transfer of a licence without the prior approval of the Authority. Project finance depends on the lender's ability to take security over the project's revenue contracts and, in a default, to step into the borrower's position and continue operating. If that step-in requires a discretionary regulatory approval obtained after the default has occurred, the security is substantially less valuable and the debt is priced accordingly or not offered at all. This is a legal restriction with a legal remedy: a pre-approval mechanism for registered projects, granted at financial close rather than at enforcement.

9.7 Data Governance

A data centre is a data facility and the legal regime governing the data it holds is part of the investment decision. Pakistan does not yet have an enacted comprehensive personal data protection statute. The Personal Data Protection Bill, prepared by the Ministry of Information Technology and Telecommunication and approved by the Federal Cabinet in its updated form, has not completed passage through both Houses.^{45,46} In the meantime the applicable criminal provisions sit in the Prevention of Electronic Crimes Act, 2016, which is not a general processing framework.

Policy has moved faster than legislation. The Digital Nation Pakistan Act, 2025 established the Pakistan Digital Authority, and the National AI Policy was adopted in 2025.⁴⁶ The Ministry released the draft National Data Governance Policy 2026 for consultation in late June 2026, closing on 10 July 2026, declaring government data a strategic national asset, asserting that government data must remain under Pakistan's lawful authority and

effective control, and introducing rules on cross-border transfer, privacy impact assessments and AI explainability. The draft states that it will be updated once a comprehensive personal data protection law is enacted.^{47,48}

For a hyperscaler two questions follow. The first is whether it can lawfully serve foreign customers from Pakistani infrastructure without triggering localisation or transfer restrictions, which depends on the final form of the data protection statute rather than on the policy. The second is whether Pakistani and regional customers can lawfully place regulated data in a Pakistani facility, which is the demand-side question and which an adequacy-grade statute would answer positively. The absence of the statute is currently a deterrent in both directions, and enacting it is the single cheapest thing Pakistan could do to improve the proposition.

The OECD's Guidelines Governing the Protection of Privacy and Transborder Flows of Personal Data, first adopted in 1980 and revised in 2013, remain the most widely used international baseline for what such a statute should contain: lawful and proportionate collection, purpose limitation, security safeguards and accountability, applied consistently to data that crosses borders.⁴⁹ The European Union's adequacy framework supplies a more specific benchmark still. Article 45 of Regulation (EU) 2016/679 allows the European Commission to recognise not only a whole third country but also a territory or one or more specified sectors within it as ensuring an adequate level of protection, and the European Data Protection Board applies the same substantive test, refined after the Schrems II judgment, to whichever transfer mechanism a business actually uses.^{50,51,52} Pakistan does not need a bilateral adequacy arrangement to benefit from the comparison. The point is narrower: a data protection regime bounded by territory or sector is not a foreign or improvised idea, it is the same unit of analysis the European Union itself applies when assessing adequacy, and it is available to Pakistan whether or not the comprehensive national statute has yet been passed.

This report proposes going further still. Pakistan could designate strategic data centre and compute zones, the same zones proposed on land, environmental and permitting grounds in Section 10.5, each carrying a purpose-built statutory and regulatory regime covering data protection, privacy, cybersecurity, cross-border data transfer, enforcement, jurisdiction and investor protection. The zones would sit inside Pakistan geographically while operating, for data protection purposes, outside Pakistan's ordinary statutory governance: a delimited legal space created by statute, with its own data protection law, its own regulator and its own enforcement mechanism, rather than the general law of Pakistan simply extended to a smaller footprint. Because the regime would apply only within the designated zones rather than to every controller and processor in the country, it is a materially smaller legislative undertaking than the comprehensive statute, and correspondingly faster to enact.

The reason for confining the law this way is commercial rather than symbolic. A hyperscaler's exposure is not to Pakistan's data protection position in the abstract; it is to whatever regime governs the specific facility holding its customers' data. A statute that covers designated data centre zones only, and does not extend to the rest of Pakistan, gives that facility a defined, adequacy-grade legal position without requiring Parliament

to first settle data protection for the whole of the economy. What makes a hyperscaler comfortable investing is the certainty that its zone is legally covered for data protection purposes; whether the same law applies outside the zone is, for that decision, largely beside the point.

International precedent for this kind of delimited zone already exists. The Dubai International Financial Centre operates as a financial free zone within the United Arab Emirates under its own law, including its own Data Protection Law, DIFC Law No. 5 of 2020, independent of the UAE's federal civil and commercial law and modelled closely on the GDPR.⁵³ A business operating within the DIFC is subject to the DIFC's data protection regime rather than the wider UAE framework, a ring-fenced legal position that has been central to the DIFC's ability to attract international financial institutions that would not otherwise have committed to the surrounding jurisdiction's general law. Pakistan already has a domestic precedent for zone-specific treatment, though a lighter one: the Special Technology Zones Authority Act, 2021 gives designated technology zones their own fiscal and regulatory incentives under a single statute.⁵⁴ A lighter-touch version of the delimited-zone idea also operates in China's free trade zones, which run their own cross-border data transfer rules that do not apply nationwide, though without the DIFC's fuller separation from the surrounding legal system.⁵⁵ The proposal here follows the DIFC model most closely: a zone that is part of Pakistan geographically, but carries its own substantive data protection law rather than a variant of the general law of Pakistan.

A workable design would vest enforcement in a dedicated data protection authority for the zone, created by the same statute, with its own tribunal or dispute resolution mechanism for data protection matters arising within the zone, on the model the DIFC uses for its own courts, and would provide expressly that the zone regime is absorbed into the comprehensive statute once that statute is enacted, in the same way the draft National Data Governance Policy 2026 already provides for its own revision once the comprehensive law is passed. On that design, the zone regime is a bridge across the interval in which the national statute remains pending, confined to the facilities this report addresses, and retired automatically once the reform it anticipates is finally enacted.

THE SPECIAL ZONE PROPOSAL, IN ONE SENTENCE

Enacting a comprehensive Personal Data Protection law remains this report's first recommendation; the special statutory regime for designated compute zones is a faster, interim measure confined to those zones and to the period before that law is passed, giving those zones a delimited data protection law of their own rather than waiting for the comprehensive national statute.

9.8 Environmental Law, Water and Land

Environmental approval is a provincial competence following the Eighteenth Amendment, exercised through the provincial environmental protection agencies under provincial environmental protection legislation. A campus spanning power generation,

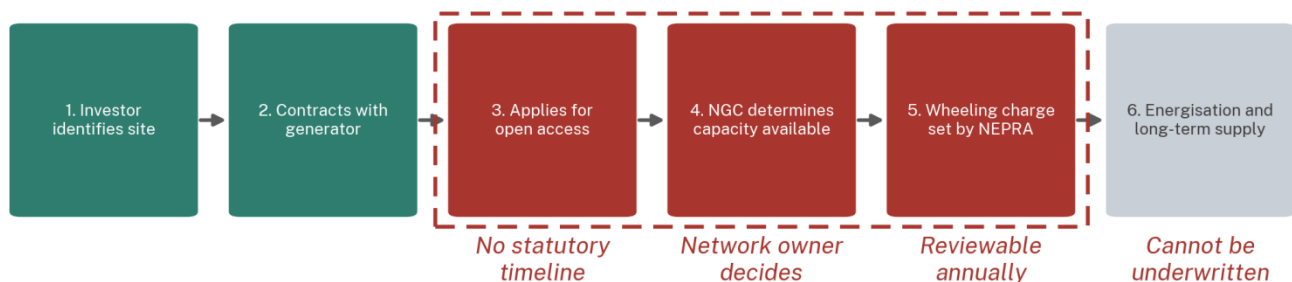
high-voltage connection and large-scale construction will require an initial environmental examination or environmental impact assessment with public hearing. The timeline is not fixed by statute in a way an investor can rely upon.

Water abstraction, discussed in Section 5.5, engages provincial irrigation departments and, for basin water, the Indus River System Authority under the 1991 Water Apportionment Accord. Land acquisition engages provincial land revenue administration, and title verification in Pakistan is a recognised diligence burden.

Jurisdictional status is the sharpest version of this problem. Azad Jammu and Kashmir and Gilgit-Baltistan are not provinces of Pakistan within the meaning of Article 1(2) of the Constitution and are governed under separate constitutional instruments. For a northern hydro-linked site, that affects which legislature may enact the applicable law, which courts have jurisdiction, how land is acquired, how a security interest is registered and perfected, and whether investor protections operate identically. These are answerable questions. They are not currently answered in any published form, and an investor will not do that work speculatively.

Figure 25. Where the Transaction Breaks

Current state process map for a bilateral power arrangement serving a large load; schematic



The transaction breaks here: three consecutive steps depend on decisions the investor cannot predict or price

Source: ARS analysis of section 14C(2) and section 22 of the NEPA Act, 1997, the NEPA (Wheeling of Electric Power) Regulations, 2016, and the Framework Guidelines for Wheeling Auctions 2025.

THE FINDING OF PART IV

Of the fifteen questions a project finance lender asks, Pakistan today answers none of them in a form the lender would accept without a sovereign guarantee. That is not because the law forbids the transaction. It is because the law permits it without committing to it, and a permission that can be withdrawn is not an asset a lender can take security over.

WHAT HAS TO CHANGE?

10. The Legal Reform Package

This is the operative section of the report. It identifies what must change, who must change it, and by which instrument. The distinction between instruments is not procedural pedantry. A reform placed in the wrong instrument either fails to bind or fails to be adjustable, and both failures are expensive.

STATUS OF THE DRAFTING IN THIS SECTION

The illustrative statutory language set out below is an ARS policy drafting proposal. It is not enacted legislation, it is not a bill before either House, and it is not a legal opinion. It is offered to demonstrate that the reform is capable of being expressed in ordinary statutory form, and to give Parliament, the Power Division and NEPRA a concrete starting point rather than a statement of principle.

10.1 What Parliament Must Change

Four amendments to the NEPRA Act, 1997 are required. Each addresses a defect identified in Section 9, and each must sit in primary legislation because each is a commitment a lender needs to be immovable.

The first converts section 22 from a permission into a right for a defined class. The current provision states that the Authority may permit sale to bulk power consumers. The legal problem is that a discretion cannot be underwritten. The proposed amendment inserts a new section 22A creating a right to procure electric power from any licensed generator, trader or supplier, exercisable by a consumer meeting published objective criteria, subject only to payment of determined use of system charges and to technical compliance. The expected effect is that eligibility becomes a question of fact rather than of favour.

ARS POLICY DRAFTING PROPOSAL: SECTION 22A

22A. Right of strategic large loads to competitive procurement. (1) Notwithstanding section 22, a consumer registered under subsection (3) shall have the right to purchase electric power from any generation, transmission, trading or supply licensee of its choice, and to have such power transported over the transmission and distribution systems of any licensee, on payment of the use of system charges determined under section 31 and subject to compliance with the applicable grid and distribution codes. (2) The right conferred by subsection (1) shall not be refused, suspended or made conditional except on grounds of technical infeasibility recorded in writing with reasons and supported by the capacity register maintained under section 23I. (3) The Authority shall by regulation prescribe objective criteria for registration as a strategic large load, which shall include a minimum contracted demand, a minimum contract term and a demonstrated capability to be metered and settled on an interval basis, and shall register any applicant meeting those criteria within sixty days. (4) Where the Authority does not decide an application under subsection (3) within sixty days, the applicant shall be deemed to be registered. Note: ARS policy drafting proposal. Not enacted legislation and not a legal opinion.

The second addresses the conditionality in section 14C(2). The current provision subjects the right of open access to a determination of transmission availability by the network owner, with no timeline and no default outcome. The proposed amendment imposes a statutory period within which the determination must be made, provides for deemed availability if it is not, and requires the grid company to maintain and publish a capacity register at each node. The model is the fifteen-day deemed approval mechanism in India's Green Energy Open Access Rules, 2022,^{42,43} adapted to a longer period appropriate to the scale of connection contemplated here.

The third addresses section 27. The current prohibition on assignment without prior approval defeats lender step-in. The proposed amendment permits the Authority, at the time it registers a strategic large load or issues a related licence, to grant a conditional pre-approval of assignment to identified financing parties, exercisable on a defined default event with notice to the Authority. The expected effect is that the security package becomes enforceable on the timetable a lender requires.

The fourth addresses the captive levy. The Off the Grid (Captive Power Plants) Levy Act, 2025 is sound policy in its own context, which is the recovery of industrial load to the grid. Applied to generation dedicated to a designated compute zone, it penalises exactly the investment the state says it wants. The proposed amendment inserts an exemption in section 3 of Act XIV of 2025 for generation dedicated to a load registered under the new section 22A, conditioned on that load also taking a minimum contracted quantity from the grid, so that the exemption cannot be used to exit the system entirely.

10.2 What NEPRA Must Change

Six regulatory instruments are required and none of them requires parliamentary time. This is the fastest-moving part of the reform and it should begin immediately.

Eligibility regulations under the proposed section 22A(3), prescribing the objective criteria and the registration procedure. A long-term tariff and use of system methodology, published and binding, under which the wheeling charge applicable to a registered load is fixed or indexed to a published formula for the term of the contract rather than revised at each determination. Amended wheeling regulations replacing the 2016 instrument, incorporating the standard wheeling agreement, interval metering and settlement requirements, and defined connection timelines. A standby and replacement power product, licensed and tarified, so that supplier default has a priced remedy. Standard forms of power purchase agreement, wheeling agreement and connection agreement approved by the Authority. And a determination on how the benefit of improved fleet utilisation is shared between the new load and the general consumer body, which is the distributional question flagged in Section 2.7. That determination should take the form of a fixed-cost contribution per unit, prescribed within the eligibility regulations and fixed or formula-indexed for the term of the registered contract, so that the amount the new load carries toward existing capacity charges is a published term of registration rather than a negotiated outcome. Section 15.2 sets out the relationship between that contribution and the tariff relief available to every other consumer, and it is a relationship NEPRA must settle under its section 31(2) obligations rather than leave to negotiation.

10.3 What the Market and System Operator Must Change

The Independent System and Market Operator, once the licences under sections 23A, 23B, 23G and 23H are consolidated,²⁵ must publish four things a hyperscaler will demand before it commits: a market access procedure with defined timelines for registration as a market participant; a transmission capacity register showing available capacity at each node and the queue of applications against it; a curtailment ranking specifying the order in which load and generation are constrained, together with the compensation payable; and scheduling and settlement rules capable of handling a flat 24/7 load contracted against a seasonal generator, which is a settlement problem Pakistani market design has not yet had to solve.

10.4 What the Grid Must Provide

The network requirements are engineering requirements and they cannot be legislated away. Restoration of the Matiari to Lahore HVDC link to a throughput approaching its 4,000 MW design rating is the single highest-value intervention available, because it directly relieves the constraint that strands southern generation.²⁰ The Asian Development Bank's USD 330 million Second Power Transmission Strengthening Project, comprising a 500 kV line of approximately 290 km and grid upgrades serving Islamabad and Faisalabad to enable evacuation of up to 3,200 MW from northern generation clusters, is the corresponding northern measure.²⁸ Beyond these, a hyperscale campus requires a dedicated substation with N+1 transformation, two independent feeds from separate grid stations, and a connection timeline the grid company is contractually bound to meet.

10.5 What Government Must Provide

Five measures follow. None lies with NEPRA; each requires a decision elsewhere in the federal apparatus. Designation of strategic compute zones with pre-cleared land, pre-completed environmental assessment and a single-window permitting authority, whether under the Special Technology Zones Authority framework or a purpose-built instrument. Enactment of the Personal Data Protection legislation, which Section 9.7 identifies as the cheapest available improvement to the proposition. A published position on cross-border data transfer sufficient for a foreign operator to serve foreign customers from Pakistani infrastructure. A State Bank framework confirming the repatriation treatment of service revenue and of investment returns. Lastly, an inter-governmental protocol resolving the interaction between federal electricity law under section 45 and provincial land, water and environmental competence, including an express answer on the position in Azad Jammu and Kashmir and Gilgit-Baltistan.

Table 11. Who Must Change What

The master reform matrix: institution, instrument, action and sequencing

Institution	Instrument	Action required	Timing	Dependency
Parliament	Amendment to Act XL of 1997	Insert s. 22A conferring a right to procure on registered strategic large loads	0-9 months	None. This is the critical path
Parliament	Amendment to Act XL of 1997	Amend s. 14C(2): statutory determination period with deemed availability	0-9 months	None
Parliament	Amendment to Act XL of 1997	Amend s. 27: conditional pre-approval of assignment to financing parties	0-9 months	None
Parliament	Amendment to Act XIV of 2025	Exempt generation dedicated to a registered strategic large load	0-9 months	Coordinate with IMF programme commitments
Parliament	Personal Data Protection legislation	Complete passage of the pending Bill through both Houses	0-18 months	Independent; can proceed in parallel
NEPRA	Regulations under s. 47	Eligibility criteria and registration procedure for strategic large loads	6-12 months	Requires s. 22A
NEPRA	Regulations under s. 47	Published, binding long-term tariff and use of system methodology	6-15 months	None. Can begin immediately
NEPRA	Amended wheeling regulations	Replace SRO 549(I)/2016; standard agreement, metering, settlement, timelines	6-15 months	None. Can begin immediately

Institution	Instrument	Action required	Timing	Dependency
NEPRA	Regulations and determination	Create a licensed standby and replacement power product with a set tariff	9-18 months	Requires supplier engagement
NEPRA	Standard contract approval	Approve standard PPA, wheeling and connection agreements	9-20 months	Requires the wheeling regulations
ISMO	Market rules	Market access, capacity register, curtailment ranking, settlement rules	9-18 months	Requires licence consolidation
National Grid Co.	Investment and code	HVDC restoration, northern evacuation, node capacity publication	6-36 months	Financing secured for ADB tranche
Federal Government	Notification	Designate strategic compute zones with single-window permitting	3-12 months	Requires provincial concurrence
State Bank	Circular	Confirm repatriation treatment for compute service revenue	3-9 months	None
Federal and provincial	Inter-governmental protocol	Resolve land, water, environmental and AJK/GB jurisdictional questions	6-24 months	Council of Common Interests engagement

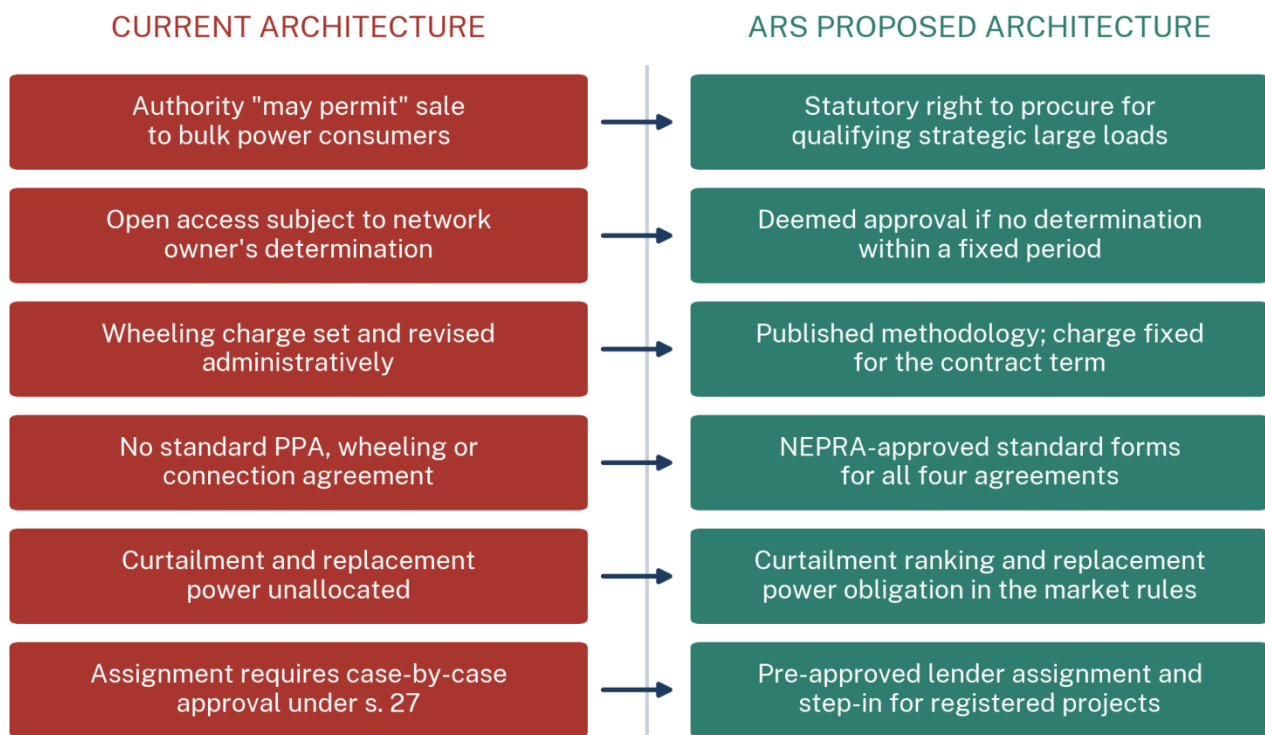
Source: ARS policy proposal. Not enacted legislation and not a legal opinion. Timing is indicative and measured in months from commencement of the programme.

11. The ARS Proposed Legal Model

Taken together, the measures in Section 10 amount to a single structural change: the replacement of a permission architecture with a rights architecture. This section states the model as thirteen propositions and shows the shift in a single diagram.

Figure 26. From a Permission Architecture to a Rights Architecture

The six changes that convert a discretionary transaction into a bankable one



Source: ARS policy drafting proposal. Not enacted legislation and not a legal opinion. Each right-hand item requires a specific instrument identified in Sections 10 and 11 of this report.

11.1 The Thirteen Elements

A statutory right to competitive power procurement, conferred by the proposed section 22A on any consumer meeting published criteria, replacing a discretion exercised case by case. Objective eligibility for strategic large loads, defined by minimum contracted demand, minimum contract term and interval metering capability, so that eligibility is a matter of fact. Non-discriminatory open access with a statutory determination period and deemed availability where the network owner does not respond, supported by a published capacity register.

Transparent wheeling, with the use of system charge determined under a published methodology and fixed or formula-indexed for the term of the registered contract, so that the single largest variable in the delivered price ceases to be a policy variable. A long-term tariff methodology under section 31, giving a fifteen to twenty-five year contract a determinable price path. A standard power purchase agreement, a standard wheeling

agreement and a standard connection agreement, each approved by the Authority, so that negotiation is confined to price and volume rather than to the allocation of every risk from first principles.

A curtailment and replacement power regime, in which the ranking is published, compensation is defined, and a licensed supplier is obliged to provide replacement energy at a determined price when a contracted generator underdelivers. Change-of-law protection, expressly provided in the standard contracts, allocating the economic effect of a discriminatory change in law to the party best able to control it. Assignment and financing provisions permitting pre-approved lender step-in under the amended section 27.

Dispute resolution on a commercial timescale, with a fast-track determination procedure before the Authority for network access and wheeling disputes, subject to a fixed period, before the ordinary appellate route under section 12G applies. And investor protection, comprising the repatriation confirmation, the data governance framework and the single-window permitting described in Section 10.5.

11.2 Why This Combination and Not a Smaller One

It is reasonable to ask whether a lighter package would suffice, and the answer is no, for a structural reason. A project finance lender does not apply a weighted average across risks. It identifies the weakest link and prices the transaction against that. A statutory right to procure that is delivered over a network whose curtailment ranking is unpublished is worth very little; so is a published curtailment ranking attached to a right that can be withdrawn. The elements are complements rather than substitutes, which is why Section 12 scores them as a set.

12. The Bankability Test

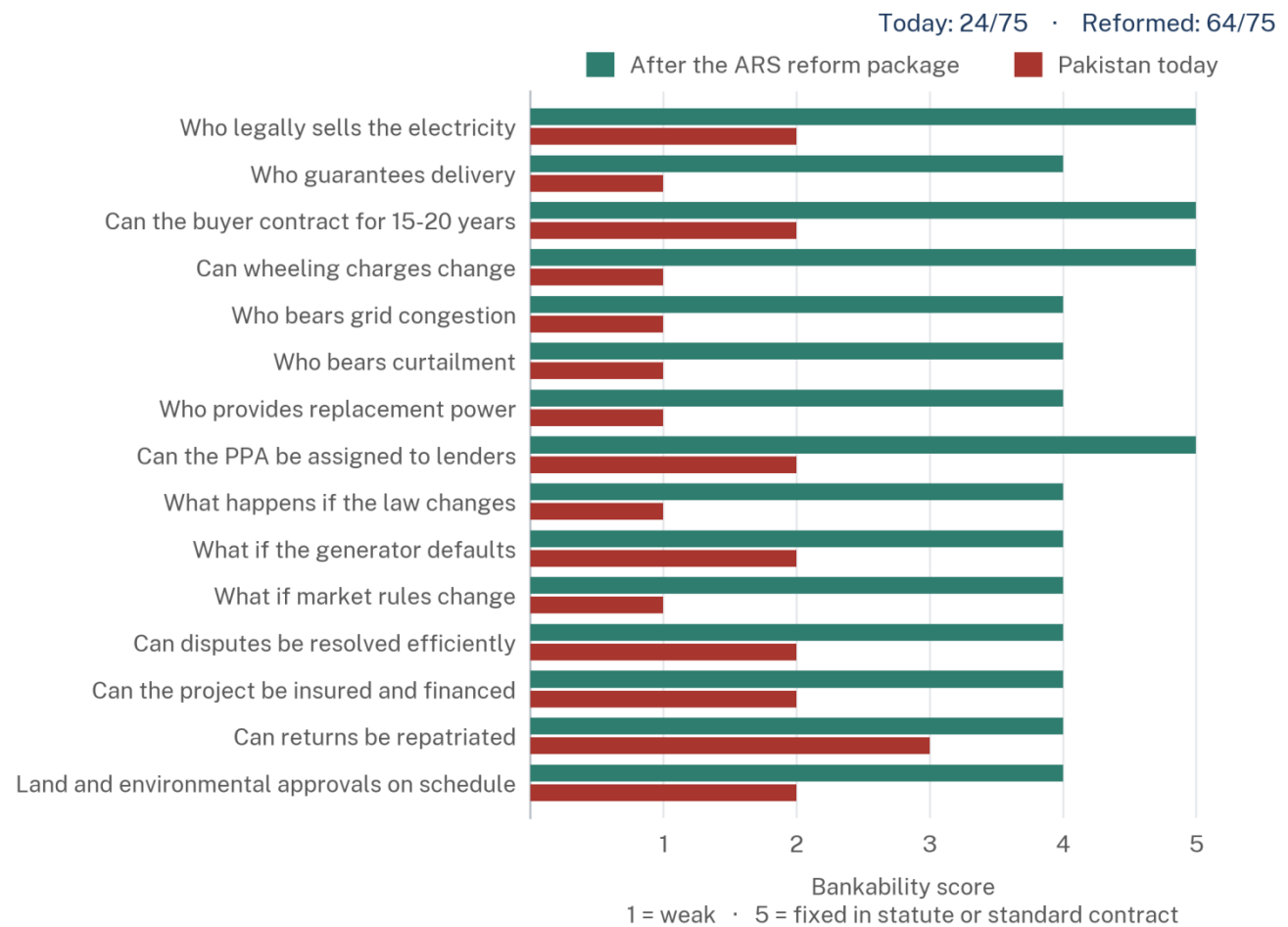
This section applies the questions a hyperscaler's investment committee and its lenders' counsel would actually ask. The methodology is stated in full so that a reader who disagrees with a score can identify precisely which judgement is in dispute.

SCORECARD METHODOLOGY

Each of fifteen questions is scored on a five-point scale. A score of 1 means there is no answer a project finance lender would accept. A score of 2 means an answer exists in principle but rests entirely on administrative discretion with no criteria or timeline. A score of 3 means an answer exists in subordinate regulation and is subject to unilateral change. A score of 4 means the answer is fixed in regulation or in an approved standard contract and cannot be changed without process. A score of 5 means the answer is fixed in primary legislation or in a NEPRA-approved standard contract and cannot be changed unilaterally. Scores are analytical judgements applied consistently, not measurements. The reformed-state column assumes the full package in Section 10 is implemented.

Figure 27. Pakistan Scores 24 Out of 75 on the Questions a Hyperscaler's Lenders Actually Ask

ARS Hyperscaler Bankability Scorecard, current state against the reformed state proposed in Section 11



Source: ARS assessment. Each of the 15 questions is scored 1 to 5, where 1 means there is no answer a project finance lender would accept, 3 means an answer exists but rests on administrative discretion, and 5 means the answer is fixed in primary legislation or in a NEPRA-approved standard contract and cannot be changed unilaterally. Scores are analytical judgements applied consistently, not measurements.

Pakistan scores 24 out of a possible 75 today. Under the reform package proposed in Section 10, the same fifteen questions score 64. The gap between those two numbers is the entire content of this report, and it is worth noting what the gap is not made of. It is not made of generation capacity, which Pakistan has in surplus. It is not made of hydrological resource, which Pakistan has in abundance. It is not made of demand, which exists globally in quantities that dwarf anything Pakistan could supply. It is made of legal instruments that cost nothing to draft and require political decision rather than capital.

Table 12. The Hyperscaler Bankability Scorecard

Fifteen questions, the current Pakistani answer, and the answer after the ARS reform package

Question	The Pakistani answer today	Today	Reformed	Instrument that closes the gap
Who legally sells the electricity?	A generator may, if the Authority permits under s. 22	2	5	New s. 22A
Who guarantees delivery?	Nobody. Dispatch is a system function under s. 14B(4)	1	4	ISMO market rules; standard PPA
Can the buyer contract for 15 to 20 years?	Nothing prevents it; nothing supports the price path	2	5	Long-term tariff methodology
Can wheeling charges change?	Yes, by administrative determination, without limit	1	5	Published binding methodology
Who bears grid congestion?	Undefined. The corridor is constrained and unallocated	1	4	Capacity register; curtailment ranking
Who bears curtailment?	Undefined. Ranking unpublished, no compensation regime	1	4	ISMO market rules
Who provides replacement power?	No such product exists in the Pakistani market	1	4	New licensed standby product
Can the PPA be assigned to lenders?	Only with the Authority's approval under s. 27, case by case	2	5	Amended s. 27 pre-approval
What happens if the law changes?	No express protection. Recent record is adverse	1	4	Change-of-law clause in standard contracts
What if the generator defaults?	Fall back to grid tariff; no priced remedy	2	4	Replacement power obligation
What if market rules change?	No grandfathering of registered transactions	1	4	Transitional protection in market rules
Can disputes be resolved efficiently?	NEPRA, then Appellate Tribunal, then High Court. Years	2	4	Fast-track network access procedure
Can the project be insured and financed?	Possible with sovereign or political risk cover at a cost	2	4	The package as a whole
Can returns be repatriated?	Yes under the existing regime, but not confirmed for this case	3	4	State Bank circular

Question	The Pakistani answer today	Today	Reformed	Instrument that closes the gap
Land and environmental approvals on schedule?	No statutory timeline; provincial competence; AJK/GB unclear	2	4	Zone designation and single window
TOTAL		24 / 75	64 / 75	

Source: ARS assessment applying the methodology stated above. Not a measurement and not investment advice. The reformed column assumes full implementation of the package in Section 10.

Figure 28. Reform Would Move Pakistan Past the Regional Benchmark on Legal Right and Lender Security

ARS composite bankability assessment, scored 1 to 5



Source: ARS assessment. The India benchmark reflects the Electricity Act, 2003 section 42 open access regime as amended by the Green Energy Open Access Rules, 2022, including the 15-day deemed approval mechanism, and does not imply that Indian implementation is uniform across states.

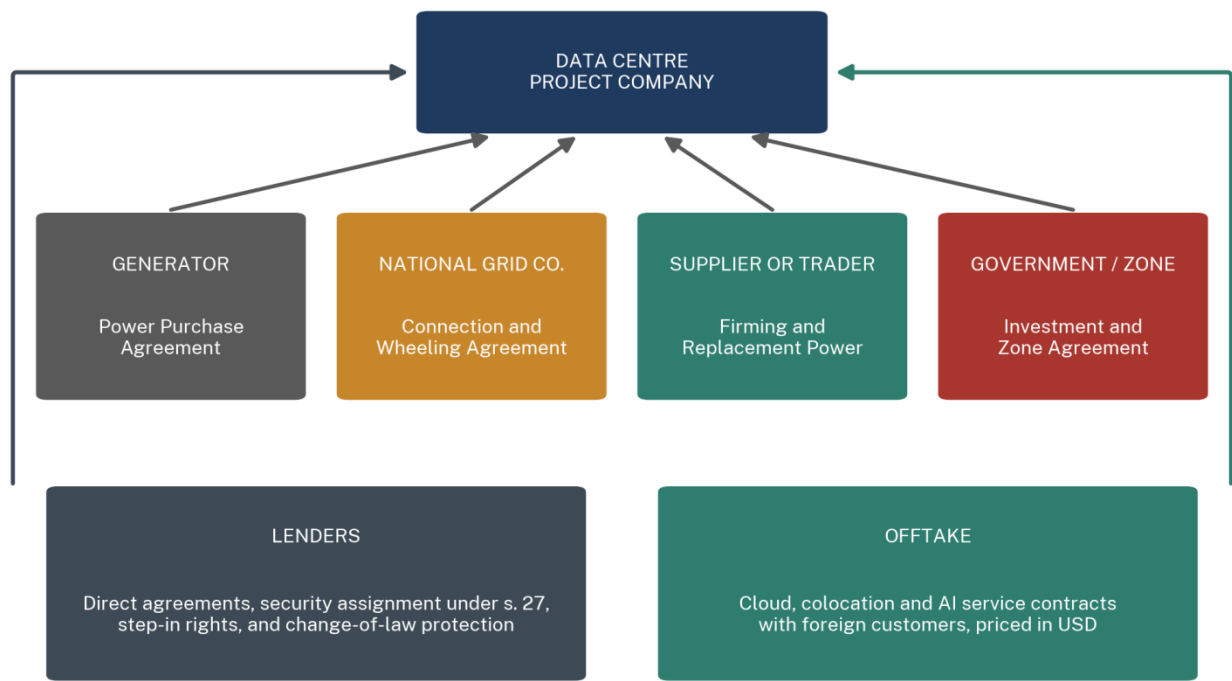
The regional comparison in the preceding figure should be read carefully. It does not assert that India is easy. Indian open access varies materially between states, several states have declined to adopt the central 100 kW threshold, distribution companies have resisted the fifteen-day timeline, and additional state-level charges have been imposed.⁴⁴ What the comparison establishes is that a neighbouring jurisdiction with a comparable institutional inheritance made the structural move from discretion to time-bound

entitlement, and that the move is therefore neither unprecedented nor beyond the administrative capacity of a South Asian electricity regulator.

13. Contracts That Make the Model Bankable

Statute confers the right. Regulation sets the terms. Contracts allocate the risk, and it is the contracts that a lender actually reads. Six agreements must interlock.

Figure 29. Six Agreements Must Interlock Before a Single Rack Is Installed
Contract architecture for a hyperscale data centre in Pakistan



Source: ARS analysis. None of the four upper agreements currently exists in Pakistan as a NEPRA-approved standard form. Section 13 of this report sets out the clauses each would need to contain.

Table 13. Required Contractual Architecture and the Clauses Each Agreement Must Contain

Six agreements, their parties, and the provisions without which the transaction is not financeable

Agreement	Parties	Essential clauses	Status in Pakistan today
Power Purchase Agreement	Generator and project company	Term of 15-25 years; price and indexation formula; availability guarantee with liquidated damages; scheduled and unscheduled outage regime; change-of-law; assignment to lenders; termination and buyout	No approved standard form for bilateral consumer sale
Wheeling Agreement	Project company and network licensees	Use of system charge fixed or formula-indexed for the term; capacity reservation; curtailment ranking and compensation; metering and settlement on interval basis; force majeure; dispute fast-track	No approved standard form; charges administratively revisable
Connection Agreement	Project company and National Grid Company	Connection date with liquidated damages for delay; two independent feeds from separate grid stations; N+1 transformation; voltage and frequency standards; maintenance coordination; security of supply standard	No approved standard form; no binding connection timeline
Supply or Trading Agreement	Project company and licensed supplier or trader	Standby capacity commitment; replacement power obligation at a determined price; balancing and imbalance settlement; credit support; step-down on generator default	Product does not exist; requires the new NEPRA instrument
Government or Zone Agreement	Project company and Federal Government or zone authority	Land tenure and access; single-window permitting with timelines; environmental clearance; fiscal treatment; repatriation confirmation; stabilisation of the applicable regime; jurisdiction and governing law	No standard instrument for compute-specific investment
Backup and Firming Agreement	Project company and firming provider	Committed firm capacity in MW; response time; seasonal profile matched to the hydro contract; availability payment structure; testing regime	Does not exist as a market product

Source: ARS analysis. Not a legal opinion and not a substitute for transaction counsel. None of the four network-facing agreements currently exists in Pakistan as a NEPRA-approved standard form.

13.1 The Risk Allocation That Makes It Work

Underlying the six agreements is a single allocation principle: each risk should sit with the party best able to control it, and where no party can control it, it should be priced explicitly rather than left to fall where it lands. Hydrological risk sits with the generator, which selects and operates the plant. Network availability and congestion risk sit with the network company, which builds and operates the network. Demand risk sits with the data centre, which sells the compute. Change-of-law risk sits with the state, which makes the law. Fuel and commodity price risk sits with the generator, subject to indexation. Currency risk on the electricity leg sits with the consumer, whose revenues are in dollars and whose costs are in rupees, which is one of the few places where Pakistan's position is structurally favourable.

That last point is worth drawing out because it is a genuine and underappreciated advantage. A hyperscale data centre in Pakistan earns dollars and pays a substantial part of its costs in rupees. Rupee depreciation, which damages almost every other participant in the Pakistani power sector, improves this project's economics. In a country whose electricity problem is at root a foreign currency problem, an investor whose revenue is denominated in foreign currency is not simply another consumer. It is a source of the currency the system needs.

MAKING DATA CENTRES A REALITY

14. Where Should Pakistan Build?

Site selection is where the energy analysis, the network analysis and the legal analysis converge, and where the trade-offs become concrete. This section scores six candidate locations against ten criteria.

Figure 30. No Pakistani Site Scores Well on Both Power and Connectivity, Which Is the Central Siting Trade-Off

ARS site selection scorecard, each criterion scored 1 (weak) to 5 (strong), with the total out of 50

	Power availability	Power cost	Climate / cooling	Water	Fibre and latency	Transmission	Land	Seismic risk	Security	Permitting	
Karachi coastal	5	3	2	2	5	4	3	4	3	4	35/50
Islamabad / Rawalpindi	3	3	4	3	4	3	3	2	4	4	33/50
Lahore / central Punjab	4	3	2	3	4	3	4	3	4	4	34/50
Northern hydro corridor	4	5	5	4	2	2	4	1	2	2	31/50
Gwadar	2	3	3	2	3	2	5	3	2	3	28/50
Thar / southern Sindh	5	4	1	1	2	3	5	4	2	3	30/50

Source: ARS assessment. Scores are analytical judgements applied consistently across sites on a common scale, not measurements. Power cost reflects proximity to low-marginal-cost generation; transmission reflects the capability of the evacuation corridor serving the site; permitting reflects the clarity of the applicable land, environmental and jurisdictional regime.

14.1 The Central Trade-Off

The scorecard states the problem in one line: no Pakistani location scores well on both power and connectivity. Every submarine cable lands at Karachi, every large hydro asset sits in the north. The cheap thermal generation sits in the south but cannot be evacuated northward at scale because of the corridor constraint described in Section 2.6. A site optimised for power cost is badly placed for latency and backhaul diversity; a site

optimised for connectivity pays the full grid tariff and sits in the country's hottest, most humid cooling environment.

14.2 The Northern Hydro-Linked Corridor

Highest score on power cost and climate, lowest on fibre, transmission, seismic risk, security and permitting. Free cooling for a substantially larger part of the year materially reduces PUE and therefore the electricity bill for a given IT load. Proximity to generation avoids the north-south corridor. Land is inexpensive.

Against that: no diverse fibre without a major backhaul investment from Karachi; the collision zone between the Indian and Eurasian plates; the constitutional position of Azad Jammu and Kashmir and Gilgit-Baltistan described in Section 9.8; and the hydro firmness problem set out in Section 4. The northern corridor is where Pakistan should build in the 2030s, and it is not where Pakistan should build first.

14.3 Karachi

Highest score on fibre and power availability, weakest on cooling and water. Every international system lands here, K-Electric operates a separate and comparatively better-performing distribution network, industrial land and skilled labour are available, and the Transworld Tier III facility already demonstrates that the basic proposition works at small scale.³⁷ Coastal humidity raises cooling load and therefore PUE, and water for evaporative cooling is scarce in a city with a chronic municipal supply problem, which pushes the design toward closed-loop or air-cooled systems and raises electricity consumption further.

14.4 Islamabad and Rawalpindi

The balanced option and, on the ARS assessment, the right location for a first demonstration project. Adequate fibre, adequate power, a moderate climate by Pakistani standards, proximity to the federal institutions whose cooperation the project requires, an existing technology sector presence, and a jurisdictional position that raises none of the questions the northern corridor raises. It scores highest on nothing and lowest on nothing, which for a first project of this kind is the correct profile.

14.5 Lahore, Gwadar and Southern Sindh

Lahore offers good connectivity and a large customer base but sits at the receiving end of the constrained corridor and has severe summer conditions. Gwadar has land and strategic attention but the weakest power and transmission position of the six, and its security profile is a material constraint on foreign investment of this class. Southern Sindh scores highest on raw power availability, sitting adjacent to the Thar coal complex and the southern wind corridor, and lowest on climate, water and fibre. It is a candidate for a power-anchored campus if, and only if, the fibre backhaul is solved.

14.6 The ARS Site Selection Recommendation

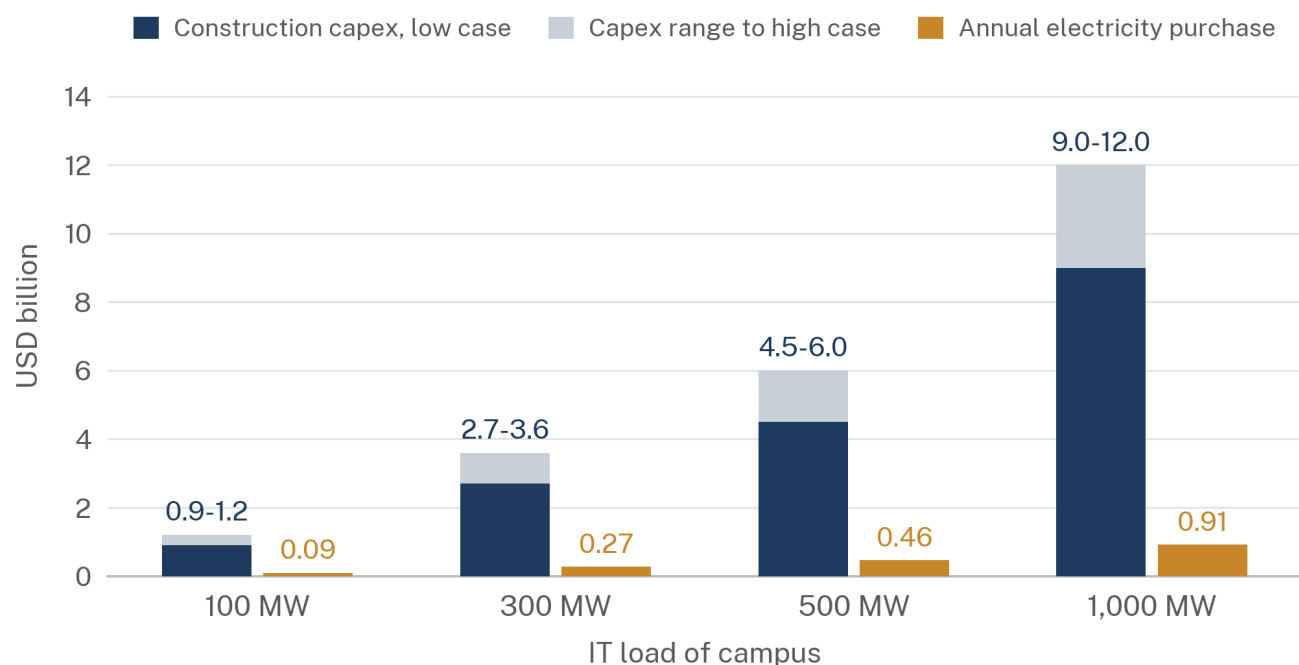
Build the first demonstration project at 50 MW to 100 MW in the Islamabad and Rawalpindi corridor, where every constraint is moderate and none is disqualifying, and use it to test the legal instruments proposed in Section 11 in a real transaction with a real lender. Develop Karachi in parallel as the connectivity-anchored location for latency-sensitive workloads, taking advantage of the three cable systems added since early 2025. Treat the northern hydro corridor as the target for the 2030s, conditional on the ADB-financed northern evacuation reinforcement,²⁸ on resolution of the jurisdictional questions, and on a serious answer to fibre diversity. Do not begin in the north because the electricity is cheap there. The electricity is cheap there because nothing else is.

15. What Could the Opportunity Be Worth?

This section sets out scenario arithmetic. It is not a forecast and it should not be quoted as one. Every figure below is derived from stated assumptions that are themselves stated, and a reader who disagrees with an assumption can recompute the result.

Figure 31. A 300 MW Campus Implies Roughly USD 3 Billion of Construction and USD 273 Million a Year of Electricity Purchases

Illustrative capital and electricity scenarios by campus size, USD billion



Source: ARS illustrative scenario, not a forecast. Capex assumes USD 9 million to USD 12 million per MW of IT capacity, a planning band ARS applies for greenfield hyperscale build; actual costs vary widely with design, cooling technology and site. Annual electricity purchase assumes PUE 1.30, 8,760 hours and a delivered price of 8 US cents per kWh.

Table 14. Scenario Arithmetic by Campus Size

Observed data, ARS-derived calculations and illustrative scenarios, distinguished by column

IT load	Annual energy	Share of FY25 generation	Construction capex (illustrative)	Annual electricity purchase	Direct jobs (order of magnitude)
100 MW	1.14 TWh	0.84%	USD 0.9-1.2 bn	USD 91 m (Rs 25 bn)	Low hundreds
300 MW	3.42 TWh	2.53%	USD 2.7-3.6 bn	USD 273 m (Rs 76 bn)	Several hundred
500 MW	5.69 TWh	4.22%	USD 4.5-6.0 bn	USD 455 m (Rs 127 bn)	Under one thousand
1,000 MW	11.39 TWh	8.43%	USD 9.0-12.0 bn	USD 911 m (Rs 255 bn)	Around one thousand
2,000 MW	22.78 TWh	16.86%	USD 18.0-24.0 bn	USD 1.82 bn (Rs 510 bn)	Low thousands

Source: ARS calculation and illustrative scenario. Facility load at PUE 1.30; annual energy at 8,760 hours. Capex band of USD 9-12 million per MW of IT capacity is an ARS planning assumption for greenfield hyperscale build, not an observed Pakistani cost. Electricity purchase at 8 US cents per kWh delivered. Rupee conversion at Rs 280 to the US dollar, the same delivered price and conversion basis as Table 7. None of these figures is a forecast.

15.1 Power Cost Sensitivity

The sensitivity shown at Figure 22 is the number that decides the investment, and it is worth stating precisely. On a 300 MW campus drawing 3.42 TWh a year, each additional US cent per kWh in the delivered price costs approximately USD 34 million a year, or roughly USD 510 million over a fifteen-year term. The difference between supply at the notified industrial tariff and supply under a well-structured bilateral contract is plausibly 4 to 6 cents. On a 300 MW campus that difference is worth USD 136 million to USD 205 million a year, which exceeds the annual electricity bill of most Pakistani industrial sectors and is larger than the entire capital cost of a 100 MW facility over the term.

What that difference decides, however, depends on which layer of the offer set out at Table 8 the Pakistani project occupies, and the layers are not interchangeable. For a powered shell leased by the megawatt to a foreign operator, electricity is the principal input cost and the delivered price is close to the whole commercial case. For cryptocurrency mining, electricity is the dominant operating cost, commonly placed at between 70% and 80% of it, and a movement of 6 cents is the difference between a viable operation and a closed one. For the sale of compute itself, which is the layer that captures the most value and earns the most foreign exchange, the proportion is materially different.

That proportion is worth calculating rather than assuming. An accelerator of the class used for AI training draws in the region of 1.4 kW once cooling and power conversion are included, which is approximately 12.3 MWh a year at continuous operation, or roughly USD 980 of electricity a year at a delivered price of 8 cents per kWh. Published on-demand

rates for that class of accelerator span a wide band, and one large provider lists NVIDIA H100 capacity at USD 3.99 per GPU-hour.⁵⁶ Taking USD 2 per hour, at the conservative end of the observed band, gross revenue is on the order of USD 17,500 per accelerator per year at continuous utilisation. Electricity is therefore in the region of 6% of gross revenue at a delivered price of 8 cents, and approximately 8% at 12 cents. Utilisation is an assumption, and the capital cost of the accelerator itself dominates the cost structure, but neither point changes the direction of the result: for the sale of compute, the delivered electricity price is a material cost and not the determining one.

Two conclusions follow, and both bear on how Pakistan makes its offer. The first is that the Federal Government's allocation of 2,000 MW names two uses, cryptocurrency mining and AI data centres, whose sensitivity to the electricity price differs by an order of magnitude.²⁴ A tariff designed to attract mining is a tariff designed for the use that captures the least value, earns the least foreign exchange, employs the fewest people and can relocate within a quarter. The second is that Pakistan cannot win a siting decision for compute on price alone, because the price advantage it is able to offer is worth a few per cent of a hyperscaler's revenue, while the constraints identified in Parts IV and V, contractual durability, deliverability and the ability to finance the asset, are worth all of it. The delivered price must clear a hurdle. It does not need to win a race, and Pakistan should not enter one.

This is why the legal reform matters more than any subsidy, and the distinction is not merely rhetorical. A concessional tariff is a fiscal transfer, it is contestable under the Authority's obligations at section 31(2) to eliminate economic distortions, and it has already been resisted. In July 2025 the International Monetary Fund declined a proposal to supply energy-intensive users, including data centres and cryptocurrency mining, at a marginal-cost-based rate in the region of Rs 22 to Rs 23 per unit, on the grounds that sector-specific pricing of that kind resembles a tax holiday and distorts the market.⁵⁷ That rate is approximately the 8 cents this report uses in its scenario arithmetic, which is precisely the point. The price Pakistan needs to offer is achievable as a commercial outcome and is not available as a concession. A statutory right to buy electricity bilaterally at a cost-reflective price transfers nothing from the exchequer and raises none of those objections. The state does not need to pay a hyperscaler to come to Pakistan. It needs to allow one to buy electricity at a price that reflects the cost of producing it, which is a legal question rather than a fiscal one, and which is available to Pakistan on terms a Fund programme can accommodate.

15.2 What This Is Worth to the Electricity Sector

A 1,000 MW IT load would purchase 11.39 TWh a year at the campus meter. Against FY 2024-25 grid sales of approximately 111 TWh, that is roughly a 10% increase in volume, from a single creditworthy customer paying in a currency the country needs, with a load shape that requires no new peaking capacity. A campus of that size would connect at transmission voltage, so the generation required to serve it exceeds metered consumption by transmission losses alone, in the region of 3%, rather than by the system average loss carried by distribution-connected sales.

The effect on the fixed-cost recovery problem is the point, and it must be stated as a function rather than as a single number. Capacity charges are recovered across the power purchase volume rather than across grid sales: CPPA-G's projected charge of Rs 2.03 trillion to Rs 2.16 trillion, against a capacity component of Rs 16.04 to Rs 16.45 per unit, implies a recovery volume in the region of 127 TWh.³ What the general consumer body receives therefore depends entirely on how much of that charge the new load itself carries in the price it pays. Every Rs 1 per unit of fixed-cost contribution embedded in the delivered price of an 11.39 TWh load yields approximately Rs 0.09 per unit of relief for every other consumer in the country. That single relationship governs the domestic case for this policy, and it is a policy variable rather than a property of the transaction.

Table 15. What the New Load Contributes and What Every Other Consumer Receives

Relief to the general consumer body as a function of the fixed-cost contribution embedded in the delivered price, 1,000 MW IT load

Fixed-cost contribution (Rs/kWh)	Delivered price (Rs/kWh)	Delivered price (US cents)	Annual contribution (Rs bn)	Relief to every other consumer (Rs/kWh)
0	19.0	6.8	0	0
2	21.0	7.5	23	0.18
4	23.0	8.2	46	0.36
8	27.0	9.6	91	0.72
16.04	35.0	12.5	183	1.44

Source: ARS calculation. Illustrative delivered prices assume a variable energy cost of Rs 10 per unit and a use of system charge of Rs 9 per unit, consistent with Sections 2.7 and 8.5, converted at Rs 280 to the US dollar. The contribution is the portion of the delivered price available to recover fixed capacity costs. Relief is calculated against an 11.39 TWh load and a capacity recovery volume of approximately 127 TWh. Illustrative, and not a tariff determination.

The table states a trade-off that Pakistani discussion of this subject has so far avoided. At the bottom of the range the new load pays short-run marginal cost plus the cost of using the network, which is an attractive price for an investor and returns nothing measurable to anyone else. At the top of the range it carries its full pro-rata share of the capacity charge, which returns Rs 1.44 per unit to every other consumer and implies a delivered price of approximately 12.5 cents, which is the notified industrial tariff and the price at which, on the evidence of Section 7.1, hyperscale investment is not made anywhere. Both ends of the range are stable and neither is useful. The workable answer lies in the middle, where a contribution of Rs 4 to Rs 8 per unit implies a delivered price in the region of 8 to 10 cents, competitive for a training-weighted workload, and returns Rs 0.36 to Rs 0.72 per unit to the general consumer body, obtained without subsidy and without new generation.

Because the benefit is proportionate to the contribution, the contribution cannot be left to the negotiation. It should be prescribed as a term of registration under the proposed section 22A, expressed as a fixed-cost contribution per unit and fixed or formula-indexed

for the term of the registered contract in the same instrument that fixes the use of system charge. That serves both sides of the transaction. The investor obtains the determinable price path over fifteen to twenty-five years that Part IV establishes it cannot obtain today. The Authority discharges its obligation under section 31(2) to protect existing consumers against a transfer of costs, which is the objection distribution companies and consumer representatives will otherwise raise against open access, and which was raised in the comparable Indian debate, where the instrument is the cross-subsidy surcharge under section 42(2) of the Electricity Act, 2003.⁴⁴ Pakistan should adopt the function of that instrument without importing its principal defect, which is revision between determinations, precisely the discretion this report argues must be removed.

Two conditions qualify the arithmetic, and both are stated elsewhere in this report. The first is that the relief is available only where serving the new load creates no new capacity obligation, the condition set out at Section 2.7 and treated as a risk control at Section 17.3. If new plant is contracted on capacity payment terms in order to serve a data centre, the numerator and the denominator rise together and the benefit disappears. The second is that the figures above are a fixed-cost calculation and not a tariff determination, which is the Authority's function under section 31 and not this report's.

Stated on that basis the domestic case stands, and it stands more credibly for being bounded. A 1,000 MW load contracted at a delivered price in the region of 8 to 10 cents would contribute between Rs 46 billion and Rs 91 billion a year toward a capacity charge that is at present recovered entirely from domestic consumers, would reduce the capacity component of every other consumer's tariff by Rs 0.36 to Rs 0.72 per unit, would do so without a rupee of subsidy and without a megawatt of new plant, and would pay in dollars. That, and not the export receipts, is the strongest domestic political argument for this policy. It is a smaller number than a promoter would quote, and unlike that number it survives examination.

WHAT ARS DOES NOT CLAIM

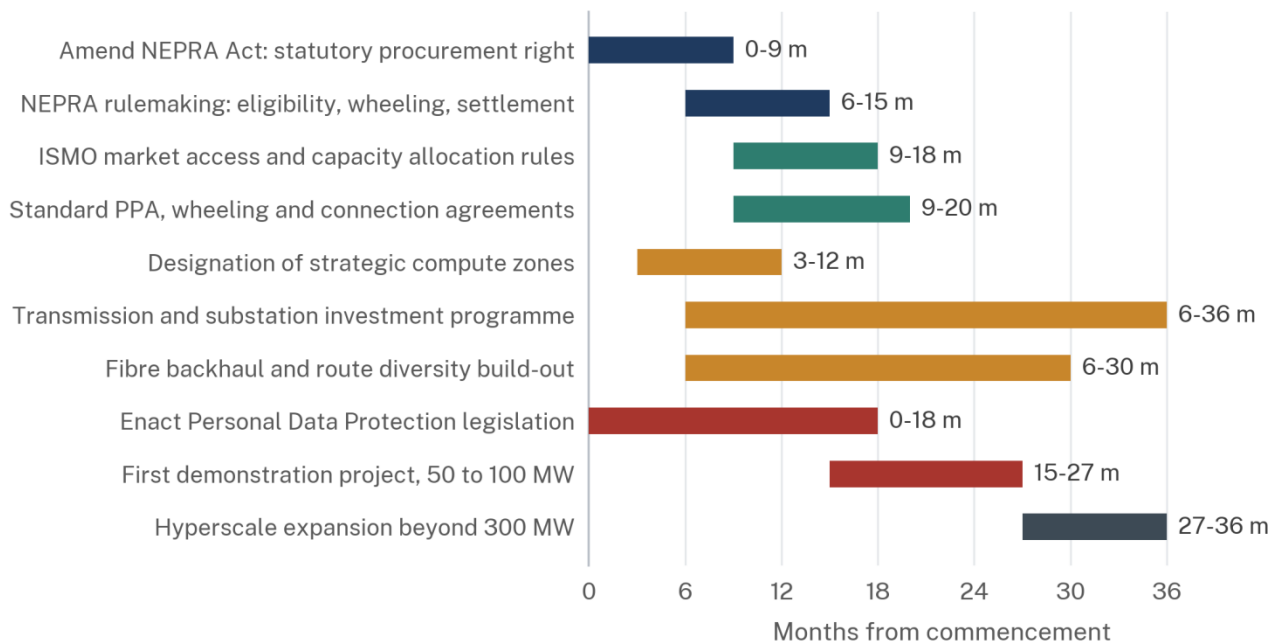
This report contains no estimate of digital service export revenue from data centre capacity, no multiplier analysis, and no employment forecast beyond an order of magnitude. Revenue depends on commercial terms with customers who do not yet exist and on a global compute market whose pricing is volatile. Data centres are capital-intensive and employment-light; anyone presenting large direct job numbers for a hyperscale campus is describing the construction phase, not the operating phase. The honest case for this policy rests on foreign exchange earnings and on the fixed-cost recovery arithmetic above, not on employment.

16. A Roadmap from Law to Data Centre

The sequencing matters as much as the content, for one reason: the legal work must be complete before construction begins, not alongside it. An investor will not commit capital against a statutory amendment that is expected. It will commit against one that has been enacted.

Figure 32. The Legal Work Must Finish Before the Concrete Starts, Not Alongside It

ARS implementation roadmap, from legislative amendment to hyperscale operation, months from commencement



Source: ARS policy proposal. Sequencing is indicative and assumes political commitment at the outset. The critical path runs through the statutory amendment and the standard contract package; transmission and fibre investment run in parallel but do not gate the first demonstration project.

16.1 Months 0 to 6

Introduce the amendment bill to the NEPRA Act, 1997 covering the new section 22A, the amendment to section 14C(2) and the amendment to section 27. In parallel, and requiring no legislation, NEPRA begins work on the long-term tariff and use of system methodology and on the replacement wheeling regulations. The Federal Government begins the process of designating strategic compute zones and secures provincial concurrence. The State Bank issues the repatriation circular. Consolidation of the market and system operator licences into ISMO proceeds.

16.2 Months 6 to 18

Enactment of the NEPRA Act amendments and of the Personal Data Protection legislation. NEPRA notifies the eligibility regulations, the tariff methodology and the replacement wheeling regulations, and begins consultation on the standby and replacement power

product. ISMO publishes the market access procedure, the transmission capacity register and the curtailment ranking. The grid company publishes node capacity and the connection procedure. Zone designation is notified. The inter-governmental protocol on land, water, environment and jurisdiction is agreed through the Council of Common Interests.

16.3 Months 18 to 36

NEPRA approves the standard PPA, wheeling agreement and connection agreement. The first demonstration project, at 50 MW to 100 MW in the Islamabad and Rawalpindi corridor, reaches financial close using those standard instruments and a real project finance structure, which is the only genuine test of whether the reform worked. Transmission reinforcement continues, including HVDC restoration and the ADB-financed northern evacuation programme. Fibre backhaul diversity is procured. Hyperscale expansion beyond 300 MW begins only once the demonstration project has been financed on the new instruments without a sovereign guarantee.

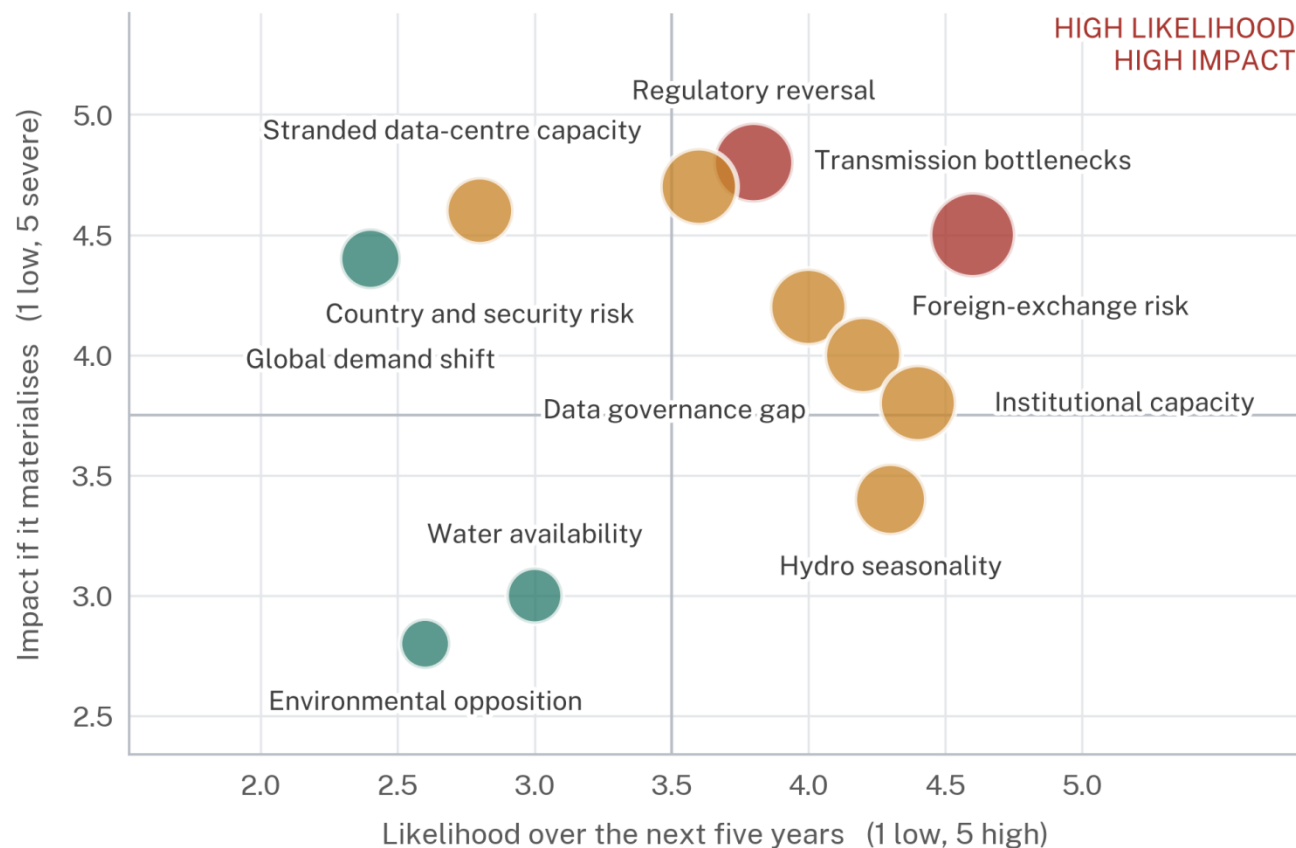
THE TEST THAT MATTERS

The reform has succeeded when an international lender finances a Pakistani data centre power contract without a sovereign guarantee. Until that happens, every other indicator, including announcements, memoranda of understanding and allocated megawatts, measures intention rather than achievement.

17. Risks, Limits and the Possibility of New Stranded Assets

Figure 33. The Two Risks That Would Kill the Proposition Are Regulatory Reversal and the Transmission Corridor

ARS risk assessment; bubble area is proportional to likelihood multiplied by impact



Source: ARS assessment. Likelihood and impact are analytical judgements applied consistently across risks on a common five-point scale, not probabilities. Stranded data centre capacity is the risk that Pakistan builds compute capacity it cannot sell, replacing one stranded asset problem with another.

17.1 The Two Risks That Would Defeat the Proposition

Regulatory reversal ranks highest on impact because the entire proposition is a bet on the durability of a legal commitment. Pakistan's recent record supplies the reason for concern: IPP contracts covering 2,829 MW terminated or renegotiated in a single year, roughly Rs 260 billion of circular debt reduction attributed to renegotiation, and a new statutory levy imposed on an existing category of investment with a legislated escalation to 20%.^{2,12,23} Each measure has a defence. Their cumulative signal to a prospective investor is that Pakistani contractual commitments are revisable when fiscal pressure requires it. The only answer is to place the commitments that matter in primary legislation and to accept the constraint that entails.

Transmission bottlenecks rank highest on likelihood because they are already occurring. The HVDC link operating at roughly 35% of design rating, a northward HVAC ceiling of approximately 4,500 MW and 1,845 MW of southern generation constrained off daily are

present tense facts, not projections.²⁰ No legal instrument relieves them, and the reinforcement programmes are multi-year.

17.2 The Remaining Risks

Table 16. Risk Register

Category, mechanism, mitigation and the party best placed to bear it

Risk	How it would materialise	Mitigation	Best borne by
Regulatory reversal	Amendment, levy or renegotiation alters the economics after financial close	Primary legislation; change-of-law clauses; stabilisation in the zone agreement	The state
Transmission bottleneck	Contracted energy cannot be delivered to site; curtailment	Site selection near generation; capacity register; corridor reinforcement	Network company
Institutional capacity	NEPRA, ISMO and the grid company cannot execute the programme on schedule	Sequenced roadmap; technical assistance; a single accountable programme office	The state
Hydro seasonality	Winter output falls below contracted volume	Firming and replacement power contracts; hydro as energy not capacity	Generator and firming provider
Foreign exchange risk	Rupee movement alters the rupee cost of dollar-indexed tariff components	Partly a natural hedge: the investor earns dollars	Shared, and structurally favourable
Country and security risk	Security incident or macroeconomic instability affects operations or insurance	Political risk insurance; site selection; phased commitment	Investor and insurers
Water availability	Cooling water unavailable or politically contested	Closed-loop or air-cooled design; disclose the trade-off early	Investor, by design choice
Environmental opposition	Assessment or public hearing delays or blocks the project	Pre-cleared zones with completed assessment	The state, through zone designation
Data governance gap	No enacted data protection statute deters both operators and customers	Enact the pending Bill; publish a cross-border transfer position	Parliament
Global demand shift	AI compute demand slows or capacity moves to cheaper markets	Phased build; contract capacity before constructing it	Investor

Risk	How it would materialise	Mitigation	Best borne by
Stranded compute capacity	Pakistan builds capacity it cannot sell	Anchor tenant commitments before construction; build in phases	Investor and the state jointly

Source: ARS assessment. Likelihood and impact ratings correspond to the positions plotted at Figure 33 and are analytical judgements applied on a common scale, not probabilities.

17.3 Could Pakistan Solve One Stranded Asset Problem Only to Create Another?

This question deserves a direct answer because it is the strongest objection to everything this report proposes, and because Pakistan has made precisely this mistake before.

The parallel is exact enough to be uncomfortable. In the 1990s and 2000s Pakistan built generation capacity against a demand forecast, guaranteed the revenue, and discovered that the demand did not arrive. The result is the capacity charge described in Part I. A state that now guarantees offtake, land, tariff concessions or minimum revenue to data centre developers against a forecast of global compute demand would be running the same experiment with different equipment.

Three features of the proposal in this report are designed to prevent that, and they should be treated as conditions rather than as commentary.

The first is that the state guarantees nothing. The reform package in Section 10 contains no offtake guarantee, no minimum revenue undertaking, no capacity payment and no sovereign wrap. It confers rights and removes discretion. If the compute market softens, the investor bears the loss, which is the correct allocation because the investor selects the market.

The second is that the load is contracted before the plant is built rather than after. The stranded capacity problem in Pakistani generation arose because plant was contracted against forecast demand. A data centre that signs anchor tenant commitments before construction is contracting against actual demand. The roadmap in Section 16 places the demonstration project after the legal instruments precisely so that this discipline is tested at small scale.

The third is that the electricity contracts must not create new capacity obligations. This is the condition stated at Section 2.7 and it bears repeating as a risk control: if serving a data centre requires new generation to be contracted on capacity payment terms, Pakistan has repeated the original error at a new address. The load must be served from existing available capacity, and the reform should be drafted so that a registered strategic large load cannot be used to justify new take-or-pay generation.

Subject to those three conditions, the answer is that the risk is real but is materially lower than in the generation case, because the party bearing it is a private investor selecting into a global market rather than a public utility responding to a national forecast. Without those three conditions, the answer is that Pakistan would be building the same problem again, and this report would not support it.

Conclusion

Pakistan does not need to invent a data centre electricity market from scratch. It needs to complete, clarify and operationalise the competitive electricity architecture it has already begun to build.

The components are in place and have been for some years. Sections 23A to 23H of the NEPRA Act create market operator, trader, supplier and system operator licences. Section 22 contemplates sale to bulk power consumers. Section 14C confers a right of open access. The Wheeling Regulations of 2016 permit nationwide bilateral sale above 1 MW. The Competitive Trading Bilateral Contract Market was launched in January 2026 and a first competitive auction of approximately 800 MW is scheduled for September 2026. Wheeling charges have fallen from above Rs 22 per unit to below Rs 9 per unit. Installed capacity has been rationalised from 45,888 MW to 41,121 MW. This is a country that has been doing the work.

What it has not done is convert that work into something an international lender can take security over. Section 22 says the Authority may permit. Section 14C(2) says the right exists subject to a determination by the counterparty. The wheeling charge falls and rises administratively. There is no standard contract, no published curtailment ranking, no replacement power product, no long-term tariff methodology and no pre-approved assignment. Pakistan has built a permission architecture where a rights architecture is required, and permissions cannot be financed.

The division of labour that would fix this is straightforward and can be set out in full below.

Parliament creates the legal right, through the new section 22A, the amendment to section 14C(2), the amendment to section 27 and the completion of the Personal Data Protection legislation. NEPRA creates the rules, through eligibility regulations, a binding long-term tariff and use of system methodology, replacement wheeling regulations, a standby and replacement power product and approved standard contracts. ISMO operates the market, publishing access procedures, a transmission capacity register, a curtailment ranking with defined compensation, and settlement rules capable of matching a flat load to a seasonal generator. The grid makes the right physically deliverable, through HVDC restoration, northern evacuation reinforcement, published node capacity and binding connection timelines. Contracts make it bankable, through six interlocking agreements whose risk allocation places each exposure with the party best able to control it. Government makes the sites investable, through zone designation, single-window permitting, a cross-border data transfer position, a State Bank repatriation confirmation and an inter-governmental protocol resolving land, water, environmental and jurisdictional questions. And data centres turn electricity into digital exports, converting a fixed charge that is currently a pure liability into revenue denominated in the currency the country most needs.

The scale of what is at stake justifies the effort. A 1,000 MW campus would purchase 11.39 TWh a year, roughly a tenth of current grid sales, from a single creditworthy customer

paying in dollars, with a load shape requiring no new peaking plant. Contracted at a delivered price of 8 to 10 US cents, it would carry between Rs 46 billion and Rs 91 billion a year toward a capacity charge presently recovered entirely from domestic consumers, reducing the capacity component of every other consumer's tariff by Rs 0.36 to Rs 0.72 per unit, with no subsidy and no new generation. That is less than this policy is usually sold with, and it is the version that holds.

None of this is guaranteed, and this report has been careful not to present it as though it were. The hydro is seasonal and one major station will be out of service until 2028. The corridor is constrained and 1,845 MW of southern generation is presently constrained off daily. Every submarine cable lands at a single city. The data protection statute is not enacted. The recent record on contractual durability is adverse. Any of these could defeat the proposition on its own.

But the binding constraint is none of them. On the ARS bankability assessment, Pakistan scores 24 out of 75 on the questions a hyperscaler's lenders actually ask, and the reform package set out in this report would take it to 64. What separates those two numbers is not generation capacity, hydrology, geography or global demand. It is a set of legal instruments that cost nothing to draft and require a political decision rather than capital. Pakistan is currently paying more than two trillion rupees a year for the availability of plant it does not use, and declining to make the legal changes that would allow someone to use it. That is the choice, and it is not a technical one.

Annex A: Statutory and Regulatory Provisions Referenced

The consolidated legal matrix appears at Table 10 in Section 8.7 and is not repeated here. This annex records the principal instruments cited in the report, in the form in which they were consulted.

Regulation of Generation, Transmission and Distribution of Electric Power Act, 1997 (Act XL of 1997), as amended, principally by the Regulation of Generation, Transmission and Distribution of Electric Power (Amendment) Act, 2018 (Act XII of 2018). Sections referenced: 2(ii), 2(iia), 7(4), 7(5), 12G, 14B, 14C, 14D, 20, 21(2)(a), 22, 23A to 23H, 27, 31, 32, 45, 46, 47.

NEPRA (Wheeling of Electric Power) Regulations, 2016, notified vide SRO 549(I)/2016. Off the Grid (Captive Power Plants) Levy Act, 2025 (Act XIV of 2025), sections 2(e), 3, 4, 5, 6 and 9. Oil and Gas Regulatory Authority Ordinance, 2002, sections 8 and 43B, as referenced in Act XIV of 2025. Public Finance Management Act, 2019, as referenced in section 6 of Act XIV of 2025. Prevention of Electronic Crimes Act, 2016. Digital Nation Pakistan Act, 2025. Personal Data Protection Bill (pending). Framework Guidelines for Wheeling Auctions 2025. Competitive Trading Bilateral Contract Market design as approved by NEPRA. Water Apportionment Accord, 1991.

Comparator jurisdiction: Electricity Act, 2003 (India), sections 42, 49, 176 and 181; Electricity (Promoting Renewable Energy Through Green Energy Open Access) Rules, 2022 (India), notified 6 June 2022, together with the amendment rules of 2023.

Annex B: Definitions, Units and Conventions

Capacity is a rate, measured in megawatts (MW) or gigawatts (GW). Energy is a quantity, measured in megawatt-hours (MWh), gigawatt-hours (GWh) or terawatt-hours (TWh). One GW sustained for one hour is one GWh. One thousand GWh is one TWh. A billion units, the term used in WAPDA reporting, equals one TWh.

Installed capacity is the aggregate nameplate rating of all connected plant. Available capacity is the portion that can be dispatched at a given moment after outages, fuel constraints and hydrology. Dependable capacity is the capacity that can be relied upon under defined conditions and is generally lower than nameplate. Load factor is average load divided by peak load, or equivalently energy delivered divided by the energy that would be delivered at continuous full output. Capacity factor is the equivalent measure applied to a generating unit. Availability is distinct from both: it is the proportion of time a unit is capable of generating, irrespective of whether it is dispatched and irrespective of whether fuel or river flow is present. This report uses three availability figures for three different things: the design availability of a data centre campus at its own meter, the availability of supply contracted at the point of delivery, and the availability a generator commits to under a power purchase agreement. Power usage effectiveness (PUE) is total facility electrical load divided by IT load; water usage effectiveness (WUE) is the corresponding water measure.

Capacity payment remunerates availability. Energy payment remunerates units delivered. Take-or-pay, in Pakistani PPA practice, describes an obligation to pay for declared availability rather than an obligation to take a specified volume. Wheeling is the transport of electricity over a network owned by a third party. Open access is the right to use that network on non-discriminatory terms. Use of system charges are the charges levied for that transport. Captive power is generation established primarily for the owner's own use. Trading, under sections 23C and 23D, is the licensed purchase and resale of electric power; supply, under sections 23E and 23F, is the licensed sale of electric power to consumers.

All rupee figures are Pakistani rupees unless otherwise stated. Financial years run from 1 July to 30 June and are stated as, for example, FY 2024-25. WAPDA generation figures are reported on a calendar year basis and are identified as such. Where a figure is an ARS calculation from published data, the chart or table source note says so. Where a figure is an illustrative scenario, the chart or table source note says so. No figure in this report is presented as an empirical forecast. Where the report proposes statutory language, that language is identified as an ARS policy drafting proposal and is neither enacted legislation nor a legal opinion.

References

- 1 NEPRA, State of the Industry Report 2025 (net generation series 2021-22 to 2024-25, as reported). See also "Affordable power", The News, June 2026.
<https://www.thenews.pk/magazine/money-matters/1421798-affordable-power>
- 2 "NEPRA releases 2025 State of Industry and 2024-25 Annual Reports for federal review", Profit by Pakistan Today, 16 January 2026 (circular debt stock, recovery ratio, T&D losses). <https://profit.pakistantoday.com.pk/2026/01/16/nepra-releases-2025-state-of-industry-and-2024-25-annual-reports-for-federal-review>
- 3 "Neptra determines power purchase price of Rs3,342.36 billion for FY2025-26", The Nation, 24 June 2025 (power purchase price determination and CPPA-G capacity charge scenarios). <https://www.nation.com.pk/24-Jun-2025/nepra-determines-power-purchase-price-of-rs3-342-36-billion-for-fy2025-26>
- 4 "Power sector mired in inefficiency: Neptra", The News, January 2026 (installed capacity, peak demand, plant retirements, PLAC, Guddu and Neelum-Jhelum performance). <https://www.thenews.pk/print/1393452-power-sector-mired-in-inefficiency-neptra>
- 5 International Energy Agency, Energy and AI, Executive Summary (global data centre electricity consumption of 415 TWh in 2024 rising to around 945 TWh by 2030). <https://www.iea.org/reports/energy-and-ai/executive-summary>
- 6 International Energy Agency, Key Questions on Energy and AI, Executive Summary (updated projection of 485 TWh in 2025 rising to 950 TWh in 2030). <https://www.iea.org/reports/key-questions-on-energy-and-ai/executive-summary>
- 7 Regulation of Generation, Transmission and Distribution of Electric Power Act, 1997 (Act XL of 1997), as amended, principally by Act XII of 2018. Ministry of Energy (Power Division) consolidated text. <https://power.gov.pk/siteimage/policy/3-nepraact1997.pdf>

- 8 "Neptra allows wheeling of electricity regulations", The News, June 2016 (NEPRA Wheeling of Electric Power Regulations, 2016, notified vide SRO 549(I)/2016). <https://www.thenews.com.pk/print/126759-Neptra-allows-wheeling-of-electricity-regulations>
- 9 "The CTBCM launch", Business Recorder editorial, 2026 (initial trading tranche and movement in wheeling charges). <https://www.brecorder.com/news/40409581/the-ctbcm-launch>
- 10 "Pakistan's power sector reforms deliver Rs143bn savings", Mettis Global, 12 June 2026 (formal CTBCM launch in January 2026; first competitive auction of approximately 800 MW scheduled for September 2026). <https://mettisglobal.news/Pakistans-power-sector-reforms-deliver-Rs143bn-savings-61088>
- 11 "800MW of electricity to be allocated in auction", The Express Tribune, 25 December 2025 (Framework Guidelines for Wheeling Auctions 2025; NEPRA to determine uniform wheeling charges; ISMO to run the auction). <https://tribune.com.pk/story/2583839/guidelines-for-electricity-wheeling-approved>
- 12 Off the Grid (Captive Power Plants) Levy Act, 2025 (Act XIV of 2025). The Pakistan Code. <https://pakistancode.gov.pk/english/UY2FqaJw1-apaUY2Fqa-apaUY2Npa5tlaA==sg-jjjjjjjjjjjj>
- 13 "Captive power gas: Govt extends monthly levy to third-party suppliers", Business Recorder, January 2026 (sections 2(e), 3, 4 and 9 of Act XIV of 2025; escalation schedule). <https://www.brecorder.com/news/amp/40400447>
- 14 "Pakistan raises gas levy on captive power plants", ARY News, April 2026 (levy raised to Rs 1,406 per MMBtu). <https://arynews.tv/gas-levy-increased>
- 15 Water and Power Development Authority, "2025: WAPDA Generates 33.12 Billion Units Hydel Electricity", 30 December 2025 (station-by-station generation and average cost per unit). <https://wapda.gov.pk/2025-wapda-generates-33-12-billion-units-hydel-electricity/>
- 16 "Neelum-Jhelum set to resume generation by March 2028", Dawn, 11 July 2026 (Senate Standing Committee on Water Resources briefing). <https://www.dawn.com/news/2014508>
- 17 "Neelum-Jhelum revival delayed until 2028", The Express Tribune, 10 July 2026 (WAPDA evidence on tunnel failure and cost recovery). <https://tribune.com.pk/story/2617645/neelum-jhelum-revival-delayed-until-2028>
- 18 Renewables First, Pakistan Electricity Review 2026 (grid sales, consumer growth, thermal and hydel capacity movement in FY 2024-25). <https://uploads.renewablesfirst.org/Pakistan%20Electricity%20Review%202026.pdf>
- 19 REN21, Global Status Report 2025, Pakistan snapshot (net-metered rooftop solar deployment and grid demand series). <https://www.ren21.net/gsr-2025/snapshots/pk/>
- 20 Khalid Waleed, Sustainable Development Policy Institute, "Pakistan's Power Grid Development Status", presented at UNESCAP (dependable capacity, HVDC utilisation, constrained southern generation, duck curve dispatch). <https://www.unescap.org/sites/default/d8files/event-documents/Pakistan%27s%20grid%20Khalid%20Waleed,%20SDPI.pdf>
- 21 Renewables First, Pakistan Electricity Review 2025 (capacity payment growth and generation mix). https://uploads.renewablesfirst.org/Pakistan_Electricity_Review_2025_89f0b613d6.pdf

- 22 NEPRA, State of the Industry Report 2023 (Part Load Adjustment Charges, surplus take-or-pay capacity, capacity purchase price). <https://nepra.org.pk/publications/State%20of%20Industry%20Reports/State%20of%20Industry%20Report%202023.pdf>
- 23 "PMO steps in after NEPRA raises concerns over circular debt, capacity payments in State of Industry Report", Profit by Pakistan Today, 12 February 2026 (Power Division response; decomposition of the circular debt reduction; surplus capacity of about 8,700 MW). <https://profit.pakistantoday.com.pk/2026/02/12/pmo-steps-in-after-nepra-raises-concerns-over-circular-debt-capacity-payments-in-state-of-industry-report/>
- 24 "Govt allocates 2,000MW for Bitcoin mining and AI data centres", Dawn, 25 May 2025 (Finance Division statement; Africa-2 cable landing). <https://www.dawn.com/news/1913238>
- 25 "Nepra backs ISMO for streamlining wholesale electricity market", Business Recorder, September 2024 (transfer of market operator and system operator licences under sections 23A, 23B, 23G and 23H). <https://www.brecorder.com/news/40321311/nepra-backs-ismo-for-streamlining-wholesale-electricity-market>
- 26 "Govt asks for Rs1.15/unit cut in electricity rate", Dawn, 30 June 2025 (national average tariff determination for FY 2025-26 and subsidy allocation). <https://www.dawn.com/news/1921131>
- 27 NEPRA, Performance Evaluation Report: Transmission, FY 2024-25 (congestion and overloading of 500 kV and 220 kV assets; causes of project delay). <https://nepra.org.pk/M&E/PER/Transmission/2025/PER%20Transmission%20FY%202024-25.pdf>
- 28 "ADB approves \$330m package to modernise Pakistan's power transmission network", The Express Tribune, 21 November 2025. <https://tribune.com.pk/story/2578311/adb-approves-330m-loan-to-strengthen-pakistans-power-transmission-network>
- 29 World Bank, Pakistan power sector programme document (NTDC unbundling and transmission system operator creditworthiness). <https://documents1.worldbank.org/curated/en/099070425094019580/pdf/P504536-834e23a9-f4d5-43e9-be47-4f9ebbbf0533.pdf>
- 30 "Pakistan imposes Rs238 per MMBtu levy on captive power plants to meet IMF condition", The Nation, 26 July 2025. <https://www.nation.com.pk/26-Jul-2025/pakistan-imposes-rs238-per-mmbtu-levy-on-captive-power-plants-to-meet-imf-condition>
- 31 NEPRA, Hydel Potential in Pakistan (provincial distribution of installed and prospective hydropower capacity). <https://nepra.org.pk/Policies/Hydel%20Potential%20in%20Pakistan.pdf>
- 32 "Neelum-Jhelum project 'a failure of planning, execution'", Dawn, 3 September 2025 (Auditor-General of Pakistan report; PC-I revisions and cost overrun). <https://www.dawn.com/news/1939326>
- 33 NEPRA, State of Industry Report 2025, company-wise generation data for FY 2024-25 (monthly generation by station, with installed and dependable capacity). <https://nepra.org.pk/publications/State%20of%20Industry%20Reports/SOI%20Data/List%20of%20Companies%20Geneneration%20wise%202024-25%20updated.htm>
- 34 "Arbitration court upholds Pakistan's stance on Indus Waters Treaty", The Express Tribune, 17 May 2026 (Supplemental Award on maximum pondage, 15 May 2026; India's response). <https://tribune.com.pk/story/2608624/arbitration-court-upholds-pakistans-stance-on-indus-waters-treaty>

- 35 "India says Indus Waters Treaty to remain in abeyance until Pakistan ends support to terrorism", News on AIR, 20 March 2026. <https://www.newsonair.gov.in/india-says-indus-waters-treaty-to-remain-in-abeyance-until-pakistan-ends-support-to-terrorism>
- 36 Aceris Law, "Indus Waters Treaty Arbitration: Can India Put the Treaty in Abeyance?", May 2026 (procedural history and the abeyance question). <https://www.acerislaw.com/indus-waters-treaty-arbitration-can-india-put-the-treaty-in-abeyance/>
- 37 "Pakistan Officially Connects to One of the World's Fastest Undersea Internet Cables", Silver Lining, December 2025 (Transworld Associates Tier III data centre in Karachi connected to the subsea systems). <https://silverlining.org.pk/pakistan-officially-connects-to-one-of-the-worlds-fastest-undersea-internet-cables/>
- 38 "How Pakistan joins the world's fastest undersea internet network", Communications Today, 24 November 2025 (SEA-ME-WE 6 landing; initial 4 Tbps addition against a national total of 13.2 Tbps). <https://www.communicationstoday.co.in/how-pakistan-joins-the-worlds-fastest-undersea-internet-network/>
- 39 Submarine Networks, "Transworld Associates to Land 2Africa Subsea Cable System in Pakistan" (2Africa Pearls branch; 45,000 km, 46 landing stations, 33 countries, up to 180 Tbps design capacity; existing Pakistani systems). <https://www.submarinenetworks.com/en/systems/asia-europe-africa/2africa/transworld-associates-to-land-2africa-subsea-cable-system-in-pakistan>
- 40 "IT exports hit record USD4.6bn", Business Recorder, 18 July 2026 (State Bank of Pakistan data: FY 2025-26 IT and telecom export remittances of USD 4.6 bn against USD 3.814 bn in FY 2024-25). <https://www.brecorder.com/news/40430631/it-exports-hit-record-usd46bn>
- 41 "Pakistan Targets \$25.1 Billion ICT Export Revenue by 2030", TechJuice, August 2026 (ICT export series FY 2020-21 to FY 2025-26 and the official 2030 target). <https://www.techjuice.pk/pakistan-targets-25-1-billion-ict-export-revenue-by-2030/>
- 42 Ministry of Power, Government of India, Press Information Bureau release on the Electricity (Promoting Renewable Energy Through Green Energy Open Access) Rules, 2022, notified 6 June 2022 under section 176 of the Electricity Act, 2003. <https://www.pib.gov.in/PressReleaselframePage.aspx?PRID=1842737>
- 43 Trilegal, "Electricity (Promoting Renewable Energy Through Green Energy Open Access) Rules, 2022" (reduction of the open access threshold from 1 MW to 100 kW; 15-day deemed approval timeline; central nodal agency and single-window registry). https://trilegal.com/knowledge_repository/electricity-promoting-renewable-energy-through-green-energy-open-access-rules-2022/
- 44 Institute for Energy Economics and Financial Analysis, "Impact of Green Energy Open Access Rules, 2022" (uneven state adoption; procedural bottlenecks; additional state-level charges). On the jurisdictional point see the Karnataka High Court ruling on sections 42(2) and 181 of the Electricity Act, 2003. <https://ieefa.org/resources/impact-green-energy-open-access-rules-2022>
- 45 Chambers and Partners, Data Protection & Privacy 2026: Pakistan (status of the Personal Data Protection Bill 2023; cross-border transfer and localisation provisions; absence of an enacted statute as at January 2026). <https://practiceguides.chambers.com/practice-guides/data-protection-privacy-2026/pakistan>

- 46 Regulations.ai, Pakistan AI Regulation Overview, January 2026 (Digital Nation Pakistan Act 2025; National AI Policy 2025; Pakistan Digital Authority; status of the Personal Data Protection Bill and the Regulation of Artificial Intelligence Bill). <https://regulations.ai/regulations/RAI-PK-NA-SUMMARY-2026>
- 47 "IT ministry unveils draft data governance policy", Dawn, 1 July 2026 (draft National Data Governance Policy 2026 released for consultation to 10 July 2026; alignment with a future Personal Data Protection law). <https://www.dawn.com/news/2011804>
- 48 "Pakistan unveils National Data Governance Policy 2026 to transform digital governance", Business Recorder, 29 June 2026 (data sovereignty, cross-border transfer, privacy impact assessments and AI explainability requirements). <https://www.brecorder.com/news/amp/40427660>
- 49 OECD, Recommendation of the Council concerning Guidelines Governing the Protection of Privacy and Transborder Flows of Personal Data, OECD/LEGAL/0188, adopted 23 September 1980 and revised 11 July 2013 (collection limitation, purpose specification, security safeguards and accountability, applied to data that crosses borders). <https://legalinstruments.oecd.org/public/doc/114/114.en.pdf>
- 50 Regulation (EU) 2016/679 (General Data Protection Regulation), article 45 (adequacy may be recognised in respect of a third country, a territory or one or more specified sectors within it). <https://eur-lex.europa.eu/eli/reg/2016/679/oj>
- 51 European Commission, adequacy decisions (the process by which a non-EU country, territory or sector is recognised as ensuring an adequate level of protection). https://commission.europa.eu/law/law-topic/data-protection/international-dimension-data-protection/adequacy-decisions_en
- 52 European Data Protection Board, Recommendations 01/2020 on measures that supplement transfer tools to ensure compliance with the EU level of protection of personal data, version 2.0, adopted 18 June 2021 (the essential equivalence assessment applied after Court of Justice of the European Union, Case C-311/18, Data Protection Commissioner v Facebook Ireland and Schrems, judgment of 16 July 2020). https://www.edpb.europa.eu/our-work-tools/our-documents/recommendations/recommendations-012020-measures-supplement-transfer_en
- 53 Data Protection Law, DIFC Law No. 5 of 2020, Dubai International Financial Centre legal database (zone-specific data protection regime, independent of the surrounding federal framework and modelled on the GDPR). <https://www.difc.com/business/laws-and-regulations/legal-database/difc-laws/data-protection-law-difc-law-no-5-2020>
- 54 Special Technology Zones Authority Act, 2021 (zone-specific fiscal and regulatory regime for designated technology zones). Special Technology Zones Authority. <https://stza.gov.pk/storage/2022/01/STZA-Act-2021.pdf>
- 55 Provisions on Promoting and Regulating Cross-Border Data Flows, Cyberspace Administration of China, effective 22 March 2024 (pilot free trade zones may formulate their own negative lists for data export, applicable within the zone rather than nationwide); English translation. <https://www.chinalawtranslate.com/en/Provisions-on-Promoting-and-Regulating-the-Cross--Border-Flow-of-Data/>
- 56 Lambda, published on-demand GPU cloud pricing (NVIDIA H100 SXM listed at USD 3.99 per GPU-hour; rates across providers span a wider band). <https://lambda.ai/service/gpu-cloud>
- 57 "Crypto mining, other sectors: IMF rejects subsidised power tariffs proposal", Business Recorder, 3 July 2025 (proposed marginal-cost-based rate of Rs 22 to Rs 23 per unit for

data centres, cryptocurrency mining and metal melting; market distortion and tax holiday objections). <https://www.brecorder.com/news/40370863/crypto-mining-other-sectors-imf-rejects-subsidised-power-tariffs-proposal>

About the Authors

Reza Khan

Reza Khan is a researcher with an M.Phil. in Government and Public Policy from the National Defence University, Islamabad, and an MBA. His professional experience spans business strategy, international trade, economics and organizational leadership. His analytical interests include public policy, governance, strategic communication and the interaction between technology, institutions and society.

Waqas Aziz Qureshi, Barrister

Waqas Aziz Qureshi is a Barrister with extensive experience in dispute resolution, corporate advisory, regulatory matters and infrastructure-related legal practice. He has been recognised by Chambers & Partners and The Legal 500 Asia as a leading individual in dispute resolution and corporate advisory. His practice encompasses complex commercial disputes, arbitration, regulatory and constitutional matters, corporate structuring, infrastructure projects and public sector engagements. He has also contributed to discussions on Pakistan's investment and regulatory environment, including matters relating to CPEC, special economic zones and economic development.